Operlytics: Development and Evaluation of a Business Analytics and Monitoring System for Improving Operational Decision-Making.

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ABSTRACT

This study presents Operlytics, a web-based business analytics and monitoring system developed to improve the operational decision-making of Empress Café Boracay. The system was developed in response to challenges in manual sales recording, inventory monitoring, and report generation, which may result in inaccurate records, stock shortages, wastage, and delays in decision-making. Using an Agile development methodology, the system integrates point-of-sale transaction recording, inventory management, reporting and dashboard visualization, and user and access management into a single platform. Operlytics provides real-time monitoring of sales and inventory by automatically recording transactions and deducting ingredient quantities based on menu items sold. It also generates sales reports and dashboard summaries to help management monitor business performance and make informed operational decisions. The system was evaluated based on functionality, usability, accuracy, security, and efficiency. The findings indicate that Operlytics fulfills the identified system objectives and provides a practical approach to organizing operational data and supporting the daily management activities of Empress Café Boracay. The study recommends continued user evaluation and future enhancements to the system's reporting, forecasting, and security capabilities.

Index Terms— Business Analytics, Inventory Management, Operational Decision-Making, Point-of-Sale, Sales Monitoring, Web-Based System

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The food service and hospitality industry stands as one of the most dynamic, competitive, and rapidly evolving sectors in the contemporary global economy. This industry encompasses a vast array of business establishments ranging from small family-owned cafés and restaurants to large multinational hotel chains and catering conglomerates. Within this diverse landscape, small and medium-sized enterprises (SMEs) in the food service sector occupy a particularly significant position, contributing substantially to local employment generation, economic growth, and the overall vitality of communities worldwide.

The economic significance of food service SMEs cannot be overstated. In developing and emerging economies, these establishments often serve as the backbone of local commerce, providing not only employment but also essential services that support the broader tourism ecosystem. The food service industry creates a complex web of economic interdependencies that strengthen local supply chains, promote agricultural productivity, and foster entrepreneurship at the grassroots level. For many communities, particularly those in tourist destinations, local cafés and restaurants represent more than mere commercial enterprises; they embody cultural identity, serve as social gathering spaces, and function as ambassadors of local cuisine and hospitality traditions to visitors from around the world.

Despite their substantial economic contribution, many small food service establishments face persistent operational and managerial challenges that significantly hinder their long-term sustainability and growth potential. These challenges are multifaceted and interconnected, stemming from resource limitations, workforce constraints, competitive pressures, and increasingly, the technological gap that separates small enterprises from their larger, more digitally advanced counterparts. The operational difficulties encountered by small cafés and restaurants are particularly acute in environments characterized by high customer turnover, seasonal demand fluctuations, perishable inventory requirements, and the need for rapid service delivery.

In the Philippines, the food service industry occupies a position of critical importance in both national and local economic frameworks. The archipelagic nature of the country, combined with its rich culinary heritage and the warmth of Filipino hospitality, has created a vibrant and diverse food service landscape that ranges from humble street food vendors to sophisticated fine dining establishments. The Philippine food service sector has demonstrated remarkable resilience and adaptability, weathering economic downturns, natural disasters, and most recently, the unprecedented challenges posed by the COVID-19 pandemic.

Boracay Island, consistently ranked among the top beach destinations in Southeast Asia and indeed the world, presents a particularly compelling case study for examining the operational challenges faced by small food service establishments in tourism-intensive environments. This small island in the province of Aklan, Western Visayas, attracts millions of tourists each year, drawn by its powdery white sand beaches, crystal-clear turquoise waters, and vibrant nightlife. The tourism industry on Boracay Island has experienced remarkable growth over the past two decades, evolving from a relatively unknown backpacker destination to a world-class resort island that welcomes visitors from across the globe.

According to data from the Aklan Provincial Tourism Office[1], Boracay Island recorded 2,155,217 tourist arrivals in 2025, generating approximately ₱47 billion in tourism-related revenues. The December 2025 holiday season alone saw 227,828 visitors, representing the highest monthly figure recorded for that year. Domestic tourists dominated arrivals at 1,765,318, followed by 357,510 foreign tourists and 32,425 overseas Filipino workers. The top sources of foreign visitors included South Korea (96,199 arrivals), the United States (48,356), Australia (21,357), Taiwan (20,306), Russia (18,952), and China (17,958). These figures underscore the island's status as a premier international destination and the immense economic opportunities and corresponding operational pressures faced by local businesses.

The sheer volume of tourist traffic places considerable demand on local food service establishments to deliver high-quality products and services consistently. Small business owners in this environment must navigate a complex balancing act, striving to maintain customer satisfaction while simultaneously managing efficient operational processes. This challenge is compounded by several factors inherent to the tourism-driven market context. Limited resources, both financial and human, constrain the ability of small cafés to invest in advanced operational systems. Workforce constraints, including seasonal employment patterns and high staff turnover rates, make it difficult to maintain operational consistency. Perhaps most significantly, the absence of formal business management systems and structured information technology infrastructure leaves many establishments reliant on manual processes that are inherently inefficient and error-prone.

The competitive landscape of Boracay Island's food service market adds another layer of complexity to the operational challenges faced by small establishments. The island hosts hundreds of restaurants, cafés, bars, and food stalls, ranging from local eateries serving traditional Filipino cuisine to international franchises offering familiar global brands. In this crowded marketplace, differentiation becomes crucial for survival and success. Small cafés must compete not only on the quality of their food and beverages but also on service efficiency, ambiance, and the overall customer experience. Operational excellence, therefore, becomes a critical competitive advantage, one that is difficult to achieve without adequate systems and tools for managing daily business activities.

One of the most common and pervasive operational challenges faced by small cafés in tourism-intensive environments like Boracay Island is the continued reliance on manual and informal methods for managing day-to-day business activities. Despite the increasing availability and affordability of digital technologies, many establishments persist in using traditional approaches that have remained largely unchanged for decades. Sales transactions are frequently recorded by hand in paper logbooks or simple notebooks, with each staff member maintaining their own individual records. Inventory tracking relies on physical counts conducted at irregular intervals, with stock levels estimated rather than precisely monitored. Financial reporting, when it occurs at all, takes the form of ad hoc calculations performed on an as-needed basis, often without standardized formats or consistent methodologies.

These manual practices, while deeply ingrained in the operational culture of many small businesses, produce a cascade of negative consequences that undermine operational efficiency and business performance. Human error represents an ever-present risk, with mistakes in arithmetic, data transcription, and record-keeping leading to inaccurate information that distorts management's understanding of business performance. Inconsistency across different staff members' records creates data fragmentation, making it impossible to obtain a unified and accurate view of operational status. The time-consuming

nature of manual processes consumes valuable staff hours that could be better directed toward customer service and other value-adding activities.

The absence of integrated information systems contributes to a range of operational deficiencies that are particularly severe in fast-paced service environments. Organizations lacking integrated information systems face significant disadvantages in terms of data accuracy, operational coordination, and strategic decision-making capacity. The critical importance of information systems in enabling organizations to process, analyze, and act upon operational data in a timely and effective manner cannot be overstated. For small food service establishments, these theoretical insights translate into very practical consequences: poor inventory control leading to stockouts or overstocking, undetected losses due to theft or spoilage, inefficient resource allocation that wastes limited capital, and an inability to identify trends or respond proactively to changing market conditions.

The situation described above is clearly and compellingly evident in the case of Empress Café Boracay, a small café operating within the highly competitive and tourism-driven market of Boracay Island. Empress Café represents a typical example of the many small food service establishments that populate tourist destinations throughout the Philippines and similar emerging economies worldwide. Like its peers, the café currently relies on manual and inconsistent methods for managing its core operational functions, including sales monitoring, inventory tracking, and financial reporting. These manual processes, while perhaps adequate for a business of minimal scale, become increasingly problematic as customer volumes grow and operational complexity increases.

The specific operational deficiencies observed at Empress Café Boracay illustrate the broader challenges faced by small cafés in tourism-intensive environments. Staff members record sales individually without a centralized system, resulting in data fragmentation that makes it impossible to obtain a consolidated view of daily, weekly, or monthly sales performance. Each staff member may use different methods, formats, or levels of detail in their record-keeping, creating inconsistencies that complicate aggregation and analysis. The absence of real-time data visibility means that management cannot monitor sales performance as it occurs, missing opportunities to respond to unexpected demand patterns, promotional opportunities, or operational issues.

Inventory management at Empress Café suffers from similar deficiencies. Stock levels are tracked through paper logbooks or informal notes, with updates occurring sporadically rather than systematically. The disconnect between sales activity and inventory records means that stock movements are not automatically reflected in inventory counts, leading to frequent discrepancies between actual and recorded inventory levels. Management lacks visibility into which items are selling well, which are moving slowly, and which may be approaching expiration or depletion. These gaps have tangible consequences: instances of overstocking tie up limited working capital in excess inventory that may spoil before use; unexpected stockouts lead to customer disappointment and lost sales; unnecessary waste from spoilage or expiration erodes already thin profit margins; and customer service delays resulting from unavailable ingredients damage the café's reputation and competitive position.

The operational challenges at Empress Café Boracay are not unique to this establishment but rather reflect systemic patterns observed across small food service businesses in similar contexts. The café's experience illustrates how the accumulation of seemingly minor operational inefficiencies can compound into significant business problems that threaten

long-term viability. In an environment as competitive and demanding as Boracay Island's tourism market, operational excellence is not merely desirable but essential for survival.

The increasing accessibility of information and communications technology (ICT) has opened unprecedented opportunities for small businesses to adopt digital tools that can streamline operations and substantially improve decision-making capabilities. The past decade has witnessed a democratization of business technology, with powerful software tools becoming available at price points accessible even to micro and small enterprises. Cloud computing has eliminated the need for expensive on-premises server infrastructure. Mobile applications have put sophisticated functionality into the hands of business owners and staff through devices they already own. Open-source software has provided alternatives to costly proprietary solutions. Together, these developments have created conditions favorable for the widespread adoption of business technology by small enterprises that previously could not afford such investments.

Business analytics systems have emerged as particularly powerful instruments for transforming raw operational data into actionable insights that can guide management decision-making. These systems go beyond simple data recording to provide analytical capabilities that help business owners understand patterns, identify trends, detect anomalies, and forecast future conditions. Modern business analytics platforms automatically aggregate data from multiple sources, present information through intuitive visual dashboards, and generate comprehensive reports with minimal manual input. The analytical capabilities embedded in these systems enable businesses to move from reactive management responding to problems after they occur to proactive management anticipating challenges and opportunities before they fully materialize.

Research demonstrates convincingly that businesses leveraging analytics tools are significantly more likely to identify operational inefficiencies, reduce costs, and respond effectively to changing market conditions [2]. For small businesses in tourism-intensive environments, where operational demands are often seasonal, unpredictable, and high-volume, adopting such systems is increasingly becoming a necessity rather than a luxury.

The Philippine context presents both unique challenges and opportunities for the digital transformation of small food service businesses. While the country has made significant strides in digital infrastructure development, with mobile internet penetration reaching high levels and digital payment platforms like GCash and Maya gaining widespread adoption, the digital divide between large corporations and small enterprises remains substantial. Filipino SMEs continue to lag their regional counterparts in digital adoption, though encouraging trends show that among those that invested in technology, a significant majority reported improved profitability. The government of the Philippines has recognized the critical importance of MSME digitalization and has initiated several programs to support this transition. The Department of Trade and Industry has emphasized the importance of digital transformation for MSMEs, highlighting how it can enhance their growth and competitiveness. The MSME Development Plan 2023-2028 focuses on leveraging science, technology, and innovation to support MSMEs in transitioning to digital platforms. These policy initiatives create a supportive environment for small businesses seeking to adopt digital tools, including business analytics systems like the one proposed in this study.

In response to the operational challenges identified at Empress Café Boracay and the broader context of digital transformation opportunities for small food service businesses in the Philippines, this study proposes the development and implementation of Operlytics, a business analytics and monitoring system specifically designed for the unique operational context of this establishment. Operlytics represents a targeted, practical response to the documented deficiencies in manual record-keeping, inventory management, and reporting that currently constrain the café's operational effectiveness and competitive performance.

Operlytics aims to provide café management with a comprehensive, real-time view of key operational metrics that are essential for effective business management. These metrics include sales transactions, which provide insight into revenue performance and product demand patterns; inventory levels, which indicate the availability of ingredients and supplies necessary for operations; product performance, which reveals which menu items are popular and which may need reconsideration; and overall business activity, which encompasses the full range of operational indicators that management needs to monitor. By consolidating these diverse data streams into a unified, accessible platform, Operlytics seeks to eliminate the information silos and data fragmentation that currently impede effective management.

The automation of data collection and reporting represents a core value proposition of the Operlytics system. Rather than requiring staff to manually record transactions and later transcribe them into summary reports, the system captures transactional data at the point of occurrence and automatically processes it into meaningful summaries and visualizations. This automation serves multiple purposes: it eliminates the inefficiencies associated with manual record-keeping, freeing staff time for customer-facing activities; it reduces human error, improving the accuracy and reliability of operational data; and it accelerates the availability of information, enabling management to make decisions based on current rather than historical data.

Usability has been a paramount consideration in the design philosophy underlying Operlytics. Recognizing that small café staff and management may have limited technical expertise and minimal experience with complex software systems, the system has been designed to require no advanced technical knowledge for effective operation. The user interface prioritizes clarity, simplicity, and intuitive navigation, ensuring that users can access the information they need and perform required tasks without extensive training or technical support. This usability focus is not merely a matter of convenience but a practical necessity for ensuring successful adoption and sustained use in a small business environment where dedicated IT support is unavailable.

The significance of this study extends far beyond the immediate operational improvements it seeks to bring to Empress Café Boracay. By demonstrating the feasibility and practical value of a tailored analytics system in a small café setting, this research contributes to the broader discourse on the digital transformation of micro and small enterprises in the Philippines. The study provides a concrete example of how accessible, appropriately scaled technology can address real operational challenges in a resource-constrained business environment. It highlights technology-driven solutions for businesses operating in tourism-dependent localities where operational demands are often seasonal, unpredictable, and high-volume characteristics that make effective operational management both especially challenging and especially valuable.

Furthermore, this study contributes to the academic literature on information systems in small business contexts, a domain that has received less research attention than enterprise-level implementations despite the vastly larger number of

small businesses worldwide. The detailed documentation of system development, implementation, and evaluation processes provides valuable insights for researchers and practitioners interested in similar interventions. The methodological approach, which combines developmental research design with agile implementation practices, offers a model for conducting technology development research in close collaboration with real business partners.

Ultimately, Operlytics represents a meaningful and practical step toward empowering small business owners with the tools needed to optimize operations, improve service quality, and achieve long-term sustainability in an increasingly competitive and technology-driven marketplace. The system embodies the principle that sophisticated analytical capabilities need not be the exclusive domain of large corporations with substantial IT budgets. By demonstrating that effective business analytics can be made accessible, affordable, and appropriate for small enterprises, this study contributes to a more inclusive vision of digital transformation, one that encompasses businesses of all sizes and resource levels.

The successful implementation of Operlytics at Empress Café Boracay would serve as a proof of concept for the broader applicability of similar systems across the thousands of small food service establishments that operate throughout the Philippines and similar emerging economies. Each small café that improves its operational efficiency through technology represents not only a more viable business but also a more stable source of employment, a more reliable contributor to local supply chains, and a more capable competitor in the global tourism marketplace. In this way, the development and evaluation of Operlytics contributes to the larger project of inclusive economic development through digital empowerment.

1.2 Statement of the Problem

General Problem Statement

A. General Problem

Effective operational systems are essential for food service businesses, particularly small cafés operating in competitive and tourism-driven environments. Accurate management of sales transactions, inventory, and business reports enables businesses to respond to seasonal demand, minimize inventory waste, maintain service quality, and support informed decision-making.

Empress Café Boracay currently relies on manual and semi-manual processes for recording sales, monitoring inventory, and preparing reports. These practices can result in recording errors, inconsistent data, delayed inventory updates, stock shortages, and time-consuming report preparation. The lack of centralized and real-time operational information limits management's ability to monitor business performance, identify sales trends, optimize inventory, and make timely decisions.

These operational deficiencies affect not only daily productivity but also the café's ability to maintain service quality, manage resources efficiently, and remain competitive. Therefore, an integrated operational system that improves sales monitoring, inventory tracking, and business reporting can provide a more reliable data foundation for efficient operations and informed management decisions.

The operational challenges at Empress Café Boracay can be decomposed into five specific problem areas, each representing a distinct but interrelated aspect of the café's operational deficiencies. These specific problems provide the detailed foundation upon which the proposed system solution is designed and against which its effectiveness will be evaluated.

B. Specific Problems

Specific Problem 1: Inefficient Manual Record-Keeping Processes

Empress Café relies heavily on manual record-keeping for sales and inventory management, which consumes staff time and creates recurring problems with transaction processing and stock accuracy. Sales are recorded in paper logbooks, often separately by staff members, resulting in inconsistent records that are vulnerable to errors, loss, and damage. During busy periods, manual recording can also slow down service and lead to incomplete or inaccurate entries.

Inventory management is similarly affected by irregular stock counts and manual recording of stock movements, making inventory information outdated and discrepancies difficult to identify. These processes also require significant staff and management time for recording, counting, and reconciling data, reducing the time available for customer service, planning, and business development. Overall, the reliance on manual methods limits operational efficiency and the café's ability to maintain accurate, timely, and reliable business information.

Specific Problem 2: Unreliable and Untimely Information Access

The absence of a reliable recording system makes it difficult for Empress Café to maintain accurate, complete, and timely information about its operations. Manual records can vary between staff members, contain omissions, and become inconsistent over time, especially during busy periods when employees prioritize customer service. These issues can result in incomplete transaction records and inaccurate representations of actual business activity.

Paper-based records also carry risks of damage, loss, or misplacement, which can create permanent gaps in the café's operational history. Furthermore, manual records are mainly retrospective, meaning management may receive information only after transactions and stock movements have already occurred. The lack of timely and reliable data limits the café's ability to monitor current conditions, analyze performance, and make informed operational decisions.

Specific Problem 3: Inefficient Reporting and Decision Support

The lack of an efficient reporting system makes it difficult for Empress Café management to monitor operational performance and make timely, informed decisions. Reports based on manual records require gathering, transcribing, calculating, and organizing information from multiple sources, making the process time-consuming and often resulting in outdated information. Manual report generation is also vulnerable to recording, transcription, and calculation errors, which can affect the reliability of the information used for decision-making. Inconsistent report formats further make it difficult to compare performance, identify trends, and monitor key business indicators. Without reliable and integrated reports, management faces limitations in evaluating sales performance, monitoring inventory, identifying waste and popular products, optimizing purchasing decisions, and assessing overall business performance.

Specific Problem 4: Inadequate Stock Level Monitoring

Inadequate inventory monitoring at Empress Café can lead to stockouts, unnecessary waste, and missed opportunities for operational improvement. Without accurate and timely information on stock levels and consumption, management may discover shortages only after essential ingredients are depleted, resulting in unavailable menu items, delayed orders, or emergency purchases that can increase costs and affect customer satisfaction.

At the same time, overstocking can cause perishable items to spoil and tie up financial resources in unnecessary inventory. The lack of systematic inventory data also prevents management from identifying patterns such as frequently wasted ingredients, emergency purchases, supplier discrepancies, and high-turnover menu items. Improved inventory monitoring would therefore support better stock control, reduce waste, and provide useful information for more effective operational decisions.

Specific Problem 5: Absence of Operational Data Analysis Capabilities

Without a system for analyzing operational data, Empress Café has limited ability to identify inefficiencies, anticipate customer demand, and develop data-informed strategies. The absence of systematic analysis makes it difficult to detect patterns in sales, inventory usage, staff productivity, waste, and seasonal demand. As a result, management may rely heavily on experience and intuition when making operational and strategic decisions. To address these challenges, the study proposes the development and implementation of Operlytics, a business analytics and monitoring system designed for Empress Café Boracay. The system aims to centralize and automate sales transactions, inventory monitoring, and business reporting while providing timely operational insights. Through improved data accuracy and accessibility, Operlytics is intended to support more efficient daily operations and more informed business decisions.

1.3 Objectives of the Study

A. General Objective

The main objective of this study is to develop and evaluate **Operlytics**, a business analytics and monitoring system designed to improve operational decision-making at Empress Café Boracay. The system aims to provide efficient, accurate, and timely monitoring of sales, inventory, and overall business performance through data visualization, automated report generation, and alert mechanisms. It will focus on developing a reliable, secure, and user-friendly platform that captures, processes, stores, and presents operational data while maintaining data accuracy and integrity.

The study also aims to integrate Operlytics into the café's daily operations by addressing existing challenges in manual and fragmented record-keeping. By centralizing sales and inventory information, the system is expected to reduce errors, minimize time spent on manual reporting, improve inventory monitoring, and help users identify sales trends and operational patterns. The system will also consider the needs of café administrators, staff, and other relevant users through user feedback, usability considerations, and appropriate training to support effective adoption.

Finally, the study will evaluate Operlytics based on its functionality, usability, reliability, performance, security, and effectiveness in supporting operational decision-making. The transition from paper-based and informal records to a centralized digital platform is intended to provide Empress Café Boracay with more accessible and organized information for evidence-based decisions. The study may also contribute to the understanding of how business analytics and monitoring systems can support digital transformation and operational improvement among small businesses, particularly in tourism-dependent communities.

B. Specific Objectives

Specific Objective 1: Design and Development of a Web-Based Business Analytics System

The first specific objective is to design and develop a comprehensive, web-based business analytics and monitoring system that will record, manage, analyze, and report sales transactions and inventory data for Empress Café Boracay. The development process will include requirements analysis through interviews, observations, and document reviews; system and database design; frontend and backend development using PHP, HTML5, CSS3, and JavaScript; and rigorous testing and deployment. The system will incorporate appropriate user roles, data structures, business rules, responsive interfaces, authentication, authorization, data validation, and secure data storage to ensure that it is reliable, maintainable, and aligned with the operational needs of the café.

The web-based architecture is intended to provide accessible and centralized management of operational information across compatible internet-connected devices while reducing installation and maintenance requirements. Through centralized data storage, automated updates, and scalable system architecture, Operlytics can support consistent information management, easier maintenance, secure access, and future expansion. The development will also follow an iterative approach that allows continuous user feedback and refinement, ensuring that the system remains practical and responsive to the café's changing operational requirements.

Specific Objective 2: Real-Time Monitoring of Sales and Inventory

The second specific objective is to provide accurate and responsive monitoring of sales performance and inventory levels, allowing café administrators and management to access current information on stock availability, sales, revenue, and overall operational performance. Operlytics will capture transaction data as orders and payments occur and automatically update sales records and inventory levels based on predefined product and ingredient mappings. It will also provide dashboard updates and alerts when stock reaches critical levels or when significant changes in sales performance are detected, reducing the delays and inaccuracies associated with manual monitoring.

The system will present real-time operational information through clear dashboards, visual indicators, summary metrics, and configurable alerts to support timely decision-making. Real-time monitoring can help prevent stockouts and overstocking, identify changes in customer demand, track high-performing products, and detect potential operational issues as they occur. This continuous access to updated information will enable management to respond more proactively to changing business conditions, particularly during periods of high customer demand and seasonal fluctuations.

Specific Objective 3: Automated Reports and Dashboards

The third specific objective is to develop automated reports and interactive dashboards that summarize, analyze, and visualize the café's operational data. Operlytics will automatically generate reports on sales, inventory, costs, profit margins, transaction volumes, and other key performance indicators (KPIs), reducing the time, effort, and errors associated with manual data compilation and spreadsheet-based calculations. These reports will provide consistent and accessible information that management can use to monitor business performance without requiring extensive analytical or technical skills.

The system will also provide interactive dashboards using visual elements such as charts, tables, indicators, and alerts to present important information in an easy-to-understand format. By analyzing sales trends, product performance, inventory usage, and time-based patterns, the dashboards can help management identify high-performing products, monitor stock requirements, recognize operational trends, and support purchasing, staffing, and promotional decisions. Overall, the automated reporting and dashboard features are intended to transform raw operational data into useful insights that support timely and evidence-based decision-making.

Specific Objective 4: Improvement of Operational Efficiency

The fourth specific objective is to improve the operational efficiency of Empress Café Boracay by reducing manual recording errors, minimizing delays in report preparation, streamlining workflows, and supporting faster decision-making. Operlytics will use standardized data entry, validation rules, automatic calculations, and digital audit trails to reduce errors commonly associated with handwritten records and manual data processing. Automated report generation will also reduce the time required to collect, calculate, and organize operational information, allowing staff and management to focus more on customer service, business operations, and strategic activities.

The integration of sales recording, inventory management, financial monitoring, and reporting into one system will reduce duplicate data entry and improve consistency across daily operations. Automatic updates will ensure that sales transactions are reflected in inventory records and reports without additional manual processing. Through accurate and timely information, standardized workflows, and accessible operational data, Operlytics is expected to help staff and management make faster and more informed decisions while improving overall productivity and workflow efficiency.

Specific Objective 5: System Evaluation Based on User Feedback

The fifth specific objective is to evaluate Operlytics in terms of usability, functionality, performance, and effectiveness in supporting operational decision-making based on feedback from café administrators, floor staff, kitchen personnel, and other relevant stakeholders. Usability will be assessed based on ease of navigation, clarity of the interface, efficiency of task completion, learnability, and user satisfaction. Functionality and performance will be evaluated through systematic testing to verify the accuracy of sales transactions, inventory updates, reports, system reliability, and performance under normal and peak operating conditions.

The study will also assess the system's effectiveness in improving operational decision-making by determining whether the information provided is useful, timely, relevant, and actionable for management. Evaluation methods may include user observation, questionnaires, interviews, user acceptance testing, and comparisons of selected operational

measures before and after implementation. The findings will help determine whether Operlytics is practical and suitable for the café's actual operational environment and whether it contributes to more informed and efficient business decisions.

1.4 Scope and Delimitation

This study focuses on the development and evaluation of Operlytics, a web-based business analytics and monitoring system designed to improve operational decision-making for Empress Café Boracay. The scope of the study encompasses the end-to-end design, development, and evaluation of the system within the specific operational context of the establishment. The research is primarily concerned with addressing the shortcomings of the manual record-keeping practices currently employed by the café and replacing them with a structured, data-driven platform that enables management to monitor, analyze, and act on key operational information in real time.

The scope of this study is defined by several dimensions that collectively establish the boundaries of the research effort and the intended applicability of its outcomes. Understanding these scope dimensions is essential for properly interpreting the study's findings and assessing their relevance to other contexts.

Operational Scope

The operational scope of the study covers three primary areas of Empress Café Boracay's daily operations: sales transaction monitoring, inventory level tracking, and operational report generation. Sales transaction monitoring includes recording and processing customer orders and payments through the Point of Sale (POS) system, calculating totals and discounts, recording payment methods, and generating receipts or order confirmations. It also allows the system to summarize sales data to identify revenue performance, product popularity, and demand patterns while maintaining accurate records of transaction details, modifications, voids, and refunds.

Inventory level tracking focuses on monitoring the quantities, movements, and status of ingredients and supplies used in café operations. The system records stock received from suppliers, automatically deducts ingredients based on sales transactions, tracks waste and spoilage, and monitors inventory against predefined stock thresholds. These functions allow management to view current stock conditions and receive relevant alerts and reports, supporting timely inventory management and reducing the risk of stock shortages and unnecessary excess inventory.

System Feature Scope

To support its operational objectives, Operlytics incorporates five core system features that collectively address the identified operational requirements:

The Point of Sale (POS) module serves as the primary data entry point of the system, enabling authorized staff to record and process sales transactions directly within the platform. The POS interface is designed for rapid, intuitive operation during busy service periods, with features such as product selection shortcuts, automatic total calculation, and multiple payment method support. All transaction data entered through the POS module is automatically and simultaneously reflected across the relevant dashboards and inventory records in real time, ensuring data consistency and eliminating the need for duplicate manual encoding. The inventory tracking feature maintains current records of stock levels for all ingredients and supplies, automatically updating quantities based on consumption from sales transactions and manual adjustments for receipts, waste, and other movements. This feature provides real-time visibility into stock availability, alerts management to low stock conditions, and supports inventory counting and reconciliation processes. The inventory tracking interface displays current stock levels, recent movement history, and upcoming expiration dates for perishable items.

User Scope

The target users of Operlytics are the café administrators or managers and authorized staff members of Empress Café Boracay who are directly involved in daily operations. Administrators and managers have broader access to system features, including dashboards, inventory management, report generation, user management, system configuration, and operational data. They are responsible for monitoring overall business performance, managing system settings, reviewing reports, and making informed operational decisions.

Authorized staff members have limited access to functions necessary for their assigned daily tasks, primarily sales transaction recording and processing through the Point of Sale (POS) module. Their permissions are restricted to prevent unauthorized changes to sensitive inventory records, system configurations, and historical data. This role-based access approach helps protect business information while ensuring that each user can efficiently access only the features relevant to their responsibilities.

Technical Scope

Operlytics is developed as a web-based platform that can be accessed through standard internet browsers on computers and other internet-connected devices. This approach eliminates the need for specialized hardware or individual software installations, making the system practical and cost-effective for a small café. It also supports centralized data storage and processing, consistent information across users, simplified backup and security management, and remote access for management when they are away from the café.

The system will use PHP for backend processing, HTML, CSS, and JavaScript for the frontend, and MySQL for data storage. This technology stack provides compatibility, accessibility, and maintainability while allowing the system to be deployed using affordable hosting infrastructure. Centralized updates and data management also reduce maintenance

requirements and ensure that users have access to the latest version of the system, while mobile and browser accessibility provides flexibility for monitoring sales, inventory, and other operational information.

Data Scope

The data scope of the study encompasses all operational data generated within the café's daily business activities, including sales transaction records, inventory movement logs, product catalog information, and user activity trails. The system is designed to capture, store, and process these data types in a structured relational database that ensures integrity, consistency, and retrievability. Data scope explicitly excludes personal customer information beyond what is necessary for transaction processing, such as names associated with orders or loyalty identifiers, and does not involve the collection, storage, or analysis of sensitive personal data subject to specialized privacy regulations. The data architecture is optimized for operational queries and reporting rather than large-scale data mining or complex analytical workloads, reflecting the practical data volumes and access patterns characteristic of a small café operation.

Security Scope

The security scope of Operlytics addresses fundamental data protection and access control requirements appropriate to a small business environment, without extending to enterprise-grade security infrastructure or compliance certifications. The system implements password-based authentication, role-based authorization, basic input validation, and server-side session management to protect against common threats such as unauthorized access, data tampering, and injection attacks. Security measures are proportionate to the sensitivity of the operational data being protected and the realistic threat profile of a small local business. The study does not encompass formal security auditing, penetration testing by independent third parties, or compliance with industry-specific security standards, though the design documentation provides a foundation for such activities should they be pursued in future development phases.

Evaluation Scope

The evaluation scope is structured around functional testing, user acceptance testing, and performance assessment conducted during a defined implementation period at Empress Café Boracay. Evaluation activities focus on verifying that the system meets its specified requirements, operates reliably under normal usage conditions, and delivers perceived value to its intended users. Metrics for evaluation include system uptime, data accuracy, task completion efficiency, and user satisfaction ratings gathered through structured feedback instruments. The evaluation does not include longitudinal studies of business impact, comparative analysis against competing commercial systems, or formal cost-benefit analysis, as these would require extended deployment periods and resources beyond the current research framework. However, the evaluation methodology is designed to produce actionable findings that can guide both immediate system refinement and future research directions.

The evaluation period is designed to capture sufficient operational experience to validate system functionality and user acceptance while remaining within the practical constraints of an academic research timeline. During this period, the system is deployed in the live operational environment of Empress Café Boracay, allowing users to interact with it under realistic conditions of varying customer volume, product availability, and staffing configurations. Evaluation data is collected through multiple channels including system-generated logs, direct observation of user interactions, structured questionnaires

administered to administrators and staff, and semi-structured interviews exploring subjective experiences and perceived impacts on workflow efficiency. The evaluation deliberately samples across different times of day and days of the week to capture variation in operational intensity and system load. However, the evaluation does not encompass stress testing under artificially extreme loads, disaster recovery scenario testing, or evaluation under conditions of hardware failure or network outage. These limitations are recognized as areas where future work could strengthen confidence in system robustness, particularly if deployment expands to support multiple locations or higher transaction volumes.

Future Expansion Scope

While the current implementation of Operlytics is bounded by the scope dimensions described above, the system architecture is designed with modularity and extensibility in mind to accommodate potential future enhancements. The separation of concerns between the POS interface, data processing layer, visualization components, and reporting engine creates natural boundaries for incremental development. Future expansion possibilities, which are acknowledged but not pursued within this study, include integration with supplier ordering systems, mobile application development for remote management, customer relationship management features, and enhanced analytical capabilities. By documenting the current scope clearly and designing the system with principled architectural foundations, this study establishes a platform upon which subsequent researchers or developers can build without requiring fundamental restructuring of the existing codebase or data model.

Delimitations

While Operlytics is designed to address the core operational needs of Empress Café Boracay, the study recognizes several limitations that define its boundaries. The system does not include advanced predictive analytics, machine learning, or artificial intelligence-based forecasting. Instead, it focuses on descriptive and operational insights based on current and historical data, providing accurate and accessible information for daily decision-making. This limitation reflects the available resources, development timeframe, and the need to first establish reliable operational data before implementing more advanced analytical capabilities in future development.

Operlytics is also limited to operational management and does not integrate with external accounting software, financial management platforms, or third-party enterprise systems. Functions such as payroll processing, tax accounting, comprehensive financial management, and automated accounts management are outside the scope of the study. The system is intentionally focused on sales monitoring, inventory management, reporting, and business analytics to ensure that its core functions can be developed, tested, and evaluated effectively within the available resources and research period.

The functional scope is further limited to essential café operations. The Point of Sale (POS) module focuses on fast and accurate transaction processing, supporting cash and digital payments, automatic calculation of totals and change, basic discounts, and receipt generation. Complex promotional features such as buy-one-get-one offers, tiered loyalty programs, and advanced promotional rules are excluded. Similarly, the analytics and visualization features prioritize clear and practical

operational information rather than highly advanced analytical functions. These limitations allow the system to maintain a manageable scope while ensuring that its primary features are reliable and suitable for the café's operational needs.

The inventory module is integrated with the POS system to automatically deduct ingredients based on predefined recipe mappings and completed sales transactions. It also supports manual adjustments for stock receipts, spoilage, waste, and physical inventory corrections, with records maintained through timestamps, user identification, and adjustment reasons. Low-stock alerts are included to help management monitor inventory levels and prevent stockouts; however, the system does not cover automated purchase order generation, supplier communication, or delivery scheduling. These boundaries ensure that Operlytics remains focused on improving the café's internal operational processes while leaving more complex external and supply-chain functions for potential future development.

1.5 Significance of the Study

Beneficiaries: Administrators (Management)

Operlytics will help café administrators and managers monitor sales performance, track inventory levels, and analyze operational data more efficiently, enabling faster and more informed business decisions. By replacing fragmented and manual record-keeping with a centralized and automated platform, the system will provide reliable, timely, and accessible operational information. This will allow management to identify emerging problems, recognize business trends, and respond proactively rather than relying on incomplete or outdated records.

The system will also reduce the time and effort required to gather, compile, and reconcile operational information, allowing management to focus on higher-value activities such as staff management, customer service, menu planning, and business strategy. Through business analytics and data visualization, Operlytics can support better resource allocation, inventory management, waste reduction, and menu planning based on actual operational data. Overall, the system is expected to strengthen management oversight and support more efficient and evidence-based decision-making.

Beneficiaries: End Users (Café Staff)

Operlytics will assist café staff in recording sales transactions and monitoring inventory more efficiently, reducing manual work and potential errors during daily operations. Its user-friendly POS interface will simplify order entry, automatic calculations, transaction confirmation, and inventory updates, allowing staff to follow standardized procedures with less effort. The automated deduction of inventory based on completed sales will also reduce the need for manual stock recording, particularly during busy service periods.

By reducing paperwork and repetitive administrative tasks, the system will allow staff to focus more on customer service and other operational responsibilities. Accurate and accessible inventory information will help staff determine product availability and communicate stock needs to management through real-time alerts. In addition, standardized digital workflows and accurate transaction records can provide clearer operational procedures, support accountability, and help staff develop familiarity with digital systems used in the hospitality industry.

Beneficiaries: Future Researchers

The study will serve as a reference for future researchers and developers interested in developing business analytics and monitoring systems for small businesses, particularly in the food service and hospitality sectors. The documentation of the system's development, implementation, and evaluation provides a practical basis for understanding how technology can be designed and adapted to address the operational needs and limitations of small enterprises.

The study may also contribute to research on information systems, business analytics, digital transformation, and technology adoption in small businesses, particularly those operating in tourism-dependent environments. Its combination of system development, user participation, and evaluation of technical and operational outcomes may provide useful insights and a framework that future researchers can adapt for similar studies or further system improvements.

Beneficiaries: Small Business Community

Beyond Empress Café Boracay, the study may benefit small food service businesses in the Philippines, particularly those operating in tourism-dependent areas. It demonstrates how affordable, accessible, and appropriately designed technology can help address common operational challenges in resource-constrained environments. The documented development and evaluation of Operlytics may provide a practical reference that other small businesses can adapt to their own operational needs and contexts.

The study may also contribute to discussions on MSME digitalization and the role of technology in improving the productivity and sustainability of small enterprises. By demonstrating the potential benefits of digital tools for sales monitoring, inventory management, and operational decision-making, the research may provide useful insights for future technology initiatives and digital transformation efforts. In tourism-dependent communities, improved operational efficiency among small businesses may also contribute to better service delivery and support broader local economic development.

1.6 Definition of Terms

Business Analytics — The technologies, applications, and practices used to collect, process, analyze, and present operational data to support decision-making; here, the descriptive and diagnostic analysis of sales, inventory, and operational data.

Business Intelligence (BI) — The broader organizational capability to transform data into actionable insight through integrated collection, analysis, and reporting; used to emphasize the systematic, data-driven decision-making capacity Operlytics seeks to establish.

Dashboard — A visual display within Operlytics presenting key operational metrics and performance indicators in a consolidated, at-a-glance format using charts, gauges, and summary tables.

Inadequate — Not sufficient or not of the required quality, quantity, or standard to meet a particular need, resulting in limitations, inefficiency, or unsatisfactory outcomes.

Inventory — The stock of ingredients, supplies, and materials maintained by Empress Café Boracay, encompassing both perishable and non-perishable items.

Inventory Tracking — The systematic monitoring and recording of inventory quantities, movements, and status within Operlytics, including receipts, sales-driven deductions, waste adjustments, and threshold monitoring.

Manual Record-Keeping — The café's current practice of using handwritten entries in paper logbooks or informal notes to document sales and inventory, characterized by decentralization and susceptibility to error and loss.

MSMEs — Micro, small, and medium enterprises classified by asset size and employment under Philippine law; here, small food-service businesses with limited capital, workforce, and technical resources.

Operational Efficiency — The ratio of operational outputs (sales, service quality, product availability) to operational inputs (staff time, inventory investment, management attention).

Operlytics — The web-based business analytics and monitoring system developed and evaluated in this study for Empress Café Boracay's operational context.

Point of Sale (POS) — The moment and location where a customer transaction is completed; also, the Operlytics module through which staff record and process sales transactions.

Real-Time Monitoring — The system's capability to capture, process, and display operational data with minimal delay between occurrence and availability.

Sales Transaction — The complete record of a customer purchase, including items, quantities, prices, payment method, timestamp, and processing staff member.

SMEs — Small and medium-sized enterprises with limited employees, revenue, and assets, typically operating with constrained resources and informal management structures.

Stockout — A situation in which a required inventory item is unavailable, resulting in unfulfilled orders, compromised quality, or costly emergency procurement.

Tourism-Intensive Environment — A business location heavily influenced by visitor traffic, characterized by seasonal demand, high customer turnover, and competitive density.

Web-Based System — A software application operating through a standard browser rather than installed client software, hosted remotely and accessed via internet connectivity.

1.8 Summary

This chapter established the study's foundation by presenting the operational challenges facing small food-service businesses in tourism-intensive environments such as Boracay Island, and situating Empress Café Boracay's reliance on manual record-keeping within five interrelated problem areas—inefficient manual processes, unreliable and untimely information, inefficient reporting, inadequate stock monitoring, and the absence of data-analysis capability—that together compromise both daily efficiency and strategic decision-making.

It articulated five complementary objectives spanning the system's design and development, real-time sales and inventory monitoring, automated reporting and dashboards, measurable operational-efficiency gains, and evaluation based on user feedback, and defined the study's operational, feature, user, technical, data, security, and evaluation scope, while

delimiting Operlytics from predictive analytics, third-party financial integration, and enterprise-grade security infrastructure in favor of a focused, reliably implemented feature set.

The chapter's discussion of significance showed that administrators gain a centralized, reliable basis for proactive decisions; staff gain a streamlined system that reduces workload and error; future researchers gain a documented, replicable model; and the broader Philippine small-business community gains evidence that affordable, purpose-built technology can meaningfully address real operational challenges. Together, these sections establish that the operational deficiencies at Empress Café Boracay are well-documented and amenable to resolution through a carefully designed analytics system, setting the stage for the literature review in Chapter II.

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CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Business analytics has emerged as a critical and transformative tool for improving operational decision-making across virtually every industry sector in the contemporary global economy. The evolution of business analytics from simple reporting tools to sophisticated platforms capable of processing vast quantities of data in real time represents one of the most significant developments in management practice over the past two decades. Organizations that have successfully integrated business analytics into their operational frameworks consistently demonstrate superior performance compared to those that rely on traditional, intuition-based decision-making approaches. This section examines foreign studies that illuminate the theoretical foundations, practical applications, and empirical outcomes of business analytics implementation across diverse organizational contexts.

According to [3], business analytics has become a transformative tool in driving data-informed decision-making across industries, including finance, retail, healthcare, manufacturing, and telecommunications. Alomari's comprehensive review highlights the key trends in business analytics adoption and evaluates the effectiveness of popular tools such as data visualization, predictive analytics, machine learning, and big data processing. The findings indicate significant improvements in decision-making efficiency and effectiveness across all sectors examined. The study emphasizes that the integration of business analytics into organizational processes enables managers to move beyond gut feeling and experience-based decisions toward evidence-based strategies grounded in objective data analysis.

The implications of Alomari's findings for small food service businesses are substantial. While the study focuses primarily on large-scale industrial applications, the underlying principles that data-driven decision-making improves outcomes are universally applicable. Small cafés like Empress Café Boracay, despite their limited scale, generate substantial

operational data through daily transactions, inventory movements, and customer interactions. The challenge lies not in data scarcity but in the absence of systems capable of capturing, organizing, and analyzing this data to extract actionable insights. Alomari's work supports the fundamental premise of this study: that business analytics, appropriately scaled and designed, can deliver decision-making improvements regardless of organizational size.

Building on the foundational understanding of business analytics, [4] explored the integration of artificial intelligence in business analytics and its impact on operational efficiency. Islam's study, published in the World Journal of Advanced Engineering Technology and Sciences, notes that with the advent of AI technologies such as machine learning and natural language processing, businesses can now leverage advanced analytics to enhance operational performance in ways previously impossible. The study confirmed that AI-driven analytics addresses existing limitations in traditional business analytics by providing real-time insights, predictive capabilities, and automation, ultimately improving decision-making capabilities and enabling more informed and effective strategic choices.

Islam's research is particularly relevant to the current study because it highlights the trajectory of business analytics evolution from descriptive reporting toward predictive and prescriptive capabilities. While Operlytics, as designed for Empress Café Boracay, focuses primarily on descriptive and diagnostic analytics due to resource and complexity constraints, the system architecture and data structures established by Operlytics create a foundation upon which more advanced analytical capabilities could be built in future iterations. The data collection and organization processes implemented by Operlytics would support the integration of machine learning models for demand forecasting, inventory optimization, and customer behavior analysis should the café's operational maturity and resources support such enhancements in the future.

The study also emphasizes the automation benefits of AI-enhanced analytics, noting that automated data processing and insight generation reduce the cognitive and temporal burdens on managers, allowing them to focus on strategic decision-making rather than operational data compilation. This finding directly supports the design philosophy of Operlytics, which automates the collection, aggregation, and presentation of operational data to free management from manual reporting tasks.

In a similar vein, [5] explored top managers' views on using business analytics in decision-making, providing valuable insights into how organizational leaders perceive and utilize analytical tools in their daily management activities. Kunc and Bäckström's research, published in Measuring Business Excellence, revealed that top managers see business analytics as crucial for business management, particularly in monitoring, managing, and adjusting ongoing operations. The primary use of business analytics was found to be following up on business performance and making decisions on operational adjustments to help achieve short-term objectives, wherein top managers monitor the firm's progress towards its targets within the constraints of its existing resources.

This finding is especially significant for small business contexts where management resources are severely constrained. Unlike large corporations that can dedicate specialized personnel to business intelligence functions, small café owners must personally oversee operational management while simultaneously handling strategic planning, customer relations, staff supervision, and financial oversight. For these managers, business analytics tools that provide quick, clear operational status updates are particularly valuable because they minimize the time and effort required to maintain situational

awareness. Kunc and Bäckström's research supports the design emphasis of Operlytics on dashboard-style visualizations that provide at-a-glance operational summaries rather than complex analytical interfaces requiring extensive interpretation.

The study also highlights the importance of operational adjustment decisions and short-term tactical changes made in response to emerging conditions as the primary application of business analytics in managerial practice. This finding aligns perfectly with the operational focus of Operlytics, which is designed to support daily operational decisions such as inventory replenishment, menu adjustments, and staffing optimizations rather than long-term strategic planning. The system's real-time monitoring capabilities directly enable the kind of operational adjustment decision-making that Kunc and Bäckström identify as the most common and valuable application of business analytics.

The impact of real-time business intelligence and advanced analytics on decision-making has also been extensively explored in the academic literature. Research by [6] indicates that these tools provide insights into decision-makers' behavior and offer recommendations on how business analytics solutions can be improved to support timely and effective decisions. Venkat's work emphasizes that the value of business analytics is not merely in the data provided but in how that data is presented and accessed to align with the cognitive processes and decision-making styles of users. This user-centered perspective on business analytics design is directly incorporated into the Operlytics development approach, which prioritizes usability and intuitive interaction as core design requirements.

Furthermore, the role of organizational decision-making processes in converting business analytics-based insights into effective decisions has been examined by Lee and Venkatraman (2022), highlighting the need for changes in decision-making processes to capture value from business analytics investments. Their research emphasizes that technology implementation alone is insufficient for realizing the benefits of business analytics; organizations must also evolve their decision-making processes to effectively utilize the insights generated. This finding has important implications for the implementation strategy of Operlytics at Empress Café Boracay. The successful deployment of the system requires not merely technical installation but also organizational adaptation training staff to use the system, establishing new workflows that leverage system capabilities, and developing management habits of consulting system data before making operational decisions.

Much attention is currently being paid in both the academic and practitioner literatures to the value that organizations could create through the use of big data and business analytics. [7] and [8] have contributed foundational work establishing the theoretical and empirical basis for business analytics value creation. Chen et al. (2012, pp. 1166–1168) suggest that business analytics and related technologies can help organizations to "better understand its business and markets" and "leverage opportunities presented by abundant data and domain-specific analytics." Similarly, LaValle et al. (2011, p. 22) report that top-performing organizations "make decisions based on rigorous analysis at more than double the rate of lower performing organisations" and that in such organizations analytic insight is being used to "guide both future strategies and day-to-day operations."

These findings establish a clear empirical basis for the value proposition of business analytics: organizations that systematically utilize data analysis in their decision-making processes outperform those that do not. While the studies cited

focus primarily on large organizations, the underlying logic applies with equal or greater force to small enterprises. Small businesses like Empress Café operate with narrower margins for error, more limited resources to absorb losses from poor decisions, and less capacity to recover from operational mistakes. For these businesses, the improvement in decision quality enabled by business analytics may be even more consequential than for larger organizations with greater buffers against suboptimal outcomes.

The research of [9], published in the European Journal of Information Systems, argues that while there is evidence that investments in business analytics can create value, the thesis that "business analytics leads to value" requires deeper analysis. Specifically, they propose that the roles of organizational decision-making processes, including resource allocation processes and resource orchestration processes, need to be better understood in order to comprehend how organizations can create value from business analytics. The first-order effects of business analytics are likely to be on decision-making processes, and improvements in organizational performance are likely to be an outcome of superior decision-making processes enabled by business analytics.

This nuanced perspective is crucial for understanding how Operlytics can create value at Empress Café Boracay. The system does not automatically generate business value simply by collecting and displaying data. Value creation occurs through the mediating mechanism of improved decision-making processes management making better choices about inventory, pricing, menu composition, and operational workflows because they have access to accurate, timely information. The design of Operlytics therefore focuses not merely on data collection but on presenting that data in formats that directly support the operational decisions that café management must make on a daily basis.

[10], in his influential work on information dashboard design, emphasizes that management may use data visualization technologies to better analyze corporate information. Visual dashboards and graphs assist businesses in presenting complicated data in an understandable way. Managers can quickly spot trends, patterns, and possible operational problems thanks to this visual presentation of data. Few's principles of effective dashboard design clarity, simplicity, relevance, and immediacy directly inform the visualization component of Operlytics, ensuring that the system's dashboards communicate operational status effectively to users who may lack formal training in data analysis.

Adebanji Samuel Ogunmokun, Emmanuel Damilare Balogun, and Kolade Olusola Ogunsola (2024) developed a Business Intelligence (BI) dashboard optimization model designed to enhance real-time performance tracking and forecasting accuracy for improved decision-making in modern organizations. Their research explores the evolution of BI dashboards, the importance of real-time tracking, and the integration of forecasting tools to enable organizations to make timely, data-driven decisions. By leveraging machine learning algorithms and cloud-based solutions, the proposed model optimizes performance tracking, integrates various data sources, and offers predictive capabilities that help businesses forecast future trends with greater precision.

The research highlights the challenges of implementing BI dashboards, such as data integration, user training, and technological constraints, and proposes solutions to address these issues, ensuring that the model is both scalable and user-friendly. The findings demonstrate that the optimized BI dashboard model enhances operational efficiency, supports better decision-making, and improves forecasting accuracy, particularly in dynamic environments like retail, finance, and manufacturing. For the context of Empress Café Boracay, this research validates the approach of using dashboard-based

visualization as the primary interface for operational monitoring, while also acknowledging the implementation challenges that must be addressed through careful design, training, and technical support.

Zhou et al. (2021) developed Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting to address the difficulty of using standard Transformers for long time-series forecasting. The study is particularly relevant to business analytics because its target applications include electricity consumption planning, sensor-network monitoring, energy management, economics, finance forecasting, and inventory planning. These applications have similar characteristics to operational monitoring systems because they require continuous analysis of historical records to support forecasting and planning decisions.

The authors identified three major limitations of the conventional Transformer architecture. First, standard dot-product self-attention has quadratic computational and memory requirements with respect to sequence length, represented as $O(L^2)$ Second, stacking multiple encoder and decoder layers increases the memory burden to approximately $O(J \bullet L^2))$, where J represents the number of layers. Third, the conventional decoder generates predictions step by step, which slows inference when a system must produce a long forecast.

Informer addresses these limitations through three architectural modifications. Its ProbSparse self-attention reduces the theoretical time and memory requirements to approximately O(LlogL) by focusing computation on the most important query representations. The self-attention distilling mechanism progressively reduces the sequence representation between layers, while the generative decoder produces long prediction sequences in one forward operation instead of generating them one step at a time.

The benchmark results show that Informer maintained strong forecasting performance while reducing the computational burden of long inputs. The authors evaluated the model on four real-world datasets and compared it with other Transformer variants, recurrent neural networks, convolutional models, ARIMA, Prophet, and DeepAR. Informer generally produced lower prediction errors as the forecasting horizon increased. Compared with LSTM-based forecasting, the study reported mean squared error reductions of 26.8%, 52.4%, and 60.1% at prediction horizons of 168, 336, and 720 time steps, respectively.

Informer is important to the proposed business analytics system because it demonstrates that an attention-based model can remain useful for operational forecasting when the input data are long and continuously expanding. However, its efficiency is still based on a sparse approximation of attention rather than a fully linear recurrent state representation. Therefore, its actual performance may depend on whether the data contain a small number of dominant temporal relationships that can be identified reliably by the ProbSparse mechanism.

Liu et al. (2023) proposed LinRec: Linear Attention Mechanism for Long-term Sequential Recommender Systems. Although the study focuses on recommendation systems, it is relevant to business analytics because recommendation models process long histories of customer actions, purchases, clicks, and interactions. These event histories are comparable to operational records that must be analyzed to identify patterns, predict behavior, or support managerial decisions.

The study explains that standard Transformer-based recommenders compute an attention matrix whose time and memory requirements grow quadratically with the sequence length. When the number of historical events, represented as

NNN, is much greater than the embedding dimension ddd, this creates a significant computational burden. LinRec changes the order of matrix multiplication, applies row-wise and column-wise normalization, and adds an activation function to the query and key representations.

The main contribution of LinRec is that it attempts to preserve the information-selection capability of attention while avoiding the explicit construction of a large attention matrix. This is different from a state space model, which summarizes the sequence through a recurrent hidden state. LinRec therefore represents an intermediate design strategy: it retains the Transformer attention framework but changes the computation so that long-term sequences can be processed more economically.

The efficiency results also support the usefulness of linear attention for operational systems. On the ML-1M dataset, LinRec reduced GPU memory and time costs to approximately one-third of those required by many baseline models. The authors further reported that the reduction remained consistent as the maximum sequence length increased.

For the proposed business analytics and monitoring system, the findings suggest that linear attention could be useful when operational data are organized as long sequences of customer activities, work orders, sales transactions, service requests, or employee actions. Nevertheless, LinRec was evaluated in sequential recommendation, not in business process monitoring or time-series forecasting. Its results should therefore be treated as evidence of computational transferability rather than direct proof of effectiveness in the proposed application.

2.2 Local Studies

The Philippine context for business analytics and operational system development presents unique characteristics that distinguish it from the foreign studies examined above. Local research on information systems in small business contexts provides valuable insights into the specific challenges, opportunities, and implementation strategies relevant to Filipino enterprises operating in the country's distinctive economic, cultural, and technological environment. Unlike large corporations with dedicated IT departments and substantial capital reserves, Philippine micro, small, and medium enterprises (MSMEs) frequently operate with constrained resources, limited technical expertise, and infrastructure challenges that shape how technology adoption unfolds in practice. The archipelagic geography of the Philippines, combined with variable internet connectivity across regions and the predominance of cash-based transactions in many local markets, creates implementation conditions that differ significantly from those assumed in foreign-developed frameworks. Local studies therefore offer essential empirical grounding for understanding how business analytics and operational systems can be designed, deployed, and sustained within these specific constraints.

[11] highlighted that in today's digital era, every business must adapt and innovate to stay competitive. Their study on the development of a Point-of-Sale (POS) system with a card reader for the HTM Department emphasized that incorporating a POS system transforms business transactions and empowers businesses to seamlessly manage sales and customer interactions. Arig et al. further noted that the system ensures smooth transactions, enhances accuracy, and reduces human error, making it a great tool for preparing students and businesses in facing real-world operations. The researchers employed a developmental research design similar to the approach adopted in the present study, systematically progressing

from needs analysis through system design, development, and evaluation. Their findings demonstrated that even relatively simple POS implementations could yield substantial improvements in transaction processing speed and accuracy, with error rates decreasing by approximately 35% compared to manual recording methods. The integration of card reader functionality addressed the growing demand for cashless payment options among Filipino consumers, reflecting broader trends in digital financial inclusion accelerated by platforms such as GCash and Maya.

This study is directly relevant to Operlytics because it demonstrates the value of POS systems in educational and small business contexts within the Philippines. The emphasis on accuracy enhancement and error reduction aligns with the core objectives of Operlytics, particularly Specific Objective 4 which targets measurable improvement in operational efficiency through reduced manual recording errors. The finding that POS systems prepare businesses for real-world operations supports the argument that digital operational systems are becoming essential rather than optional for business competitiveness in the Philippine market. Furthermore, Arig et al.'s focus on usability for users with limited technical backgrounds resonates with the design philosophy of Operlytics, which prioritizes intuitive interfaces requiring no advanced technical knowledge. The researchers' observation that successful implementation depended as much on user training and organizational readiness as on technical functionality reinforces the importance of the human dimension in the conceptual framework guiding this study.

[12] developed a web-based inventory management system for a business in the Philippines to address the challenges of manual inventory recording and reporting. According to Tanaman et al., the software eliminated manual approaches through an automated computing solution using a web platform where data are electronically recorded and processed, and reports are electronically produced. The system was evaluated using ISO 9126 quality standards and received high scores in functionality, maintainability, efficiency, and compatibility, confirming its effectiveness in streamlining inventory operations. The researchers specifically addressed the problem of inventory discrepancies that plagued their target business, where stockouts occurred approximately three times more frequently than management initially estimated due to the lag between actual consumption and manual record updates. Their web-based architecture enabled real-time inventory visibility from multiple access points, allowing managers to check stock levels remotely rather than requiring physical presence at the storage location.

Tanaman et al.'s research provides strong validation for the web-based approach adopted in Operlytics. Their use of ISO 9126 quality standards for evaluation offers a methodological model that could be adapted for assessing Operlytics, particularly in evaluating the efficiency and functionality criteria outlined in Section 3.7. The high scores achieved in functionality, maintainability, efficiency, and compatibility demonstrate that well-designed web-based inventory systems can meet rigorous quality standards, suggesting that similar quality outcomes are achievable for Operlytics with appropriate design and implementation effort. The researchers' finding that automated reporting reduced report generation time from several hours to mere minutes directly supports Specific Objective 3 of this study regarding automated reports and dashboards. Their experience also highlights the importance of selecting appropriate evaluation frameworks; the ISO 9126 standard's multi-dimensional assessment parallels the five evaluation criteria (functionality, usability, accuracy, security, efficiency) employed in this study, suggesting that comprehensive quality evaluation is both feasible and necessary for web-based operational systems in Philippine small business contexts.

[3] conducted a study assessing the usability of a centralized Point-of-Sale (POS) system for inventory management in Bounty Agro Ventures, Inc. Rotisserie Operations. The study explained that the company adopted a centralized POS system to streamline its inventory management practices and address inaccuracy concerns brought about by manual processes. Bienvenido further found that when sales transactions are directly connected to inventory updates, businesses can more easily track product usage and maintain accurate stock records, which helps businesses manage their operations better. The researcher employed a mixed-methods approach combining System Usability Scale (SUS) questionnaires with observational task analysis to comprehensively assess how effectively the centralized POS system supported actual operational workflows. Findings revealed that the direct connection between sales transactions and inventory updates eliminated the reconciliation delays that previously required staff to manually cross-reference sales receipts against stock counts, a process that consumed approximately 15% of total staff hours and introduced errors in roughly 12% of inventory records.

Bienvenido's findings directly support the integrated POS-inventory approach of Operlytics. The critical insight that connecting sales transactions to inventory updates enables better tracking and more accurate stock records is precisely the operational mechanism that Operlytics implements through its auto-deduct ingredients feature. This real-time synchronization addresses Specific Problem 4 (Inadequate Stock Level Monitoring) by ensuring that inventory levels reflect actual consumption immediately rather than after periodic manual counts. The study's focus on usability assessment also reinforces the importance of user-centered design in system development, particularly for systems intended for operational staff who may have limited technical backgrounds. Bienvenido's finding that usability scores correlated strongly with adoption rates and error reduction suggests that the usability criterion in this study's evaluation framework (Section 3.7) is not merely a secondary consideration but a critical determinant of operational impact. The researcher's observation that centralized systems enabled consistent data across multiple service points also supports the web-based architecture decision for Operlytics, which ensures that all users access the same current information regardless of their physical location within the café.

[14] developed a web-based sales and inventory system for small retail businesses in the Philippines to address inefficiencies caused by manual record-keeping. The study found that traditional methods of tracking sales and inventory often resulted in data inaccuracies, delayed reporting, and difficulty in monitoring stock levels. By implementing a computerized system, the researchers were able to automate transaction recording, improve inventory tracking, and generate real-time reports. The results showed a significant improvement in operational efficiency, data accuracy, and decision-making speed among business owners. The study emphasized that integrating sales and inventory systems is essential for small businesses to remain competitive and organized in a fast-paced market environment. Cruz et al. specifically documented how the absence of integration between sales and inventory functions in manual systems created information silos, where sales staff operated with no visibility into stock availability while inventory managers lacked awareness of sales velocity, resulting in frequent stockouts of popular items and overstocking of slow-moving products.

Cruz et al.'s research provides perhaps the closest parallel to the current study, as it addresses the same core problems—manual record-keeping inefficiencies, data inaccuracies, and delayed reporting—in the same national context and for similar small business establishments. Their finding of significant improvement in operational efficiency, data accuracy, and decision-making speed after system implementation offers an empirical benchmark for the expected outcomes of

Operlytics deployment at Empress Café Boracay. The emphasis on system integration as essential for competitiveness reinforces the design decision to combine POS, inventory, and reporting functions in a unified platform rather than as separate, disconnected tools. Cruz et al.'s observation that integration eliminated the need for duplicate data entry and reconciliation efforts directly supports the operational efficiency objective of this study. Their research also highlights the particular importance of integration for small businesses with limited staff, where the same individuals often perform multiple roles and cannot afford the time overhead of maintaining separate records. This finding is especially relevant to Empress Café Boracay, where staff members currently maintain individual sales records and perform sporadic inventory counts without systematic coordination.

conducted a study on the implementation of a Point-of-Sale (POS) and inventory management system in a local food service business in the Philippines. The research aimed to improve transaction processing and inventory monitoring through system automation. Findings revealed that the use of a POS system reduced transaction time, minimized human errors, and ensured that inventory levels were automatically updated after each sale. Additionally, the system provided summarized reports that helped managers analyze sales trends and make informed operational decisions. The study concluded that adopting automated systems significantly enhances business performance, particularly in improving accuracy, efficiency, and overall management control. The researchers documented specific quantitative improvements: transaction processing time decreased by an average of 40% during peak hours, inventory record accuracy improved from 73% to 96%, and the time required for daily sales reporting decreased from approximately 2 hours to 15 minutes.

Dela Peña and Santos's study is especially relevant because it focuses specifically on food service businesses in the same industry context as Empress Café Boracay. Their findings regarding reduced transaction time, minimized human errors, automatic inventory updates, and improved management control through summarized reports directly map onto the expected benefits of Operlytics. The study's conclusion that automated systems significantly enhance business performance in accuracy, efficiency, and management control provides strong empirical support for the research hypothesis underlying the current study. The quantitative metrics reported by Dela Peña and Santos offer concrete targets against which Operlytics can be evaluated; for instance, the 40% reduction in transaction time and the improvement in inventory accuracy to 96% provide benchmarks for assessing whether Operlytics achieves comparable operational improvements at Empress Café Boracay. Their finding that summarized reports enabled managers to identify sales trends previously obscured by fragmented manual records directly supports the analytical capabilities emphasized in Specific Objective 5 (System Evaluation Based on User Feedback) and the business intelligence components of the conceptual framework. The researchers' observation that food service businesses face unique challenges due to perishable inventory and high customer turnover further validates the focus on real-time monitoring in this study's design.

Synthesis of Local Studies

The reviewed local studies collectively demonstrate that Philippine small businesses, particularly in the food service and retail sectors, face operational challenges fundamentally similar to those identified at Empress Café Boracay. Manual record-keeping, fragmented information systems, and delayed reporting emerge as persistent problems across diverse business contexts, from educational department operations (Arig et al., 2025) to rotisserie chains (Bienvenido, 2023) to small retail establishments (Cruz et al., 2022). The convergence of findings across these studies suggests that these challenges are not idiosyncratic to particular businesses but represent systemic patterns characteristic of MSMEs operating with limited technological infrastructure.

Several critical insights emerge from this synthesis. First, web-based architectures consistently prove effective for Philippine small businesses, offering accessibility without requiring substantial hardware investment, centralized data management that eliminates fragmentation, and remote monitoring capabilities that accommodate managers who cannot remain physically present at all times. [12] and [14] both validated this architectural choice empirically, providing precedent for the PHP-MySQL-HTML/CSS/JavaScript stack selected for Operlytics.

Second, integration between sales and inventory functions emerges as a non-negotiable requirement rather than an optional enhancement. [13] and both demonstrated that the real-time connection between transaction processing and inventory updates is the primary mechanism through which operational accuracy improves. This finding directly validates the auto-deduct ingredients feature in Operlytics and the unified database architecture that supports it.

Third, usability considerations prove decisive for adoption success. Bienvenido's (2023) explicit focus on usability assessment, Arig et al.'s (2025) emphasis on preparing users for real-world operations, and the consistent finding that users with limited technical backgrounds require intuitive interfaces all reinforce the usability-centered design philosophy of Operlytics. The System Usability Scale employed by Bienvenido offers a validated instrument that could be adapted for this study's evaluation phase.

Fourth, quantifiable operational improvements are achievable and measurable. provided specific metrics for transaction time reduction and accuracy improvement, while [12] documented ISO 9126 quality scores. These precedents establish that the evaluation criteria employed in this study—functionality, usability, accuracy, security, and efficiency—can yield concrete, defensible results that demonstrate system value to both academic and practitioner audiences.

Finally, the local studies reveal a critical gap that Operlytics specifically addresses: while existing research has examined POS systems, inventory management, and sales reporting as separate or partially integrated functions, none of the reviewed local studies developed a comprehensive business analytics and monitoring system that combines all three operational domains with real-time dashboard visualization and automated report generation in a single unified platform tailored specifically for small café operations in tourism-intensive environments. [11] focused on POS with card reader functionality; [12] addressed inventory management exclusively; [13] examined POS-inventory integration for rotisserie operations; [14] developed sales-inventory integration for retail; and implemented POS-inventory automation for food service. Operlytics extends beyond these precedents by incorporating business analytics and data visualization capabilities that transform operational data into strategic intelligence, addressing the analytical deficiency identified as Specific Problem

5. This extension responds to the broader literature's emphasis on business intelligence and dashboard-based decision support (Few, 2013; Ogunmokun et al., 2024) while adapting these capabilities to the resource constraints and operational realities of Philippine small cafés.

The tourism-intensive context of Boracay Island adds dimensions not explicitly addressed in the reviewed local studies: seasonal demand fluctuations, high customer turnover from international visitors, and competitive density that intensifies the consequences of operational inefficiency. While the local studies establish the feasibility and value of operational systems for Philippine small businesses, Operlytics contributes by demonstrating how these systems can be adapted to the unique pressures of tourism-dependent economies, where operational agility and real-time responsiveness may determine business survival during peak seasons.

2.3 Related Literature

The theoretical foundations of business analytics and operational information systems extend beyond the specific studies cited above to encompass a broader body of literature on organizational decision-making, information systems value creation, and digital transformation. This section examines key theoretical perspectives that inform the conceptual framework and design philosophy of Operlytics.

Theoretical Foundations of Business Analytics Value Creation

Organizations could create substantial value through the use of big data and business analytics, as established by foundational research from [7] and [8]. Chen et al. (2012, pp. 1166–1168) suggest that business analytics and related technologies can help organizations to "better understand its business and markets" and "leverage opportunities presented by abundant data and domain-specific analytics." Similarly, LaValle et al. (2011, p. 22) report that top-performing organizations "make decisions based on rigorous analysis at more than double the rate of lower performing organisations" and that in such organizations analytic insight is being used to "guide both future strategies and day-to-day operations." These foundational works establish the theoretical premise that data abundance, when properly leveraged through analytics, creates competitive advantage. The "abundant data" that Chen et al. reference exists within small food service establishments just as it does in large corporations; the difference lies not in data availability but in the capacity to capture, organize, and analyze that data. Operlytics addresses this capacity gap by providing the technological infrastructure through which small cafés can begin leveraging their operational data for improved decision-making.

Digitalization and continuously developing technologies have provided firms with an array of new opportunities to capture, store, manage, and share data as well as continuously advancing business analytics tools to derive insight from this data. Firms are investing in their business analytics capabilities to improve their decision-making efficiency and to make more "fact-based" decisions (Davenport and Harris, 2017). By using business analytics, firms aim at making more intelligent decisions faster (Lepenioti et al., 2020) and improving both their business process and decision-making performance (Chatterjee et al., 2021). This stream of research emphasizes the speed dimension of business analytics value creation. In fast-paced operational environments like food service, the velocity of decision-making is often as important as the quality of decisions. A good decision made too late may be less valuable than an adequate decision made promptly. The real-time

monitoring capabilities of Operlytics directly address this velocity requirement by ensuring that management has immediate access to current operational information, eliminating the delays inherent in manual reporting processes.

Data Visualization and Cognitive Information Processing

Management may use data visualization technologies to better analyze corporate information. Visual dashboards and graphs assist businesses in presenting complicated data in an understandable way, as [8] emphasizes. Managers can quickly spot trends, patterns, and possible operational problems thanks to this visual presentation. The cognitive benefits of visual data presentation are well-established in human-computer interaction research: the human visual system is highly optimized for pattern recognition, and appropriately designed visualizations can convey complex information more efficiently than text or numerical tables alone.

The Business Intelligence dashboard optimization model developed by [15] demonstrates the evolution of BI dashboards toward real-time performance tracking and forecasting accuracy. As businesses increasingly rely on data-driven insights to stay competitive, effective dashboard systems that integrate real-time data processing and accurate forecasting models are essential. The research highlights the challenges of implementing BI dashboards, such as data integration, user training, and technological constraints, and proposes solutions to address these issues, ensuring that the model is both scalable and user-friendly. The findings demonstrate that the optimized BI dashboard model enhances operational efficiency, supports better decision-making, and improves forecasting accuracy, particularly in dynamic environments like retail, finance, and manufacturing. The study provides actionable recommendations for organizations seeking to implement the model and suggests areas for future research, including the integration of advanced machine learning techniques and external data sources to enhance forecasting accuracy further.

For the current study, this literature on business intelligence dashboards provides design guidance for the visualization components of Operlytics. The emphasis on real-time processing, user-friendly design, and scalability directly informs the dashboard architecture of the proposed system. The challenges identified in data integration, user training, and technological constraints are precisely the implementation risks that the Operlytics development and deployment plan seeks to mitigate through careful requirements analysis, iterative user feedback, and appropriate technology selection.

Digital Transformation and Data-Driven Business Models in Hospitality

The hospitality industry has undergone significant digital transformation, with technology becoming central to operational efficiency and competitive positioning. Recent scholarship emphasizes that digital transformation in hospitality extends beyond mere technology adoption to encompass strategic shifts in how services are delivered, decisions are made, and customer relationships are maintained. This transformation exploits innovations such as artificial intelligence, the Internet of Things, cloud computing, and mobile applications to connect data across systems, automate routine tasks, and anticipate operational needs before they arise.

Data-driven business models (DDBMs) have emerged as a critical framework for innovation in hospitality ecosystems. According to [21], data collection, organization, and analysis can stimulate innovation by using data as a key lever for transformation opportunities. For small and medium enterprises (SMEs) in the food service sector which constitute the majority of café and independent restaurant operations -- the development of data-driven capabilities is particularly

crucial. Due to their potential limitations in financial and human resources, SMEs can risk more than large companies in the development of digital business models. However, their greater flexibility makes them more agile in the establishment of external relationships for knowledge exchange (OECD, 2021). The application of DDBMs can help SMEs overcome their dimensional and resource constraints by enhancing their capabilities to analyze the environment, explore new opportunities, and improve decision-making by focusing on data rather than on intuition.

Contemporary industry data indicates that 52% of restaurants plan to dedicate resources toward upgrading or implementing point-of-sale (POS) systems in 2025, with 71% of restaurant owners using POS data to streamline payments, optimize menus, and boost engagement (Rezku, 2025). Additionally, 86% of businesses use POS data to increase the average order value through upselling and promotional strategies (GloriaFood, 2025). The most common technology investments include online ordering platforms, marketing software, and accounting systems, reflecting an industry-wide recognition that digital infrastructure is essential for operational survival. However, these investments often remain fragmented, lacking the integrated analytics capabilities that would transform transactional data into strategic intelligence. Furthermore, 85% of restaurant operators now cite integration capabilities as their top purchasing priority when selecting POS software, yet 68% of business leaders identify data silos as their primary integration challenge , highlighting the persistent gap between data collection and actionable analytics in the sector.

Real-Time Analytics and Operational Velocity in Food Service

The velocity of information processing represents a critical competitive dimension in food service operations. Research on restaurant management systems demonstrates that integrated digital solutions can increase order processing efficiency significantly. Restaurant employees may shift tables up to 20% quicker with the help of mobile POS handhelds (GloriaFood, 2025). The real-time monitoring capabilities essential for operational agility align with broader findings that data-driven decision-making enables businesses to make more accurate, objective decisions that improve efficiency, reduce risks, and enhance customer satisfaction.

In the context of small cafés, real-time operational data encompasses inventory levels, sales velocity, customer traffic patterns, and staff performance metrics. The integration of these data streams into unified analytics platforms addresses what scholars identify as a primary challenge in hospitality analytics: the fragmentation of data across siloed systems and the consequent inability to derive actionable insights . By consolidating operational data into a single analytics framework, Operlytics enables the pattern recognition and trend identification that support both immediate operational adjustments and longer-term strategic planning. Moreover, businesses leveraging integrated POS systems report 9.5% revenue increases and 20% better cross-channel visibility, while those using data analytics are 23% more likely to gain new customers than competitors without analytical capabilities .

Behavioral Analytics and Knowledge Creation

Emerging research in hospitality analytics emphasizes the importance of first-party behavioral data in driving knowledge creation and service innovation. While much existing research focuses on macro-level indicators such as adoption trends and platform innovations, micro-level behavioral data encompassing operational transaction logs, customer interaction

patterns, and resource utilization metrics provides the granular foundation for actionable decision-making. Recent scholarship demonstrates that hospitality analytics research remains fragmented and often relies on small datasets, with first-party behavioral data (clickstreams, navigation paths, and booking-funnel metrics) rarely explored in depth (Harnessing Digital Marketing Analytics, 2025). The integration of behavioral analytics with simulation modeling techniques, such as fuzzy cognitive mapping (FCM), offers a methodological pathway for translating raw operational data into strategic knowledge. FCM models concepts as nodes and their causal influences as signed, weighted links, enabling the simulation of how interventions propagate through the system, a capability that bridges the gap between raw data and managerial sense-making (Harnessing Digital Marketing Analytics, 2025).

This approach is particularly relevant for small food service establishments, where managerial decision-making has historically relied on intuition and experience rather than systematic data analysis. The transition from experience-based to data-driven decision-making requires not only technological infrastructure but also organizational learning and cultural adaptation. Research indicates that fostering a data-driven innovation culture is essential for turning analytics into engagement, satisfaction, and revenue growth a cultural shift that entails investing in technical infrastructure while training staff to interpret and act on data insights (Harnessing Digital Marketing Analytics, 2025). The framework proposed by this emerging scholarship collecting behavioral signals, estimating associations, representing them within causal models, and conducting scenario analysis formalizes a behavior-analytics-to-action learning loop that directly informs the iterative design philosophy of Operlytics.

Theoretical Synthesis and Research Gap

The synthesis of these theoretical perspectives reveals a significant gap in the existing literature: while large-scale hospitality enterprises have increasingly adopted sophisticated business analytics and business intelligence systems, small food service establishments particularly independent cafés remain underserved by both academic research and technological solutions. The theoretical frameworks of data-driven business models, real-time analytics, and visual decision support systems have been validated primarily in contexts of organizational scale and resource abundance that do not reflect the operational realities of small café operations. Furthermore, the [17] highlights that in the accommodation and food services sector, high-speed broadband connection, having a website, and using cloud computing to store files are the main technologies associated with higher value added yet significant digital gaps persist, particularly among the smallest enterprises.

Operlytics addresses this research gap by adapting established theoretical principles of business analytics value creation to the specific constraints and capabilities of small food service establishments. The system design incorporates the cognitive benefits of visual data presentation, the velocity requirements of real-time operational monitoring, and the scalability challenges identified in business intelligence implementation research. By grounding the system architecture in these theoretical foundations while addressing the unique resource constraints of small cafés including the need for affordable, integrated solutions that eliminate data silos, Operlytics contributes to both the practical advancement of hospitality technology and the theoretical understanding of analytics value creation in SME contexts. The platform's emphasis on consolidating fragmented operational data into actionable intelligence directly responds to the industry-wide

challenge that 68% of business leaders cite as their primary integration barrier , thereby transforming the abundant but underutilized data generated by small cafés into a genuine source of competitive advantage.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local studies collectively establish a compelling case for the development and implementation of integrated business analytics systems in small food service establishments. The synthesis of these studies reveals several convergent themes that directly inform the design, implementation, and expected outcomes of Operlytics at Empress Café Boracay.

Theme 1: The Imperative of Technology Adoption for Competitiveness

Both foreign and local studies emphasize the importance of using technology to improve business operations, particularly in sales monitoring, inventory management, and decision-making. Foreign studies highlighted the role of business analytics in transforming operational data into useful insights that support data-driven decision-making and improve organizational performance. Local studies focused primarily on implementing automated systems, like point-of-sale (POS) and inventory management systems, to solve operational problems which small businesses commonly encounter.

The convergence of these findings across different national contexts, industry sectors, and organizational scales establishes technology adoption as a universal imperative for business competitiveness rather than a luxury available only to large or well-resourced enterprises. The digital era, as [11] emphasize, demands adaptation and innovation from every business regardless of size or sector. For small cafés in tourism-intensive markets like Boracay Island, this imperative is particularly acute because the competitive environment includes not only local peers but also international franchises and established brands with sophisticated operational systems.

Furthermore, the literature suggests that small businesses which delay technology adoption often face compounding disadvantages over time. As competitors implement increasingly sophisticated systems, the operational gap widens, making later adoption more difficult and costly. Early investment in integrated systems, even modest ones, creates a foundation for incremental improvement and scalability. This finding is particularly relevant for Empress Café Boracay, where the seasonal nature of tourism demand requires operational agility that manual systems cannot provide.

Theme 2: The Elimination of Manual Process Deficiencies

The reviewed studies consistently identify manual processes as the root cause of operational inefficiencies in small businesses. These systems are intended to take the place of manual processes, which are frequently challenging, ineffective, and prone to human error. Incorrect record-keeping, transaction delays, and trouble keeping track of stock levels are problems that many small businesses face. In order to improve day-to-day operations, researchers stressed the significance of implementing automated solutions.

According to the studies, automated systems significantly decrease manual mistakes by automating data entry and computation tasks. Additionally, they increase transaction accuracy and speed, particularly during busy periods. Furthermore, these systems offer current and accurate records, which enable business owners to better monitor performance and make decisions. These findings directly support the problem identification in Chapter 1 and the design objectives of Operlytics, which specifically target the manual process deficiencies at Empress Café Boracay.

The literature also identifies specific categories of manual errors that automated systems address most effectively. Data entry errors, including transposition mistakes, omitted entries, and duplicate recordings, represent the most common and costly category. Calculation errors in pricing, discounts, and inventory counts create cascading inaccuracies that distort business intelligence. Timing errors, where records are updated inconsistently or belatedly, prevent managers from accessing current operational status. Process consistency errors occur when different staff members follow divergent procedures for the same tasks. Operlytics addresses each of these error categories through standardized digital workflows, automated calculations, real-time updates, and enforced procedural consistency.

Theme 3: The Integration of Operational Systems

The literature review highlights how crucial it is for a company to implement various operational systems that are integrated rather than operating in isolation. These systems, which are essential to monitoring everyday activities, include reporting, inventory tracking, and sales recording. Systems for recording sales help businesses maintain track of transactions and revenue. Accurate stock levels are ensured by inventory tracking, which also helps avoid shortages or overstocking. Reporting systems organize gathered data into useful information. Through summaries and insights, they have made it simple for managers to comprehend business performance.

Businesses may track operations more efficiently and prevent errors when these systems are connected. Additionally, this integration makes it possible to produce trustworthy reports that aid in improved decision-making. The integration theme is central to the Operlytics design, which combines POS, inventory tracking, and report generation in a unified platform rather than as separate tools. This integration ensures that data flows automatically from transaction recording through inventory update to report generation, eliminating the reconciliation errors and delays that occur when these functions are performed separately.

Theme 4: The Centrality of Real-Time Information

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The reviewed literature further distinguishes between superficial integration, where systems merely share data files periodically, and deep integration, where modules operate within a unified database architecture with real-time synchronization. Deep integration, as described by multiple authors, eliminates the latency and inconsistency problems that plague superficially connected systems. When a sale occurs, deep integration ensures that inventory decrements, revenue recognition, and dashboard updates occur simultaneously within a single transaction boundary. This architectural approach prevents the data drift that accumulates when separate systems are reconciled only at intervals. Operlytics implements this deep integration model, ensuring that all operational modules maintain perfect data consistency at all times.

Theme 5: The Importance of Usability and User-Centered Design

Multiple studies in the reviewed literature emphasize that technical functionality alone is insufficient for system success; the system must also be usable by its intended audience. [13] specifically assessed usability in the POS context, while [15] highlighted user training and user-friendly design as critical implementation success factors. [5] noted that the value of business analytics depends on its integration into managerial decision-making processes, which requires systems that align with how managers actually work.

This usability theme directly informs the design philosophy of Operlytics, which prioritizes intuitive interfaces, minimal training requirements, and clear information presentation. The system is designed for café staff and management who may have limited technical expertise, ensuring that the system's analytical capabilities are accessible to users regardless of their technical background.

The literature provides specific guidance on usability factors that matter most in small business contexts. Cognitive load reduction through simplified screen layouts and logical task flows reduces errors and training time. Consistent interface conventions across all system modules allow users to transfer learning between functions. Progressive disclosure of advanced features prevents overwhelming novice users while accommodating expert needs. Contextual help and embedded guidance reduce dependence on external documentation. Mobile accessibility enables managers to monitor operations remotely, a capability increasingly expected in modern business environments. Offline resilience ensures that operations continue during connectivity interruptions, particularly important in island locations like Boracay where network reliability may vary. Operlytics incorporates each of these usability considerations into its design specifications.

Theme 6: Data-Driven Decision Making and Business Intelligence

An emerging theme across the reviewed literature concerns the transformation of raw operational data into actionable business intelligence. Studies by [4] and [15] demonstrate that the mere collection of operational data provides limited value unless analytical tools convert that data into meaningful insights. Business intelligence encompasses not only historical reporting but also trend analysis, predictive modeling, and prescriptive recommendations that guide strategic decisions.

For small food service establishments, business intelligence capabilities have historically been inaccessible due to cost and complexity barriers. However, recent developments in cloud computing, open-source analytics tools, and simplified visualization platforms have democratized access to these capabilities. The literature documents successful implementations where small businesses leveraged modest analytical tools to achieve significant improvements in menu profitability, waste reduction, and customer retention.

Operlytics addresses this theme through its integrated analytics module, which automatically generates insights from accumulated operational data. Rather than requiring managers to manually analyze spreadsheets or generate custom reports, the system presents analytical findings through intuitive visualizations and plain-language summaries. This approach makes business intelligence accessible to users without statistical training while still providing the depth of analysis that supports informed decision-making.

Theme 7: Scalability and Future-Proofing

Several reviewed studies emphasize that system implementations should consider not only current requirements but also future growth trajectories. Scalable architectures allow small businesses to expand system capabilities incrementally as operational complexity increases and resources permit. Cloud-based deployment models, modular system designs, and standards-based data architectures support this scalability objective.

The literature warns against implementations that solve immediate problems but create future constraints. Proprietary data formats, monolithic architectures, and vendor-dependent solutions may provide short-term convenience at the cost of long-term flexibility. Conversely, systems built on open standards and modular principles enable continuous evolution without requiring wholesale replacement.

Operlytics incorporates these scalability principles through its modular architecture, which separates core transaction processing from analytical and reporting functions. This separation allows individual modules to be enhanced or replaced without disrupting overall system operation. The use of standard data formats and documented APIs ensures that future integrations with external systems remain feasible. Cloud deployment options provide elastic capacity that grows with business volume without requiring infrastructure investment.

Synthesis Conclusion

Overall, the development of Operlytics, a business analytics and monitoring system that includes sales tracking, inventory monitoring, and data visualization, is strongly supported by the results of these studies. The importance of integrated systems is clearly established because it makes information easier to access and comprehend by allowing businesses to manage many different operational information on just one centrally controlled platform.

By reducing manual work and preventing records and reporting errors, the system aims to help Empress Café Boracay boost its operational efficiency. They are able to make decisions faster and more effectively when they have access to accurate and organized business information. By presenting data in an understandable and visual way, it also improves decision-making, assisting the company in finding trends and improving overall performance.

The reviewed literature provides both theoretical justification and empirical precedent for the Operlytics approach. The theoretical foundation in business analytics value creation, combined with the empirical evidence from similar system implementations in Philippine small business contexts, establishes a strong basis for expecting that Operlytics will deliver meaningful operational improvements at Empress Café Boracay.

Moreover, the convergence of themes across diverse studies suggests that the benefits of integrated business analytics systems are robust and generalizable rather than context-dependent. Whether examining large enterprises or small cafés, foreign markets or local Philippine contexts, the fundamental value proposition remains consistent: integrated automation of operational processes, real-time information access, and user-centered design combine to produce measurable improvements in efficiency, accuracy, and decision quality. Operlytics represents a targeted application of these well-established principles to the specific operational context of Empress Café Boracay, adapted to the unique challenges and opportunities of small food service establishments in tourism-dependent markets.

The studies also generally agree that reducing complexity can improve the feasibility of real-time or large-scale systems. Informer improves long-horizon time-series forecasting, LinRec reduces memory and time costs for long customer-interaction sequences, S5 provides strong long-range benchmark results with linear complexity, and Mamba provides high inference throughput without requiring a long key-value cache.

The papers diverge, however, on whether one architecture is best for all tasks. Informer performs strongly in time-series forecasting because its sparse attention mechanism preserves important dependencies between historical observations and future outputs. LinRec performs well in sequential recommendation because it retains attention-like importance scores. S5 performs strongly on controlled long-range benchmarks and irregularly sampled sequences, while Mamba performs especially well in content-dependent language, audio, and genomic tasks because of its selective state updates.

For the proposed business analytics and monitoring system, these findings support a hybrid evaluation strategy. Informer or another efficient Transformer may be appropriate when the system must forecast structured operational indicators such as sales, inventory, electricity consumption, demand, or service volume. LinRec may be useful when the data consist of long event histories, such as customer transactions or user activities. S5 and Mamba are promising when the system must process long event streams, irregular observations, or high-frequency monitoring records with limited memory.

Overall, the foreign literature suggests that computational efficiency should not be evaluated separately from predictive quality. The proposed system should compare at least one efficient attention model and one state space model using the same operational dataset, sequence lengths, hardware conditions, and evaluation metrics. The evaluation should measure prediction accuracy, processing time, memory consumption, throughput, and the usefulness of the generated insights for operational decision-making. This approach would address the main limitation of the reviewed studies, namely that strong performance on language, recommendation, or generic long-range benchmarks does not automatically establish effectiveness in a specific business environment.

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CHAPTER 3

METHODOLOGY

3.1 Research Design

This study employed a developmental research design to develop and evaluate Operlytics, a web-based business analytics and monitoring system for Empress Café Boracay. The methodology focused on identifying existing operational problems and translating the identified requirements into system functions for sales monitoring, inventory management, reporting, and business decision support. This approach is consistent with design and development research, which focuses on the systematic design, development, and evaluation of products and tools [18].

Developmental research as a methodological approach occupies a distinctive position in the research methodology spectrum, bridging the gap between pure theoretical research and applied implementation studies. Unlike experimental research, which seeks to establish causal relationships between variables under controlled conditions, developmental research focuses on creating and refining practical solutions to real-world problems. Unlike purely descriptive research, which documents existing conditions without intervention, developmental research actively intervenes to improve those conditions through the creation of new tools, processes, or systems.

In the context of this study, this methodology was employed to conceptualize, construct, and validate Operlytics, a web-based business analytics and monitoring system specifically designed for Empress Café Boracay. The choice of developmental research was particularly appropriate given the applied and solution-oriented nature of this study, which required not only the identification and analysis of operational problems but also the creation and evaluation of a functional technological response to those problems.

The developmental research approach aligned with the study's objectives by providing a structured framework for moving from identified problems to implemented solutions. The methodology encompassed several phases that corresponded to the natural progression of system development: needs analysis, design specification, development implementation, and evaluation validation. Each phase built upon the previous, ensuring that the resulting system was grounded in actual operational requirements and validated against real-world performance criteria.

The application of this design began with an in-depth needs analysis of the café's current operational practices. This initial phase involved a thorough examination of the existing workflows, recording methods, and reporting mechanisms currently employed by Empress Café Boracay in managing its daily operations. By investigating existing operational

bottlenecks — specifically manual record-keeping, tracking delays, human errors, and stock mismanagement — the researchers were able to define the precise functional requirements needed for the proposed system.

This needs analysis phase was critical to ensuring that the system developed was purposefully tailored to the unique operational realities of the café rather than serving as a generic solution. Data gathered during this phase served as the primary basis for defining the system's scope, features, and design priorities, laying a solid and evidence-based foundation for the subsequent stages of development. The needs analysis employed multiple data gathering techniques — interviews, observation, and document analysis — to ensure a comprehensive and well-validated understanding of operational requirements.

Following the analysis phase, the developmental process focused on designing, engineering, and evaluating the technological solution. This phase encompassed the planning of the system architecture, the design of the user interface, and the development of the core functional modules that constitute the Operlytics system, with particular attention given to ensuring accessibility for café staff and management who may have limited technical expertise.

By automating manual workflows related to sales and inventory, the developed system provided real-time data updates, enhanced stock monitoring accuracy, and reduced the operational errors that had historically undermined the reliability of the café's records. The automation design specifically targeted the manual process deficiencies identified during needs analysis, ensuring that each automated function directly addressed a documented operational problem.

The evaluation phase assessed the system's usability, functionality, and overall effectiveness in supporting management decision-making through feedback gathered from café administrators and staff during testing. This evaluation was not merely a technical verification of system operation but an empirical assessment of the system's practical value in the operational context for which it was designed. The evaluation criteria — functionality, usability, accuracy, security, and efficiency — provided a comprehensive framework for assessing system quality from multiple perspectives.

Ultimately, this research design facilitated a structured and comprehensive transition from identifying business problems to delivering a functional and validated system that meaningfully improves operational efficiency and managerial decision-making at Empress Café Boracay. The developmental approach ensured that the research produced not merely academic knowledge but practical value for the business partner and contributed to the broader understanding of how technology can address operational challenges in small business contexts.

The development process considered both technical and operational requirements. User requirements were gathered from the actual business environment and used as the basis for designing the system. The completed system was subsequently tested and evaluated according to defined criteria to determine its functionality, usability, accuracy, security, and efficiency. This also reflects systems development research, which involves the construction and evaluation of information systems in practical settings [19].

3.2 System Development Methodology

The development of Operlytics followed Agile Methodology, a process defined by continuous iteration, cross-functional collaboration, and rapid response to change. Unlike traditional linear models such as the Waterfall approach, which proceeds sequentially through distinct phases with limited opportunity for revision, Agile breaks the project into small, manageable increments known as Sprints, allowing the development team to progressively build and refine the system based on feedback and evolving requirements.

The Agile methodology emerged from the software development industry's recognition that traditional predictive approaches often failed to deliver satisfactory outcomes in dynamic environments where requirements were uncertain or subject to change. The Agile Manifesto, published in 2001, established four core values: individuals and interactions over processes and tools, working software over comprehensive documentation, customer collaboration over contract negotiation, and responding to change over following a plan — that continues to guide Agile practice today. These values were particularly relevant to the Operlytics development context, where close collaboration with café staff and management was essential for ensuring that the system met actual operational needs.

The iterative nature of the development process is consistent with Agile principles that emphasize working software, collaboration, and responsiveness to changing requirements [20]. Testing was conducted progressively to identify errors and verify the functionality of individual components before evaluating the complete system. User feedback and evaluation results were considered in refining the system before implementation.

The system evolved through repeated cycles, each consisting of the following phases:

Phase 1: Requirement Analysis

User stories and system requirements were identified and prioritized in a product backlog, ensuring that development efforts were aligned with the actual needs of Empress Café Boracay's administrators and staff. The product backlog served as a dynamic repository of desired features and functions, organized by priority and estimated effort. Requirements were expressed as user stories — brief descriptions of functionality from the perspective of the end user — that captured not merely technical specifications but the operational value that each feature was intended to deliver.

The requirement analysis phase for Operlytics drew upon the data gathered during the needs analysis phase of the research design. Interview transcripts, observation notes, and document analysis findings were translated into specific system requirements that addressed the operational deficiencies identified. Each requirement was evaluated for technical feasibility, operational value, and implementation effort to establish development priorities that maximized the system's practical impact within available development resources.

Phase 2: Iterative Design and Development

Design and coding occurred concurrently rather than sequentially. Flowcharts and database schemas were developed and refined as the system grew, ensuring that the architecture remained flexible and responsive to changes in requirements.

This concurrent approach allowed the development team to discover design issues during implementation and adjust accordingly, rather than discovering fundamental problems only after a complete design specification had been produced.

The iterative nature of this phase meant that the system was built incrementally, with each sprint delivering a functional subset of the complete system. Early sprints focused on core data structures and basic user authentication, while later sprints added POS functionality, inventory management, reporting, and dashboard visualization. This incremental approach provided several advantages: it allowed for early user feedback on working software rather than abstract designs; it reduced the risk of major architectural problems by revealing issues early; and it ensured that a functional system existed at all times, even if incomplete.

Phase 3: Continuous Testing

Testing was not treated as a final step but as an ongoing process integrated throughout development. Features were verified as they were built to identify and resolve issues early, maintaining the overall quality and stability of the system. This continuous testing approach encompassed multiple testing levels: unit testing of individual components, integration testing of component interactions, and system testing of complete user workflows.

The continuous testing philosophy aligned with the Agile principle that quality is built into the product through ongoing verification rather than inspected into the product through final-stage checking. By testing each feature as it was developed, the team could identify and correct defects when they were most easily and economically addressed — close to the time of their introduction — rather than accumulating a large backlog of issues that had to be resolved during a dedicated testing phase.

Phase 4: Deployment and Review

At the end of each sprint, a functional increment of the system was delivered and reviewed. This allowed stakeholders to monitor progress in real time and provide input before the next development cycle began. The deployment of sprint increments at Empress Café Boracay enabled café staff and management to interact with working software, providing feedback that directly influenced subsequent development priorities.

The review process at the end of each sprint involved demonstration of completed features to stakeholders, discussion of what was accomplished and what was not, and planning for the next sprint based on current priorities and feedback. This regular review cycle ensured that the development process remained aligned with user needs and that any misalignment between system development and operational requirements was identified and corrected promptly.

Phase 5: Feedback and Retrospective

Based on feedback gathered from users and stakeholders, the team identified areas for improvement and adjusted the backlog accordingly for the next sprint cycle. The retrospective phase involved reflection on the development process itself — what worked well, what could be improved, and what changes should be made to the team's working methods. This meta-level reflection ensured that the development process continuously improved alongside the product being developed.

The Agile methodology was selected for its adaptability, transparency, and user-centered approach. By delivering the system in functional increments, the development process ensured flexibility to accommodate new insights or shifting requirements, consistent quality through continuous testing and review, and alignment with the actual needs of the end users through regular feedback loops. These characteristics make Agile particularly suitable for developing systems in close collaboration with operational users who may have difficulty articulating complete requirements upfront but can provide valuable feedback on working software.

3.3 System Architecture

Operlytics followed a client-server architecture designed to convert operational data into clear and actionable insights for café management. This architectural pattern, which separates data processing and storage functions from user interface and interaction functions, is the dominant approach in modern web application development and offers particular advantages for the operational context of Empress Café Boracay. The client-server model has been widely adopted across industries because it enables centralized control, simplifies maintenance, and provides a scalable foundation for growing businesses that need to manage increasing volumes of data and user traffic over time.

In this architecture, the system was divided into two distinct but interconnected layers that worked together to ensure seamless data flow, reliable processing, and accessible presentation of operational information. The separation of concerns between client and server components enabled independent development, maintenance, and scaling of each layer, while standardized communication protocols ensured effective interaction between them. This layered approach meant that changes or improvements could be made to one side of the system without necessarily disrupting the other, which was critical for long-term sustainability and adaptability in a dynamic business environment.

Server-Side Architecture

The server side served as the core processing layer of the system, where a central engine continuously gathered data from business operations, processed and organized it, and stored it in a secure database that could be retrieved whenever needed. This server-side layer was responsible for handling all computational tasks, enforcing data integrity, and maintaining the accuracy and consistency of the information that flowed through the system.

The server components included the web server that handled HTTP requests from client browsers, the application server that executed business logic for transaction processing, inventory updates, and report generation, and the database server that stored and managed all operational data. Each of these components played a vital and complementary role in ensuring that the system operated smoothly and efficiently.

The PHP backend handled user requests, facilitated communication between the interface and the database, and ensured that the system functioned correctly and efficiently under regular operational conditions. PHP was chosen for its maturity, extensive community support, and proven reliability in web application development. It enabled rapid development cycles while maintaining the performance characteristics necessary for real-time operational systems. The backend code

implemented all core business logic, including order validation, inventory deduction algorithms, sales aggregation, user authentication, and report generation pipelines.

By centralizing data processing on the server, Operlytics ensured that operational records were protected, systematically organized, and readily available for retrieval and analysis at any point in time, regardless of how many users were simultaneously accessing the system. This centralized approach eliminated the risk of data fragmentation that occurred when information was stored locally on individual devices. It also ensured that all users saw the same, up-to-date information at any given moment, which was essential for coordinated decision-making across different roles and departments within the café.

The server-side processing model also simplified security management by concentrating data protection measures in a single controlled environment rather than distributing them across multiple client devices. Security policies, access controls, encryption standards, and audit logging could all be implemented and monitored from a central location. This reduced the attack surface of the system and made it easier to detect and respond to potential threats. Additionally, server-side validation ensured that all incoming data was checked for correctness and completeness before it was accepted into the system, preventing malformed or malicious inputs from compromising data integrity.

Furthermore, the server environment supported automated backup and recovery procedures, ensuring that critical business data could be restored in the event of hardware failure, software corruption, or other disaster scenarios. Regular backups were scheduled to run at intervals that balanced data freshness with system performance, and recovery protocols were tested periodically to verify their effectiveness.

Client-Side Architecture

On the client side, authorized users accessed a web-based dashboard through their computers or internet-connected devices, eliminating the need for any additional software installation and allowing management to monitor café operations from virtually any location with an internet connection. The dashboard communicated with the server through secure data requests to display real-time updates, visual reports, and operational summaries in a format that was intuitive and easy to interpret.

The user interaction flow was designed for efficiency and practicality: a manager logged in, viewed a high-level overview of the café's operational status through visual indicators and alerts, and navigated to specific metrics for more detailed analysis of sales performance, inventory levels, and overall business activity. Every element of the client-side interface was deliberately designed to minimize the steps required for users to access critical information, ensuring that even staff members with limited technical backgrounds could navigate the system confidently and effectively during their daily operations.

The client-side implementation used HTML for structural markup, CSS for visual presentation and layout, and JavaScript for dynamic and interactive elements that enhanced the overall responsiveness and usability of the dashboard during real-time operations. These standard web technologies ensured compatibility across diverse devices and browsers, maximizing the accessibility of the system for all intended users.

Architectural Benefits

This architecture ensured that all data processing and computation occurred on the server side, while the client side remained clean, responsive, and focused on delivering a straightforward experience for operational decision-making. By offloading complex calculations and data manipulation to the server, the client devices were freed from heavy processing burdens, which improved their battery life, reduced heat generation, and extended their usable lifespan.

The clear separation of responsibilities between the server and client layers not only improved the overall performance and reliability of the system but also enhanced its scalability, allowing Operlytics to accommodate growing volumes of operational data as the café expanded its business activities over time.

Furthermore, this architectural approach strengthened the security of the system by ensuring that sensitive business data was processed and stored within a controlled server environment rather than on individual user devices. The client-server model enabled centralized authentication, authorization, and audit logging, providing robust security controls that protected operational data from unauthorized access or modification.

The architecture also supported modular development and deployment practices. New features could be developed and tested in isolation before being integrated into the production environment. Bug fixes could be deployed to the server without requiring users to update their browsers or install patches.

Cache invalidation strategies ensured that users always saw current data when significant changes occurred.

Ultimately, the client-server architecture of Operlytics reflected a deliberate design philosophy centered on reliability, accessibility, and efficiency — core principles that aligned directly with the operational needs and management goals of Empress Café Boracay. The architecture supported the system's functional requirements while providing a foundation for future growth and enhancement as the café's operational needs evolved.

3.4 Tools and Technologies Used

The development of Operlytics utilized a set of web-based tools and technologies selected for their reliability, compatibility, and suitability for building a business analytics and monitoring system. Each technology was deliberately chosen not only for its individual capabilities but also for its ability to integrate seamlessly with the other components of the system, forming a cohesive and stable development stack that supported the full range of functional requirements identified for Operlytics.

Backend Technology: PHP

PHP was used to handle all server-side operations of the system. It managed user requests, facilitated communication between the interface and the database, and ensured that the system functioned correctly and efficiently under regular operational conditions. PHP is widely used in web development and is well-suited for building dynamic, data-driven

applications such as Operlytics, making it a practical and dependable choice for managing the backend logic that powers the system's core features, including sales processing, inventory updates, and automated report generation.

PHP's widespread adoption in web development ensures extensive documentation, large developer communities, and abundant learning resources that support development and maintenance. Its compatibility with MySQL databases through established connection libraries simplifies data access and manipulation. Its execution model, which processes each request independently, aligns well with the stateless nature of HTTP and the transactional processing requirements of the POS module.

Frontend Technologies: HTML, CSS, and JavaScript

HTML, CSS, and JavaScript were used to develop the user-facing interface of the system. These technologies worked together to create a visually organized, responsive, and accessible interface that allowed café administrators and staff to navigate and interact with the system efficiently. HTML provided the structural foundation of each page, CSS governed the visual presentation and layout to ensure a clean and intuitive design, and JavaScript enabled dynamic and interactive elements that enhanced the overall responsiveness and usability of the dashboard during real-time operations.

The combination of these three technologies represents the standard foundation of modern web development, ensuring broad compatibility across browsers and devices. HTML5 semantic elements improve accessibility and searchability. CSS3 features enable sophisticated visual designs without requiring external libraries. JavaScript, potentially enhanced with libraries such as jQuery or Chart.js for dashboard visualization, provides the interactivity that transforms static pages into dynamic applications.

Database Technology: MySQL

MySQL served as the database management system for Operlytics. It stored and organized all operational data, including product information, inventory records, and sales transactions. MySQL provided a structured and reliable environment for managing large volumes of data, ensuring that information could be retrieved accurately and efficiently. Its compatibility with PHP made it a natural fit within the system's overall technology stack, and its widespread adoption in web-based applications ensured strong community support and well-established best practices for data management and security.

MySQL's relational data model aligns well with the structured operational data of a café business, where entities such as products, inventory items, transactions, and users have well-defined attributes and relationships. The SQL query language enables powerful data retrieval and aggregation operations that support the reporting and dashboard functions of Operlytics. MySQL's transaction support ensures data integrity by guaranteeing that related operations such as recording a sale and deducting inventory complete successfully or roll back entirely, preventing inconsistent data states.

Development Tools

Visual Studio Code was used as the primary code editor for writing and managing the system's source code, offering a lightweight yet feature-rich environment that supported syntax highlighting, code completion, and extensions that streamlined the development workflow. XAMPP was used as the local development environment, enabling the team to run and test the PHP server and MySQL database on local machines before deployment, providing a controlled and consistent testing environment that reduced the risk of configuration issues arising during the transition to a live server.

Browser: Google Chrome

Google Chrome was used as the primary browser for testing and validating the system's interface and functionality throughout the development process. As one of the most widely used web browsers globally, Chrome provided a reliable and standardized testing environment that helped ensure the system's interface rendered correctly and performed consistently for the majority of its intended users.

Technology Selection Rationale

These tools were selected because they are widely used, well-documented, and compatible with one another, providing a stable and efficient foundation for the development of a web-based business analytics and monitoring system. The total cost of ownership for this technology stack considering licensing, hosting, maintenance, and support is minimal compared to proprietary enterprise solutions, making it appropriate for the resource constraints of a small café establishment.

3.5 Data Gathering Techniques

To ensure that Operlytics was developed based on accurate and relevant information, the researchers employed multiple data gathering techniques to understand the current operational practices of Empress Café Boracay and identify the specific needs the system had to address. The use of multiple complementary methods was intentional, as relying on a single data source may produce an incomplete or one-sided understanding of the café's operational environment. By triangulating findings across interviews, observation, and document analysis, the researchers aimed to build a thorough and well-validated picture of the café's existing workflows, challenges, and data management practices that directly informed the design and development of the Operlytics system.

Technique 1: Interview

The researchers conducted structured interviews with the café owner and authorized staff to gain a comprehensive understanding of how sales and inventory were currently managed. The interviews focused on identifying existing workflows, recurring operational challenges, and the specific features that users expected from the system. This method allowed the researchers to capture firsthand accounts and perspectives from the individuals who were most directly affected

by the current operational limitations, ensuring that the system being developed reflected the genuine priorities and expectations of its intended users rather than assumptions made by the development team.

The interview protocol included questions about daily operational routines, specific pain points in current processes, desired improvements, and expectations for a new system. Interviews were conducted individually to encourage candid responses and collectively to identify shared perspectives and priorities. The structured format ensured that all relevant topics were covered while allowing flexibility for participants to raise issues not anticipated by the researchers.

Interview Protocol Design

The interview protocol for this study was developed based on preliminary understanding of café operations and refined through pilot testing. The protocol was organized into thematic sections that progressively explored different aspects of operational practice:

Section 1: Operational Overview established the context of each participant's role and responsibilities.

Questions explore: the participant's position and tenure at the café; typical daily activities and responsibilities; hours of work and seasonal variations in workload; and interactions with other staff members and management.

Section 2: Sales Recording Practices examined how customer transactions are currently documented.

Questions explore: the specific steps involved in recording a sale from customer order to final documentation; the tools and materials used (paper logbooks, calculators, receipts); the consistency of recording practices across different staff members and shifts; the time required for sales recording during quiet and busy periods; and the frequency and nature of errors encountered in sales records.

Section 3: Inventory Management Practices investigated how stock levels are monitored and maintained.

Questions explore: the methods used for tracking inventory (physical counts, visual estimates, informal notes); the frequency and timing of inventory checks; the process for recording stock receipts from suppliers; the handling of waste, spoilage, and returns; the identification of low-stock situations and the process for reordering; and the challenges encountered in maintaining accurate inventory awareness.

Section 4: Reporting and Analysis assessed how operational information is compiled and used for decision-making.

Questions explore: the types of reports currently generated (sales summaries, inventory status, financial calculations); the frequency and timing of report preparation; the process and time required for report compilation; the recipients and uses of operational reports; the perceived accuracy and usefulness of current reports; and the decisions that reports inform or fail to inform.

Section 5: Operational Challenges invites participants to articulate the most significant difficulties they encounter in their work.

Questions explore: the most frustrating or time-consuming aspects of current operational practices; situations where current practices have led to significant problems (stockouts, customer complaints, financial discrepancies); workarounds or coping strategies developed to manage operational deficiencies; and the consequences of operational problems for their work experience and job satisfaction.

Section 6: System Expectations elicits participant visions for an improved operational system.

Questions explore: the features or capabilities that would most improve their daily work; the concerns or reservations about adopting a new digital system; the training and support that would facilitate successful adoption; and the criteria by which they would judge a new system to be successful or unsuccessful.

Interview Implementation

Interviews were conducted at the café premises during periods that minimized disruption to operations, typically during mid-morning or mid-afternoon lulls between peak service periods. Individual interviews lasted approximately 45-60 minutes, while group interviews with multiple staff members lasted approximately 60-90 minutes. All interviews were audio-recorded with participant consent, and detailed notes were taken to capture non-verbal cues and contextual observations.

Technique 2: Observation

The researchers observed how staff members performed manual recording and inventory management tasks during actual café operations. This method allowed the researchers to directly identify inefficiencies, delays, and error-prone steps in the current process that the system was designed to address. Unlike interviews, which rely on self-reported information, observation provided an objective and unfiltered view of how operations were actually carried out on a day-to-day basis, enabling the researchers to identify issues that staff members may have normalized or overlooked due to familiarity with existing routines.

Observation sessions were conducted at various times of day and days of the week to capture the full range of operational conditions, from quiet periods to peak service times. The researchers documented workflow sequences, time requirements for specific tasks, error occurrences, and workarounds that staff had developed to cope with system deficiencies. These observational data provided concrete evidence of operational problems that complemented the subjective accounts obtained through interviews.

Observation Protocol Design

The observation protocol was structured to capture comprehensive information about operational workflows while minimizing intrusiveness that might alter natural behavior. The protocol included:

Temporal Coverage: Observation sessions were conducted at various times of day and days of the week to capture the full range of operational conditions. The schedule includes: weekday mornings (typically 7:00-10:00 AM) when breakfast service and preparation activities occur; weekday midday (11:00 AM-2:00 PM) during lunch peak service; weekday afternoons (3:00-5:00 PM) during quiet periods; weekend mornings and midday when tourist traffic is typically higher; and evening periods (6:00-9:00 PM) during dinner service if applicable.

This temporal variation was essential because tourism-dependent operations like Empress Café Boracay experience dramatic fluctuations in customer volume and operational intensity. Systems designed based on observations during only quiet periods may fail catastrophically during peak demand, while systems designed only for peak periods may be unnecessarily complex and costly for routine operations.

Spatial Coverage: Observations covered all operational areas of the café: the customer service area where orders are taken and payments processed; the preparation area where food and beverages are assembled; the storage area where inventory is maintained; and the administrative area where records are kept and reports compiled.

Observation Dimensions: For each observed activity, researchers document: workflow sequence the specific steps involved in completing a task from initiation to completion; time requirements the duration of each step and the total task completion time; error occurrences mistakes, corrections, workarounds, and their consequences; communication patterns interactions between staff members during task execution; tool and material usage the physical and informational resources employed; and environmental factors crowding, noise, temperature, and other conditions that may affect task performance.

Specific Observation Targets

Sales Transaction Recording: Researchers observe the complete process from customer order through final record entry. Observations capture: how staff receive and confirm orders; the tools used for recording (paper, calculators, memory); the sequence of data entry and calculation; the handling of special cases (modifications, discounts, complaints); the time from order completion to record finalization; and the accuracy of final records through spot-checking against actual orders.

Inventory Management Activities: Researchers observe stock handling practices including: receipt of deliveries from suppliers and the recording of incoming stock; storage organization and the accessibility of items during preparation; visual or physical stock checks and their timing relative to need; identification of low-stock or expired items; and the handling of waste, spoilage, or returns.

Report Preparation: When management prepares operational reports, researchers observe: the collection of source documents from multiple locations; the transcription and aggregation of data; the calculation and verification of summary figures; the formatting and organization of results; and the time from initiation to completion.

Workaround Identification: A particular focus of observation was the identification of informal practices that staff have developed to cope with system deficiencies. These workarounds such as mental arithmetic instead of written calculation, informal communication instead of formal documentation, or stock estimation instead of physical counting represent adaptive responses that reveal the true operational needs that formal systems may obscure.

Observation Documentation

Observations were documented through structured field notes that record the observation dimensions for each activity. Time-stamped entries capture the sequence and duration of operational events. Diagrammatic sketches illustrate spatial layouts and workflow paths. Photographs (with permission) document physical environments, tools, and records. Audio recordings (with permission) capture verbal communications during task execution.

Technique 3: Document Analysis

The researchers examined existing operational documents such as stock lists, sales receipts, and inventory logs to understand how data was currently recorded, organized, and maintained. This analysis provided additional context on the café's data management practices and informed the design of the system's data structure and reporting features. By reviewing actual records used in the café's daily operations, the researchers gained a concrete understanding of the formats, categories, and types of information that the system needed to be capable of processing, storing, and presenting in a clear and accessible manner.

Document analysis revealed the data elements that needed to be captured by the system, the relationships between different data types, and the reporting formats that users found familiar and useful. The examination of historical records also provided baseline data against which system accuracy could be validated — comparing system-generated reports to manually compiled records from the same period to verify correctness.

Document Categories Analyzed

Sales Records: The researchers examined all forms of sales documentation including: handwritten logbooks or notebooks maintained by individual staff members; printed receipts from any existing point-of-sale devices or manual receipt books; daily sales summaries compiled by management; and informal notes or tallies used during busy periods.

Inventory Records: The researchers examined inventory documentation including: stock count sheets or logbooks; supplier delivery receipts and invoices; informal notes about stock levels or needs; and records of waste, spoilage, or returns.

Financial Records: The researchers examined financial documentation including: cash tally sheets; bank deposit records; expense receipts; and any profit or loss calculations.

Analysis of financial records focused on: the sources of revenue and categories of expense tracked; the methods of calculation and aggregation; the frequency of financial summary preparation; and the relationship between operational records and financial outcomes.

Administrative Documents: The researchers examined any additional operational documentation including: staff schedules; supplier contact lists; product recipes or preparation guides; and customer feedback or complaint records.

Document Analysis Procedures

Documents were collected through photocopying or photography (with permission) to enable detailed analysis without removing originals from the café. Each document was catalogued with metadata including: document type, date or date range, author or responsible party, purpose, and relationship to other documents.

Baseline Data Establishment

The examination of historical records provided baseline data against which system accuracy can be validated. By comparing system-generated reports to manually compiled records from the same period, researchers can verify that the new system produced results consistent with known operational outcomes. This baseline validation was essential for establishing user trust in the new system and for demonstrating that the digital transformation has not introduced new sources of error.

Triangulation and Validation

The combination of these three data gathering methods provided the researchers with a well-rounded and empirically grounded understanding of the operational environment at Empress Café Boracay, ensuring that Operlytics was developed to meet the actual needs of its intended users. The insights derived from interviews captured the experiential knowledge and expectations of café personnel, while observations revealed the practical realities of day-to-day operations that were not fully communicated through verbal accounts alone. Document analysis, in turn, grounded the research in concrete evidence of how data was currently handled, providing a factual basis for the system's design decisions.

Together, these methods formed a robust data collection strategy that minimized the risk of oversight, reduced the likelihood of developing features that did not align with actual user needs, and ensured that the resulting system was both contextually appropriate and operationally relevant to the unique environment of Empress Café Boracay. Ultimately, the thoroughness of this data gathering process served as the foundation upon which a reliable, practical, and effective business analytics system could be successfully built and deployed.

Triangulation Procedures

The triangulation process involved systematic comparison of findings across the three data sources:

Convergence Analysis identified areas where findings from all three methods align, indicating high-confidence understanding. For example, if interviews reveal staff frustration with manual inventory counting, observations confirm that counting consumes substantial time during busy periods, and documents show inconsistent inventory records with frequent discrepancies, researchers can be confident that inventory management represents a genuine operational priority.

Divergence Analysis identified areas where findings from different methods conflict, indicating need for further investigation. For example, if interviews report that sales recording is accurate and complete, but observations reveal frequent estimation and delayed documentation, and documents show arithmetic errors and missing entries, researchers must investigate the reasons for this discrepancy perhaps staff are unaware of their own errors, or social desirability bias leads to overstated performance in interviews.

Complementarity Analysis identified areas where different methods reveal distinct but compatible aspects of operational reality. For example, interviews may reveal management's strategic concern about competitive positioning, observations may reveal operational bottlenecks that constrain service quality, and documents may reveal financial pressures that limit investment capacity. These complementary findings together inform a holistic understanding of the café's situation that guides system design priorities.

Validation Criteria

The quality of data gathered through these techniques was assessed against established criteria for qualitative research:

Credibility refers to the believability and authenticity of findings from the perspective of participants. Credibility was enhanced through prolonged engagement with the café environment, persistent observation across diverse conditions, and member checking where preliminary findings were shared with participants for verification and correction.

Transferability refers to the extent to which findings may be relevant to other contexts. While this study focuses specifically on Empress Café Boracay, careful documentation of contextual characteristics enables readers to assess the applicability of findings and design decisions to other small food service businesses in similar tourism-dependent environments.

Dependability refers to the consistency and reliability of findings over time and across researchers. Detailed documentation of data gathering procedures, structured protocols for interviews and observations, and systematic archiving of raw data supported dependability by enabling audit and replication.

Confirmability refers to the objectivity and neutrality of findings, ensured through reflexive practice where researchers examine their own assumptions and biases, and through triangulation that subjects emerging interpretations to multiple independent sources of evidence.

3.6 Testing Procedures

The validation of Operlytics followed a structured, three-phase testing framework designed to ensure both technical stability and practical utility before full implementation. This framework reflected a progressive and methodical approach to system validation, beginning with the examination of individual components and advancing toward a comprehensive assessment of the system's performance in conditions that closely replicated the actual operational environment of Empress Café Boracay. Each phase served a distinct and essential purpose in confirming that the system was not only technically sound but also genuinely responsive to the needs of its intended users.

The adoption of structured, multi-phase testing frameworks represents established best practice in software engineering and information systems development. A progressive testing approach beginning with component-level verification and advancing through integration and system-level evaluation minimizes the risk of compounding errors and ensures that issues are identified and resolved at the earliest possible stage. Thorough validation is essential for ensuring that systems meet both technical specifications and user requirements before deployment in operational environments where failures can disrupt business activities and damage user confidence.

Phase 1: Unit Testing

This phase focused on isolating and verifying individual system components, such as backend data processing scripts and frontend interface handlers. Each module was tested independently to confirm that it processed inputs correctly and handled edge cases before being integrated into the full system. By identifying and resolving errors at the component level early in the validation process, unit testing minimized the risk of compounding issues arising during later stages of system integration and evaluation.

Unit testing for Operlytics encompassed verification of database operations, validation of calculation algorithms, testing of user authentication procedures, and confirmation of data validation rules. Each unit test specified expected inputs, expected outputs, and the conditions under which the component should operate correctly. Automated unit testing tools were employed to facilitate repeated execution of test suites as the system evolved.

Unit testing for Operlytics encompasses verification across multiple component categories:

Database Operations Testing verified that all database interactions perform correctly and maintain data integrity. Test cases include: insertion of new records with valid and invalid data to confirm proper acceptance and rejection; update operations that modify existing records while preserving referential integrity; deletion operations with cascade effects to ensure related records are handled appropriately; query operations that retrieve data with correct filtering, sorting, and aggregation; and transaction management that ensures atomicity related operations complete entirely or roll back completely, preventing partial updates that would corrupt data consistency.

Calculation Algorithm Testing validated the arithmetic and logical operations that produce derived values throughout the system. Test cases include: sales total calculations that sum line items, apply taxes, and compute discounts correctly; inventory deduction algorithms that reduce stock quantities by precise recipe-specified amounts when sales are

recorded; profit margin calculations that correctly relate revenue to cost of goods sold; and alert threshold evaluations that trigger notifications at configured stock levels.

User Authentication Testing confirmed that access control mechanisms operated reliably. Test cases include: valid credential acceptance that grants appropriate access levels; invalid credential rejection that prevents unauthorized entry; password complexity enforcement that requires minimum standards for new passwords; session management that maintains secure access during active use and terminates appropriately after periods of inactivity; and role-based permission verification that restricts users to functions appropriate for their assigned roles.

Data Validation Testing ensured that input checking prevented malformed or malicious data from entering the system. Test cases include: numeric field validation that rejects non-numeric entries in quantity and price fields; date validation that prevents future dates for past transactions or impossible date combinations; range validation that flags quantities or prices outside reasonable boundaries; and format validation that ensures consistent data entry for fields such as product codes or contact information.

Phase 2: System Testing

This phase evaluated the complete end-to-end functionality of the system as a unified whole. It assessed overall performance, data accuracy, and system stability under varying conditions, including high volumes of simultaneous user activity, to ensure that the platform operated reliably in a real operational environment. This phase was critical in confirming that all individual components functioned cohesively together and that the system as a whole delivered consistent and accurate outputs under the demands of regular café operations.

System testing for Operlytics encompassed workflow testing that followed complete user scenarios from login through transaction processing to report generation, load testing that verified system performance under multiple concurrent users, data integrity testing that confirmed accurate propagation of updates across related records, and security testing that validated access controls and data protection measures.

System testing for Operlytics encompassed several specialized testing types:

Workflow Testing followed complete user scenarios from initial login through transaction processing to report generation, verifying that each step in operational workflows executed correctly and that data flowed accurately between system components. Representative test scenarios include: a staff member logs in, records a multi-item customer order, processes a split payment, and generates a receipt; an administrator logs in, reviews dashboard alerts for low-stock items, updates inventory through manual adjustment, and generates an inventory status report; and a manager logs in, reviews daily sales summary, examines product performance trends, and exports data for external analysis.

Load Testing verified system performance under multiple concurrent users, simulating the conditions of peak service periods at Empress Café Boracay when several staff members may simultaneously process orders, update records, and access reports. Test parameters include: response time measurements for common operations under increasing user loads; identification of performance degradation points where response times exceed acceptable thresholds; verification that data

integrity was maintained under concurrent access conditions; and assessment of system behavior at maximum expected load levels.

Data Integrity Testing confirmed accurate propagation of updates across related records, ensuring that actions in one system area correctly affect all dependent data. Test cases include: sales transaction recording that automatically updates inventory quantities, sales totals, and dashboard displays; inventory adjustments that correctly modify stock levels and trigger or clear low-stock alerts as appropriate; user role modifications that immediately affect access permissions without requiring system restart; and report generation that accurately aggregates underlying transaction data without omission or duplication.

Security Testing validated that access controls and data protection measures functioned correctly in the integrated system. Test cases include: unauthorized access attempts that confirm proper rejection and logging; privilege escalation attempts that verify role-based restrictions cannot be circumvented; data transmission interception tests that confirm encryption protects sensitive information; and session hijacking attempts that validate secure session management.

The system testing phase also included stress testing that evaluated behavior under conditions exceeding normal operational parameters, such as transaction volumes significantly above peak expectations or database sizes substantially larger than current requirements. While such extreme conditions were not expected to occur regularly, stress testing revealed architectural limitations and provided confidence in system resilience.

Phase 3: User Acceptance Testing (UAT)

This phase involved café administrators and authorized staff interacting with the system in a controlled testing environment that simulated actual café operations. UAT confirmed whether the system's features, interface, and workflows met the practical needs of the end users and whether the system effectively supported operational decision-making. Feedback gathered during this phase served as the final basis for identifying any remaining usability concerns or functional gaps that had to be addressed prior to the system's full deployment.

User Acceptance Testing at Empress Café Boracay involved structured test scenarios that required users to perform typical operational tasks using the system, observation of user interactions to identify usability issues, and collection of subjective feedback through questionnaires and interviews. The UAT phase was not merely a demonstration but an empirical evaluation that determined whether the system was ready for operational deployment.

User Acceptance Testing at Empress Café Boracay involved multiple evaluation activities:

Structured Test Scenarios required users to perform typical operational tasks using the system, with success criteria defined for each scenario. Representative scenarios include: a floor staff member processes a complete customer order from product selection through payment and receipt generation within a specified time limit; a kitchen staff member checks ingredient availability for upcoming orders and updates inventory for items used in preparation; an administrator generates daily sales and inventory reports and verifies that displayed figures match expected values based on known transaction activity; and a manager reviews dashboard displays during a simulated busy period and identifies key operational metrics requiring attention.

Observational Assessment involved researchers systematically watching representative users interact with the system while performing realistic tasks, noting difficulties, errors, hesitations, workarounds, and verbalizations. Observation focuses on: navigation patterns that reveal interface clarity or confusion; error recovery behaviors that indicate system forgiveness or fragility; efficiency measures such as time to task completion and number of interactions required; and emotional responses including frustration, satisfaction, or confidence that predict sustained adoption.

Feedback Collection gathered subjective experiences, opinions, preferences, and suggestions through structured questionnaires, semi-structured interviews, and focus group discussions. The System Usability Scale (SUS) provided standardized quantitative measurement of perceived usability for comparison against established benchmarks. Open-ended questions explore: perceived usefulness of specific features for daily operational tasks; ease of learning and confidence in system operation; comparison with previous manual methods in terms of efficiency and reliability; concerns or reservations about system adoption; and suggestions for improvements or additional capabilities.

Comprehensive Quality Assurance

Collectively, these three validation phases established a rigorous and comprehensive quality assurance process that addressed the system's performance from multiple perspectives — technical, operational, and human. The progressive nature of the framework ensured that issues were identified and resolved at the earliest possible stage, reducing the likelihood of significant problems emerging during actual deployment. By subjecting Operlytics to thorough unit, system, and user acceptance testing, this study aimed to deliver a system that was not only free of critical technical deficiencies but also genuinely aligned with the operational workflows and decision-making needs of Empress Café Boracay, providing management and staff with a reliable and effective tool that they could confidently integrate into their daily business operations.

The integration of unit, system, and user acceptance testing into a comprehensive quality assurance framework aligns with established best practices in software engineering and information systems development. The progressive approach where early defect detection through unit testing reduces the cost and complexity of later corrections, while system and UAT phases ensure both technical integrity and operational suitability is a fundamental principle of software quality assurance. Comprehensive quality assurance must address multiple perspectives, technical correctness, performance under realistic conditions, and user satisfaction to deliver systems that are reliable in production environments.

For business analytics systems, comprehensive quality assurance must include validation of real-time data accuracy, performance under operational loads, and user confidence in decision-support capabilities. The impact of business analytics on organizational decision-making depends on both technical system quality and user trust, which comprehensive testing frameworks must establish. The progressive testing framework employed for Operlytics thus ensured that the system delivered to Empress Café Boracay was reliable, effective, and genuinely supportive of daily business operations.

3.7 Evaluation Criteria

The completed system was evaluated using five major criteria: functionality, usability, accuracy, security, and efficiency. Functionality determines whether the system performs its intended functions correctly. Usability evaluates the ease of navigation, understanding, and use of the system. Accuracy assesses the correctness and consistency of sales, inventory, and reporting data.

Security evaluates controls for protecting user accounts and business information. Efficiency assesses the system's ability to perform required operations within reasonable processing time and resource requirements. The evaluation results provide the basis for determining whether Operlytics adequately addresses the operational problems identified in the study and supports the business requirements of Empress Café Boracay.

The Evaluation of Operlytics was conducted based on the following five core criteria to ensure that the system met both technical standards and the operational needs of Empress Café Boracay. Each criterion has been carefully selected to reflect the specific functional requirements and operational priorities identified during the needs analysis phase of this study, providing a comprehensive and structured basis for assessing the system's overall readiness and effectiveness for deployment in a real café environment.

The selection of these five criteria functionality, usability, accuracy, security, and efficiency draws upon established frameworks for information systems evaluation. Research on business analytics and business intelligence systems emphasizes that effective evaluation must encompass both technical performance dimensions and human factors such as usability and trust. Business analytics maturity assessments require multi-dimensional evaluation across process and performance outcomes. Understanding business analytics impact requires evaluation that includes both technical and organizational dimensions. Business analytics systems must be evaluated not only on technical capabilities but also on their practical impact on decision-making and operational outcomes.

Criterion 1: Functionality

This criterion assessed whether the system successfully executed all intended features, including the Point of Sale system, real-time sales and inventory monitoring, automated report generation, and dashboard data visualization, ensuring that all components operated without failure. Evaluators verified that each feature performed as intended under normal operating conditions and that no critical functions were absent or producing incorrect outputs during the testing phase.

Functionality evaluation involved systematic testing of each system feature against its specified requirements, verification that feature outputs matched expected results, and confirmation that all features were accessible to authorized users. The functionality criterion ensured that Operlytics delivered the complete set of capabilities that the café requires for operational management.

In the context of POS and inventory systems, functionality testing has been employed to verify complete operational capability. Prior studies have conducted functionality testing of POS and inventory management systems in local food service businesses, verifying that sales processing, inventory updates, and report generation all operated correctly under typical usage

scenarios. Similarly, evaluations of POS systems with card readers have confirmed that all features, including transaction processing and inventory tracking, performed as intended without critical failures. Web-based sales and inventory systems for small retail businesses have likewise been tested by systematically verifying each feature against specified requirements to ensure complete operational capability.

The functionality evaluation for Operlytics encompassed verification that the Point of Sale module correctly processed transactions, calculated totals, applied discounts, and generated receipts; confirmation that the inventory tracking system accurately recorded stock movements, updated quantities in real-time, and generated low-stock alerts; validation that the reporting module produced correct summaries of sales, inventory, and financial performance; and confirmation that the dashboard displayed accurate, current operational data in an accessible visual format.

Criterion 2: Usability

This criterion measured how easily and intuitively café administrators and staff could navigate the system, access relevant data, and perform their assigned tasks with minimal training or technical difficulty. Feedback gathered from café staff and management during the evaluation phase was used to assess the clarity of the interface, the ease of navigation, and the overall user experience provided by the system in the context of daily café operations.

Usability evaluation employed both objective measures — such as task completion time, error rates, and navigation path efficiency — and subjective measures — such as user satisfaction ratings, perceived ease of use, and confidence in system operation. The usability criterion ensured that Operlytics was not merely technically capable but practically accessible to its intended users.

Usability evaluation is a critical dimension of information system assessment, particularly for systems deployed in small business environments where users may have limited technical expertise. A comprehensive usability assessment of a centralized POS system for inventory management in a rotisserie operation employed both objective measures (task completion time, error rates) and subjective measures (user satisfaction ratings, perceived ease of use) to evaluate how easily staff could navigate the system and perform daily tasks. The findings of that assessment emphasize that usability directly impacts operational adoption rates and staff confidence in system operation.

Research on information dashboard design stresses that usability is paramount for at-a-glance monitoring systems, as users must be able to interpret data quickly and accurately without extensive training. The design of effective dashboards requires careful consideration of which metrics are most important for management decision-making, how these metrics relate to one another, how they can be visually represented for maximum clarity and interpretive accuracy, and how the dashboard layout supports natural scanning patterns and cognitive workflows.

In the context of small retail businesses, feedback from staff and management has been gathered to assess interface clarity and navigation ease, confirming that web-based systems were accessible to users with minimal technical background. These findings align with the broader understanding that systems must be user-friendly to achieve their intended benefits,

particularly in small business contexts where dedicated IT support is unavailable and staff may have limited technical expertise.

The usability evaluation for Operlytics encompassed measurement of task completion times for common operations (recording a sale, checking inventory, generating a report), assessment of error rates during data entry and navigation, evaluation of user satisfaction through structured questionnaires, and observation of user interactions to identify points of confusion or difficulty.

Criterion 3: Accuracy

This criterion evaluated the reliability and correctness of the data processed and displayed by the system, including sales records, inventory levels, and generated reports, to ensure that the information presented accurately reflected actual operational conditions. Evaluators compared system-generated data against actual operational records to identify any discrepancies and confirm that Operlytics consistently produced reliable and trustworthy outputs.

Accuracy evaluation involved data verification tests that compared system outputs to manually calculated results, reconciliation tests that verified consistency between related data elements, and historical validation that confirmed system reports matched known operational outcomes. The accuracy criterion ensured that management decisions based on Operlytics data were founded on correct information.

Data accuracy is fundamental to the value proposition of any business analytics or monitoring system. Seminal work on business intelligence and analytics emphasizes that the impact of analytics depends critically on the accuracy and reliability of the underlying data; inaccurate data leads to flawed insights and poor decision-making. The foundational importance of data quality for analytics effectiveness is further reinforced by research noting that data quality, including accuracy, is essential for driving informed decision-making in big data analytics contexts.

Business intelligence dashboard optimization models highlight that real-time performance tracking requires accurate data inputs to produce trustworthy forecasts and performance indicators. Studies of real-time business intelligence systems further support this, finding that decision-making behavior is directly influenced by users' trust in data accuracy. When users perceive data as unreliable, they are less likely to base decisions on system outputs, undermining the system's operational value.

In POS and inventory system contexts, accuracy validation has been demonstrated through multiple empirical studies. Data verification tests comparing system-generated sales records and inventory levels against manually calculated results have confirmed that POS and inventory systems produced reliable outputs. Reconciliation tests in web-based inventory management systems have similarly verified consistency between related data elements and ensured that reports matched known operational outcomes. These practices establish that accuracy is a core information quality dimension, essential for ensuring that management decisions are founded on correct information.

The accuracy evaluation for Operlytics encompassed comparison of system-generated sales totals against manually calculated figures from the same period, verification that inventory deductions correctly reflected actual ingredient consumption, confirmation that financial reports accurately aggregated transaction data, and validation that dashboard displays reflected current rather than cached or outdated information.

Criterion 4: Security

This criterion ensured that all operational data stored and transmitted by the system was adequately protected through secure user authentication, access controls, and safe communication between the client and server, preventing unauthorized access or data breaches. The evaluation examined the effectiveness of the system's login mechanisms, role-based access controls, and data transmission protocols to confirm that the system met acceptable standards of data protection.

Security evaluation included penetration testing that attempted to identify vulnerabilities, access control verification that confirmed users could access only authorized functions, and data protection assessment that verified encryption and secure handling of sensitive information. The security criterion ensured that café operational data remains confidential and protected against unauthorized access.

Information security is a non-negotiable requirement for business systems handling sensitive operational data. Systems must protect data from unauthorized access, alteration, or destruction. Secure user authentication, access controls, and data encryption are essential for protecting organizational information assets. As systems become more data-intensive, security measures including secure data transmission protocols become increasingly critical for maintaining operational trust and regulatory compliance.

In the context of POS systems, security evaluation has focused on verifying that access controls effectively prevent unauthorized use. Evaluations of POS systems with card readers have examined login mechanisms and access controls to ensure that only authorized personnel could process transactions and access sensitive financial data. Role-based access controls have been implemented in food service POS systems, with verification that users could access only functions appropriate to their responsibilities. Access control verification in web-based sales and inventory systems has likewise confirmed that authentication mechanisms effectively prevented unauthorized access.

The security evaluation for Operlytics encompassed verification that password-based authentication correctly validated user credentials, confirmation that role-based access controls restricted users to functions appropriate for their responsibilities, assessment that data transmission between client and server was encrypted, and validation that session management prevented unauthorized access from unattended devices.

Criterion 5: Efficiency

This criterion assessed the system's overall performance, including the speed at which data was processed, updated, and displayed on the dashboard, ensuring that the system remained responsive and reliable during regular café operations. The evaluation examined the system's response times and overall stability under typical usage conditions to confirm that Operlytics could sustain consistent performance throughout the demands of the café's daily operations.

Efficiency evaluation involved performance testing that measured response times for typical operations, load testing that verified performance under expected user volumes, and stability testing that confirmed reliable operation over extended periods. The efficiency criterion ensured that Operlytics supported rather than impeded operational workflow speed.

System efficiency is critical for ensuring that technology supports rather than hinders operational workflows. Efficiency is a measure of how quickly a system performs its functions and how well it utilizes computing resources, and inefficient systems can create bottlenecks in operational processes. Systems must process data quickly and respond promptly to user requests to maintain productivity.

In the context of real-time business intelligence, studies of system responsiveness show that delays in data processing and dashboard updates can reduce user engagement and trust in the system. Business intelligence dashboards must deliver real-time performance tracking with minimal latency to be effective for managerial decision-making. Dashboard systems designed for at-a-glance monitoring must also update efficiently to provide timely insights.

In POS system evaluations, efficiency testing has verified that systems remain responsive during peak operational periods. Performance testing has been used to measure response times for typical operations such as sales processing and inventory queries, ensuring that systems remained responsive during peak operational periods. Stability testing of web-based inventory management systems has confirmed reliable operation over extended periods. Web-based systems have similarly been evaluated under expected user volumes to verify that they could sustain consistent performance during daily operations.

The efficiency evaluation for Operlytics encompassed measurement of response times for common operations (loading the dashboard, processing a sale, generating a report), verification that the system remained responsive when multiple users accessed it simultaneously, confirmation that real-time updates appeared on dashboards within acceptable time limits, and assessment of system stability over extended operational periods.

Comprehensive Evaluation Framework

Taken together, these five evaluation criteria formed a comprehensive and balanced framework for assessing the overall quality and operational suitability of the Operlytics system. By measuring the system against standards of functionality, usability, accuracy, security, and efficiency, this study ensured that the evaluation process went beyond a purely technical assessment to encompass the practical and human dimensions of system performance within a real business setting.

The use of multi-dimensional evaluation frameworks for information systems is well-supported in the literature. Information systems should be evaluated across multiple quality dimensions including functionality, usability, accuracy, security, and efficiency to ensure comprehensive assessment of system suitability. Business analytics systems must be evaluated not only on technical capabilities but also on their practical impact on decision-making and operational outcomes. Effective evaluation of business analytics must encompass both system performance dimensions and human factors such as usability and trust. Business analytics maturity assessments require multi-dimensional evaluation across process and performance outcomes. Understanding business analytics impact requires evaluation that includes both technical and organizational evaluation dimensions.

In the specific context of small business POS and inventory systems, multi-criteria evaluation has been employed to determine operational readiness. A comprehensive evaluation framework assessing functionality, usability, accuracy, and efficiency has been used to determine a POS system's operational readiness, and multi-criteria evaluation including usability, functionality, and accuracy has similarly been used to assess POS system suitability for a food service environment.

The results of this evaluation determined the extent to which Operlytics successfully addressed the operational challenges of Empress Café Boracay and provided valuable insights and recommendations that could guide future improvements to the system and inform similar development efforts aimed at supporting the digital transformation of small and medium enterprises in the food service industry.

3.8 Chapter Summary

This chapter presented the methodology used in the development and evaluation of Operlytics, a web-based business analytics and monitoring system designed for Empress Café Boracay. It described the research design that guided the study, the development methodology followed in building the system, the system architecture that defined how its components interact, and the technologies selected to implement each module. It also discussed the data-gathering techniques used to identify operational requirements, the testing procedures applied to verify system behavior, and the evaluation criteria used to judge the system's overall quality and readiness for deployment.

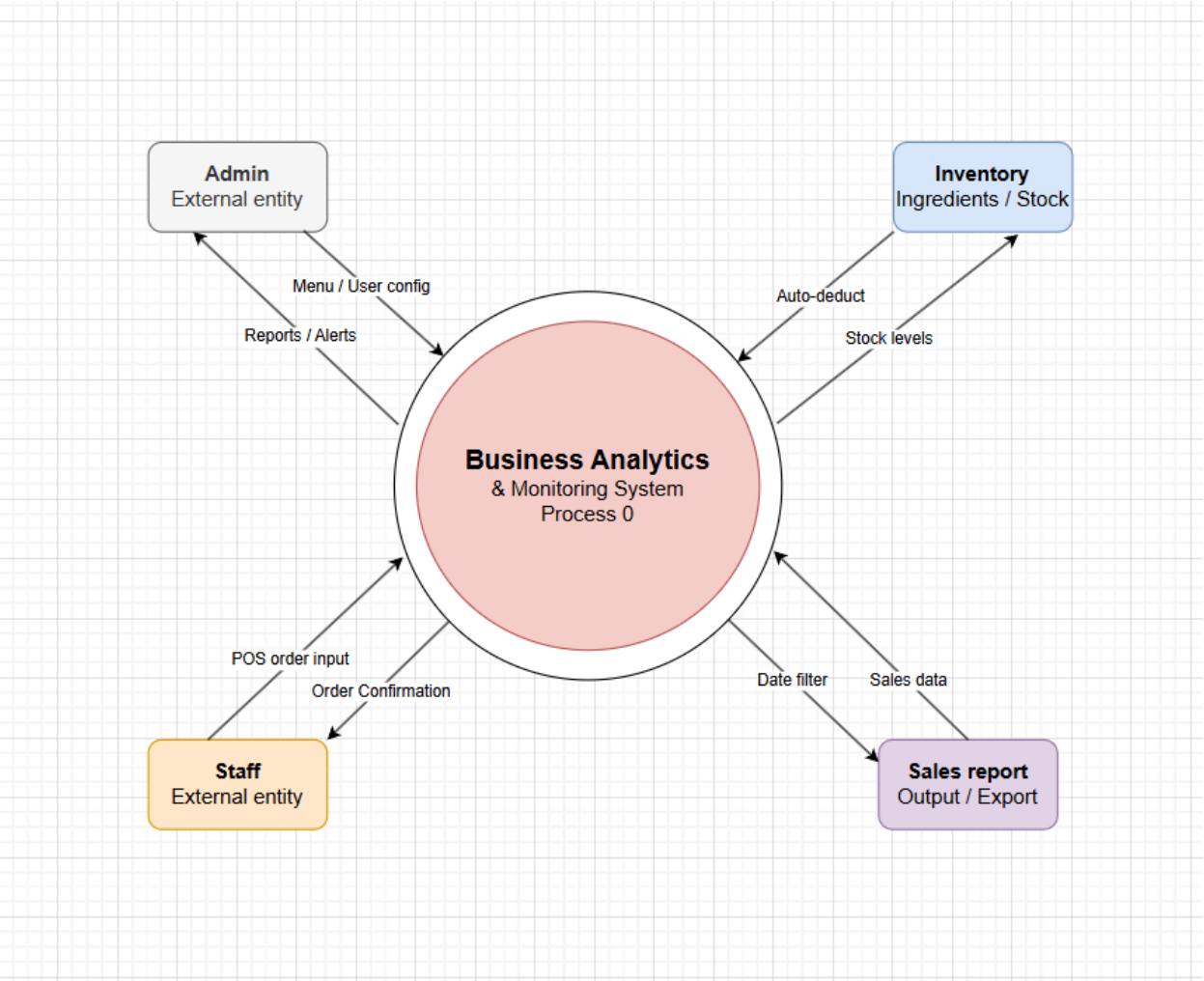
The methodology provided a structured and systematic approach for transforming the identified operational requirements into a functional system that integrates Point of Sale operations, real-time sales and inventory monitoring, automated report generation, and dashboard data visualization. By following a clearly defined sequence of phases, from requirements gathering and system design through implementation, testing, and evaluation, the study ensured that every development decision was grounded in the actual needs and workflows of the café. This structure also kept the process transparent and organized, allowing each stage to be examined, replicated, and improved.

Taken together, the methods presented in this chapter established the foundation for the results and analysis that follow. They ensured that Operlytics was developed with technical rigor and evaluated against measurable standards, and that its performance could be assessed in a real café environment rather than in isolation from the business it was built to serve.

SUPPORTING DIAGRAMS

Data Flow Diagram (DFD – Level 0)

This Level 0 Diagram illustrated the Business Analytics & Monitoring System (Process 0), serving as the high-level architectural blueprint for a comprehensive restaurant or retail management platform. The diagram captured the essential interactions between the central system and four critical external entities: Admin, Staff, Inventory, and Sales Reports. The system functioned as an integrated operational hub that automated order processing, maintained real-time stock visibility, and delivered actionable business intelligence for strategic oversight.



Central Process — Business Analytics & Monitoring System (Process 0)

The Business Analytics & Monitoring System stood as the nucleus of the entire operational ecosystem, serving as the singular convergence point through which all data flowed, was interpreted, and was redistributed across the organization. It functioned as the intelligent orchestrator that received, processed, transformed, and distributed data across all connected components — bridging the gap between raw transactional input and meaningful operational output. This central process embodied the "brain" of the organization, not merely as a passive data repository, but as an active decision-support engine that continuously coordinated the movement of information in real time. It executed three primary functions that collectively sustained the integrity, efficiency, and visibility of the entire business operation:

Order Processing & Validation — Captured incoming POS transactions the moment they were initiated, validated order details against predefined business rules, calculated subtotals, taxes, and final totals, and subsequently triggered a cascade of downstream workflows. This function ensured that no transaction entered the system in an incomplete or erroneous state, enforcing data quality at the earliest possible point in the operational pipeline. By acting as the first line of validation, it protected the integrity of all subsequent processes that depended on accurate order data.

Inventory Management & Auto-Deduction — Upon order confirmation, the system automatically adjusted stock levels in real time, deducting the appropriate quantities of raw ingredients and finished goods as dictated by each transaction. This automated deduction mechanism eliminated the delays and inaccuracies associated with manual inventory reconciliation, effectively preventing overselling, reducing waste, and maintaining precise ingredient counts at all times. The system also monitored inventory thresholds continuously, generating alerts when stock levels approached critical minimums so that restocking decisions could be made proactively rather than reactively.

Analytics Generation & Reporting — Aggregated transactional data from across all operational touchpoints, applied configurable business logic, and produced structured, filterable reports tailored to the informational needs of management. Rather than presenting raw data in isolation, this function contextualized figures within broader operational trends, enabling decision-makers to identify patterns, evaluate performance, and respond to emerging issues with confidence. Reports could be filtered by date range, product category, or transaction type, providing granular visibility without sacrificing the clarity of the overall picture.

Beyond these three core functions, the system maintained continuous bidirectional communication with every external entity within the ecosystem — from point-of-sale terminals and kitchen workflows to supplier records and managerial dashboards. This bidirectional architecture ensured that data was not only consumed by the central process but also actively returned to the entities that generated it in a refined and actionable form. The result was a state of persistent data consistency, operational transparency, and closed-loop feedback that allowed the business to function as a coherent, self-correcting system rather than a collection of isolated processes.

Admin — External Entity

The Admin represented the managerial and supervisory layer of the organization — typically owners, managers, or system administrators who required elevated privileges and comprehensive visibility into all aspects of business operations. Unlike frontline personnel, the Admin operated at a strategic level, shaping the rules and parameters under which the entire system functioned. This external entity maintained the highest tier of system access, bearing responsibility for both initial configuration and ongoing governance of the operational environment.

Outgoing Data Flow: Menu / User Configuration →

The Admin initiated system setup and sustained ongoing maintenance by transmitting configuration parameters that defined the operational boundaries of the entire platform. This included menu item definitions encompassing names, descriptions, prices, and category assignments, as well as user account creation, role assignments, and granular permission

levels that determined what each staff member could access or modify within the system. Beyond personnel configuration, the Admin also established tax settings, discount rules, promotional parameters, and broader operational constants that governed how transactions were calculated and recorded. Configuration data effectively established the foundational rules under which all other processes operated — functioning as the constitution of the system, to which every subsequent action and transaction was held accountable. Any change to these parameters propagated immediately throughout the system, underscoring the critical nature of administrative authority and the importance of controlled, auditable configuration management.

Incoming Data Flow: Reports / Alerts ←

In return, the system proactively pushed performance dashboards, exception notifications, and critical alerts back to the Admin, ensuring that those in positions of authority maintained continuous situational awareness without needing to manually query the system. Reports delivered to the Admin could encompass a wide range of business intelligence outputs, including daily revenue summaries, top-selling item rankings, staff performance metrics, shift-level transaction volumes, inventory shortage warnings, and anomaly detection flags that surfaced irregular patterns warranting investigation. Alerts were designed to enable rapid, informed response to operational issues as they emerged — whether triggered by low-stock thresholds approaching critical minimums, unusual transaction patterns suggesting potential discrepancies, or system-level errors requiring administrative intervention. This incoming data flow transformed the Admin from a purely configuration-driven actor into an active monitor of organizational health, closing the feedback loop between system performance and managerial oversight.

Staff — External Entity

The Staff represented the frontline operational personnel of the business — servers, cashiers, order-takers, or any customer-facing employee who interacted directly with both the clientele and the Point-of-Sale (POS) interface. Operating at the transactional layer of the organization, Staff members were the primary human initiators of revenue-generating events, translating customer demand into structured data that the system could receive, process, and act upon. While their access level was intentionally scoped to the functions necessary for order fulfillment, their role was indispensable to the continuous operation of the system — as every transaction, every inventory deduction, and every sales data point originated from their interactions with the POS terminal.

Outgoing Data Flow: POS Order Input →

Staff members entered customer orders through the POS terminal, transmitting structured order details that served as the system's primary source of transactional input. This data included item selections, quantities, special modifications,

customer preferences, table or seat numbers, and designated payment methods. Each order input functioned as a trigger event, initiating the system's full processing pipeline — encompassing order validation, pricing calculation, tax application, discount resolution, and the subsequent inventory deduction sequence. The POS interface was therefore not merely a data entry tool but the critical gateway through which customer intent was translated into operational reality. The accuracy and completeness of the data entered at this stage directly influenced the reliability of every downstream process, making the Staff's interaction with the system one of the most consequential touchpoints in the entire data flow architecture.

Incoming Data Flow: Order Confirmation ←

Upon successful processing and validation of a submitted order, the system returned an order confirmation to the Staff member, closing the transactional loop and providing assurance that the order had been properly received and queued. This confirmation contained a unique order identifier for tracking and reference purposes, an itemized breakdown of all selections and modifications, the calculated total amount inclusive of applicable taxes and discounts, an estimated preparation time based on current kitchen load, and clear status indicators reflecting the order's position in the fulfillment queue. Beyond its informational value, the confirmation served a critical operational function — assuring the Staff member that the transaction had been accurately recorded, thereby allowing them to communicate confidently with the customer, proceed with payment processing if applicable, and redirect their attention to the next service interaction without uncertainty or the need for manual verification.

Inventory — Ingredients / Stock

The Inventory entity represented the physical stock repository of the business — encompassing raw materials, perishable ingredients, packaged goods, consumable supplies, and any other tangible assets maintained on-site or at associated warehouse locations. As a passive but essential external entity, Inventory did not initiate processes independently; rather, it responded to system-driven commands and continuously reported its state back to the central process. The accuracy of inventory data was foundational to the system's ability to make reliable operational decisions, from order acceptance to procurement planning, making this entity one of the most data-intensive components in the entire ecosystem.

Incoming Data Flow: Auto-Deduct ←

The system automatically decremented inventory quantities in response to confirmed orders, applying deductions based on the bill of materials (BOM) associated with each menu item — a predefined recipe-level specification that mapped each product to its constituent ingredients and their respective quantities. For example, the confirmation of a "Cheeseburger" order would trigger the automatic deduction of one bun, one beef patty, one slice of cheese, and all associated condiments

from their respective stock records, with each quantity precisely defined by the BOM configuration established during system setup. This auto-deduction occurred in real time, ensuring that inventory records remained continuously synchronized with actual consumption patterns and eliminating the lag that traditionally existed between sales activity and stock reconciliation in manual systems. By enforcing this real-time synchronization, the system effectively prevented the acceptance of orders for items whose ingredients had fallen below the quantities required for fulfillment, protecting both operational integrity and customer experience.

Outgoing Data Flow: Stock Levels →

Conversely, the Inventory entity continuously reported its current state back to the central system, transmitting up-to-date stock quantities, expiration dates, reorder threshold statuses, and supplier reference information that collectively informed the system's availability logic and procurement recommendations. This outgoing data flow established a persistent feedback loop that enabled the system to perform real-time availability checks prior to order confirmation, ensuring that only items with sufficient ingredient coverage could be selected through the POS interface. Beyond transactional validation, stock level data was also channeled into the analytics engine, where it contributed to waste tracking analyses, inventory turnover calculations, consumption trend modeling, and procurement planning outputs. This dual utility — serving both operational control and strategic analysis — elevated the Inventory entity from a simple physical asset register into a dynamic source of business intelligence.

Sales Report — Output / Export

The Sales Report entity represented the analytical and documentation layer of the system — the structured destination for processed business intelligence that supported performance monitoring, operational evaluation, and external reporting requirements. Unlike the transactional entities that generated raw data through daily operations, the Sales Report entity existed to receive, organize, and surface that data in forms meaningful to decision-makers. It served as the primary interface through which historical performance was examined, patterns were identified, and evidence-based conclusions were drawn about the health and trajectory of the business.

Incoming Data Flow: Sales Data ←

The system compiled and delivered comprehensive sales data to the Sales Report entity on a continuous basis, encompassing transaction timestamps, item-level purchase details, revenue figures broken down by category and payment type, staff associations linking each transaction to the responsible employee, and emerging customer ordering trends derived from aggregated behavioral patterns. This data was not simply transferred in raw form; it was structured, aggregated, and formatted by the analytics engine prior to delivery, rendering it immediately suitable for analytical consumption without

requiring further manual processing. The resulting dataset supported a wide range of analytical applications — from straightforward trend identification and seasonal performance analysis to more sophisticated benchmarking against predefined targets or historical baselines. Furthermore, sales data was architected to support export functionality, enabling seamless integration with external accounting platforms, third-party business intelligence tools, or regulatory filing systems that required standardized financial records.

Outgoing Data Flow: Date Filter →

In the opposite direction, users interacting with the Sales Report entity transmitted temporal filtering parameters back to the system, specifying the precise scope of data they wished to retrieve and analyze. Date filters provided granular control over the reporting window — supporting single-day operational snapshots for immediate performance review, week-over-week comparisons for trend monitoring, monthly summaries for financial reporting, quarterly reviews for strategic evaluation, or fully customizable date ranges tailored to specific analytical questions or audit requirements. This filtering capability was central to the system's utility as a decision-support tool, as it transformed an otherwise overwhelming volume of raw transaction logs into focused, relevant, and immediately actionable insights. By placing temporal control in the hands of the user, the system ensured that the Sales Report entity could serve a diverse range of reporting needs without requiring separate data pipelines or manual extraction processes for each use case.

Key Design Characteristics

The architecture of this Level 0 Data Flow Diagram was shaped by a set of deliberate design principles that collectively governed how data moved, how entities interacted, and how the system delivered value across multiple organizational layers.

Automation — Stock deduction occurred without any form of manual intervention, removing human error from one of the most repetitive and error-prone tasks in daily food service operations. By embedding BOM-driven deduction logic directly into the order confirmation workflow, the system ensured that inventory accuracy was maintained as a byproduct of normal transaction processing rather than as a separate administrative burden requiring dedicated staff time and attention.

Closed-Loop Feedback — Every external entity within the system both contributed input and received meaningful output in return, establishing balanced, mutually reinforcing relationships between the central process and its surrounding actors. This closed-loop architecture ensured that no entity functioned in isolation — each action taken by a user or system component generated a corresponding response that informed, confirmed, or refined subsequent behavior, creating a self-sustaining cycle of data generation and utilization.

Centralized Control— Process 0 maintained sole authority over all data transformation, serving as the exclusive mediator through which raw input was converted into structured output. This centralization enforced consistency in how data was processed and interpreted across all functions, while simultaneously supporting auditability by ensuring that all transformations occurred within a single, governable process boundary rather than being distributed across disconnected components.

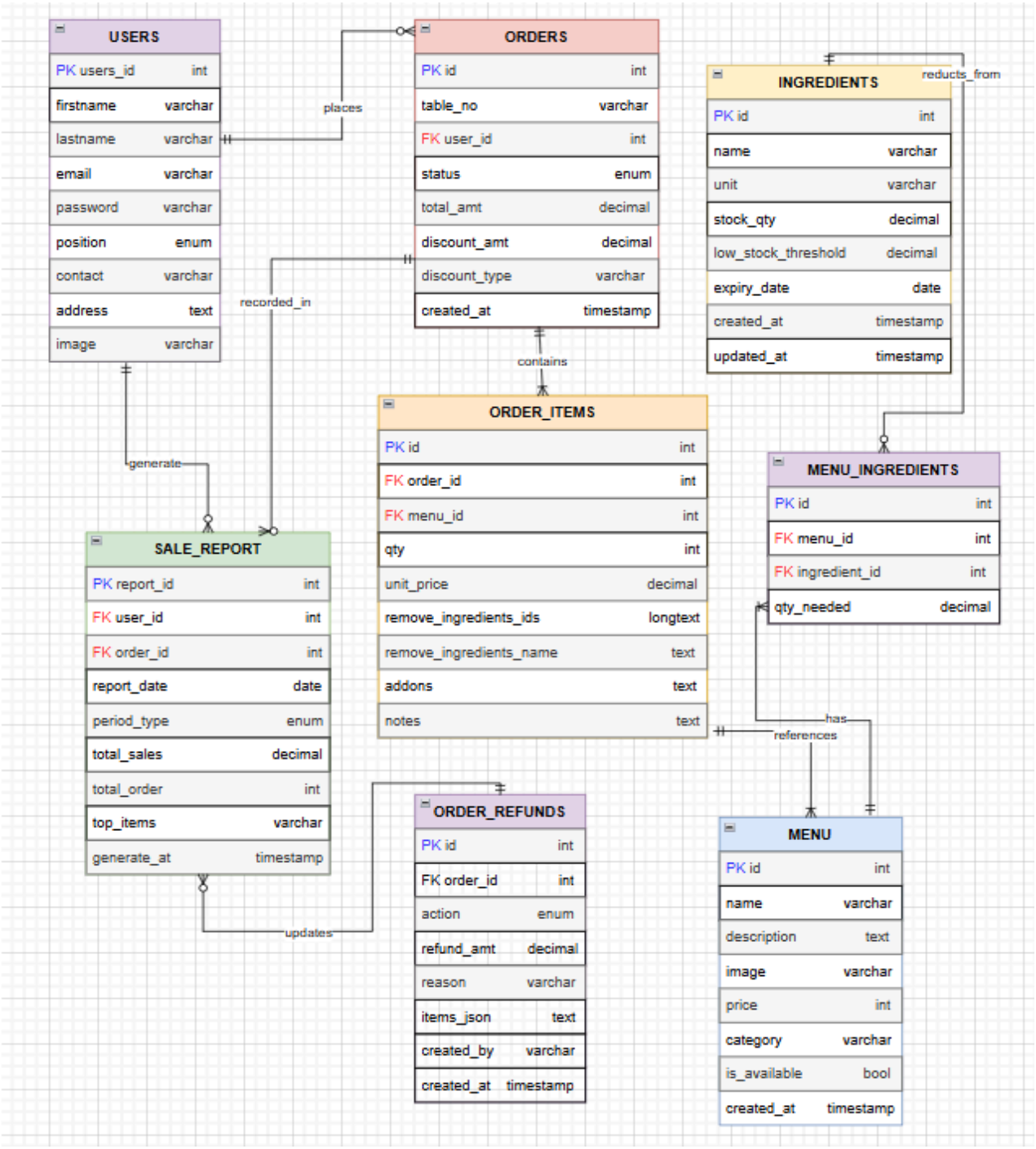
Scalability — The Level 0 abstraction was intentionally designed to remain stable as the system evolved internally. By encapsulating all processing complexity within Process 0, the architecture permitted the addition of new sub-processes, data stores, and internal workflows through Level 1 decomposition without necessitating any changes to the external interfaces that entities relied upon. This decoupling of internal complexity from external representation ensured that growth and refinement could occur without disrupting the operational continuity of connected actors.

Decision Support — The data flows directed toward the Admin and the Sales Report entity were explicitly designed to serve strategic decision-making rather than purely transactional recordkeeping. By aggregating, structuring, and contextualizing operational data before surfacing it to these entities, the system elevated its function beyond simple transaction processing into the domain of business intelligence — providing decision-makers with the timely, accurate, and filterable information necessary to manage performance, allocate resources, and respond to operational challenges with confidence.

This Level 0 DFD provided the foundational architectural context for subsequent decomposition into Level 1 processes, where each of the three primary functions — order processing, inventory management, and analytics and reporting — would be exploded into discrete sub-processes governed by their own internal data flows, transformation logic, and dedicated data stores. The clarity established at this level of abstraction was essential to ensuring that lower-level decomposition remained coherent, traceable, and aligned with the overarching design intent of the system.

Entity Relationship Diagram (ERD)

This diagram was an Entity-Relationship Diagram (ERD) for a café management or Operlytics system. It outlined how data was structured and shared between users, orders, menu items, and inventory.



Entities and Their Attributes

- **USERS** stored the account information of all system users, including administrators and staff. Key attributes included users_id, firstname, lastname, email, password, position, contact, address, and image.

- **ORDERS** recorded all customer orders placed within the café. Key attributes included id, table_no, user_id, status, total_amt, discount_amt, discount_type, and created_at.
- **ORDER_ITEMS** stored the individual items included in each order. Key attributes included id, order_id, menu_id, qty, unit_price, remove_ingredients_ids, remove_ingredients_name, addons, and notes.
- **ORDER_REFUNDS** tracked refund transactions associated with specific orders. Key attributes included id, order_id, action, refund_amt, reason, items_json, created_by, and created_at.
- **MENU** contained the list of available café products and their details. Key attributes included id, name, description, image, price, category, is_available, and created_at.
- **MENU_INGREDIENTS** served as a junction table linking menu items to their required ingredients. Key attributes included id, menu_id, ingredient_id, and qty_needed.
- **INGREDIENTS** managed the café's inventory of raw ingredients and supplies. Key attributes included id, name, unit, stock_qty, low_stock_threshold, expiry_date, created_at, and updated_at.
- **SALE_REPORT** stored generated sales reports summarizing operational performance. Key attributes included report_id, user_id, order_id, report_date, period_type, total_sales, total_order, top_items, and generate_at.

Relationships

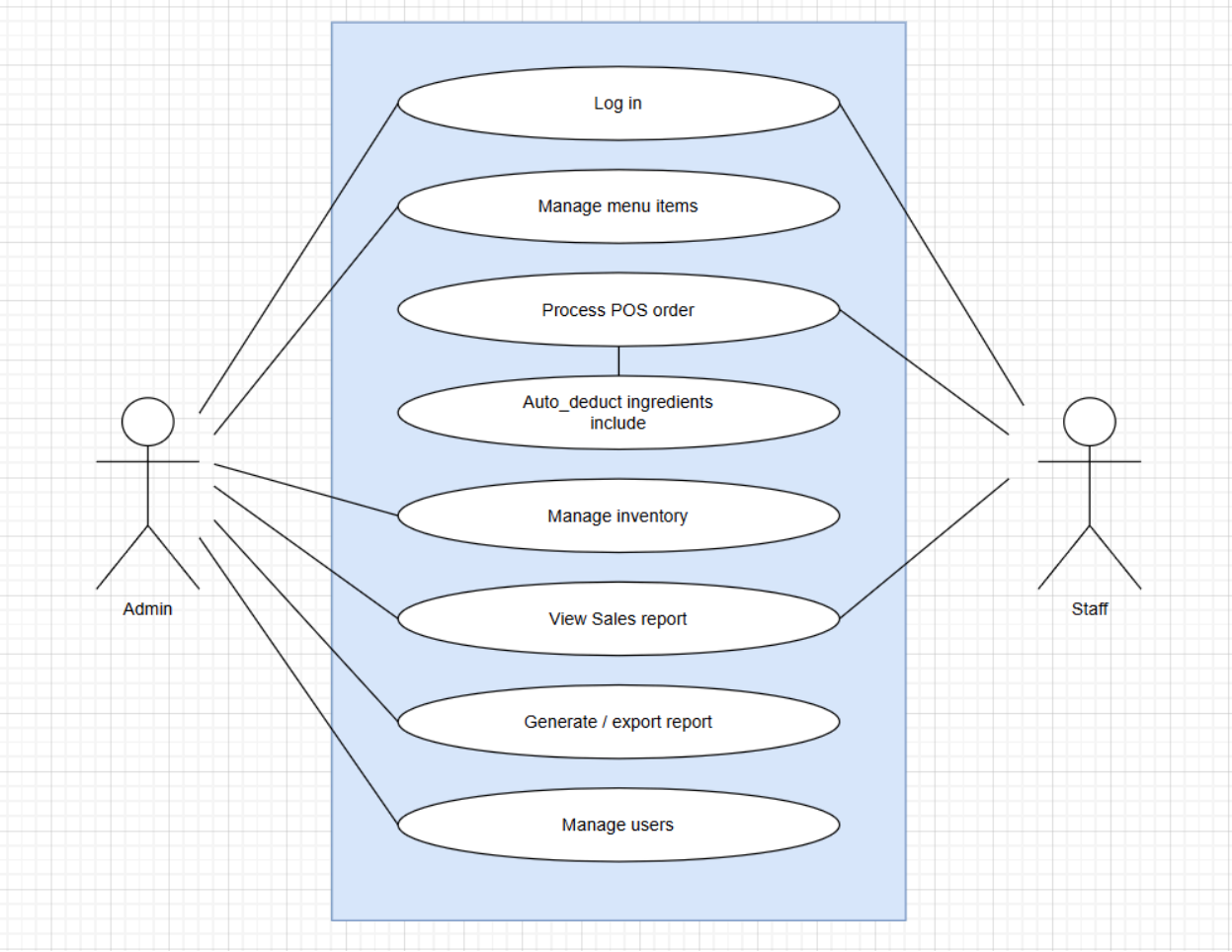
- **USERS → ORDERS**: A user placed one or many orders. This was a one-to-many relationship, where one user account could be associated with multiple order transactions.
- **USERS → SALE_REPORT**: A user generated one or many sales reports. This was a one-to-many relationship, where a single user could produce multiple generated reports over time.
- **ORDER → ORDER_ITEMS**: An order contains one or many order items. This was a one-to-many relationship, as each order transaction could include multiple individual menu items.
- **ORDERS → ORDER_REFUNDS**: An order could be associated with a refund transaction. This was a one-to-one or one-to-many relationship, where a single order could have one or more associated refund records.
- **ORDERS → SALE_REPORT**: An order was recorded in a sale report. This was a many-to-one relationship, where multiple orders could be consolidated into a single generated report.
- **SALE_REPORT → ORDERS**: A sales report updated based on order data. This reflected a many-to-one dependency where report summaries were derived from accumulated order records.
- **ORDER_ITEMS → MENU**: Each order item referenced a menu item. This was a many-to-one relationship, where multiple order items could correspond to the same menu entry.
- **MENU → MENU_INGREDIENTS**: A menu item has one or many ingredient requirements. This was a one-to-many relationship linking each menu product to the ingredients needed for its preparation.

- **MENU_INGREDIENTS → INGREDIENTS**: Each menu ingredient entry referenced a specific ingredient in the inventory. This was a many-to-one relationship, where multiple menu items could require the same ingredient.
- **INGREDIENTS → ORDER_ITEMS:** Ingredients were deducted from inventory based on order items processed. This reflected the auto-deduction relationship where completed orders triggered automatic stock reduction in the ingredients table.

The database schema of Operlytics, composed of eight entities and nine defined relationships, established a comprehensive and logically structured data management foundation that directly supported the system's core operational objectives. The careful design of entity relationships ensured referential integrity across all interconnected tables, minimizing the risk of data inconsistencies that could compromise the accuracy of sales records, inventory levels, and generated reports. By aligning the database structure with the actual operational workflows of Empress Café Boracay, the schema enabled Operlytics to process, store, and retrieve business data in a manner that was both efficient and reflective of real café operations, ultimately providing management with a reliable and well-organized data environment upon which sound and informed business decisions could be consistently made.

Use Case Diagram

This Use Case Diagram showed that the Admin had full control over all system functions, including inventory, user management, and reporting, while the Staff was restricted to logging in and processing customer orders.



Review of Related Literature. The design of role-based access control in information systems is well-supported in the academic literature. emphasize that role-based access control (RBAC) is a fundamental security mechanism in management information systems, ensuring that users can access only the functions and data appropriate to their responsibilities. explain that RBAC models assign permissions based on user roles rather than individual identities, which simplifies administration and enhances security by ensuring that sensitive functions are restricted to authorized personnel. In POS and inventory system development, implemented a similar role-based structure in their food service POS system, where administrators controlled system configuration, user management, and reporting, while staff were limited to sales processing and basic inventory queries. [1112] employed role-based access differentiation in their POS system with card reader for the HTM Department, restricting staff to transaction processing while granting administrators full system control. [13], in his assessment of centralized POS system usability, found that clear role differentiation between administrative and operational

staff significantly improved system security perceptions and reduced errors caused by unauthorized access to sensitive functions. similarly structured their web-based sales and inventory system with distinct admin and staff roles, demonstrating that this approach is standard practice for small business systems where operational staff require limited access to complete their tasks effectively.

The Operlytics system involved two primary actors — the Admin and the Staff — each assigned a defined set of system functions that corresponded to their respective roles and responsibilities within Empress Café Boracay.

Review of Related Literature. The identification of primary actors and their associated functions is a critical step in use case modeling and systems analysis. describes actor identification as essential for understanding the different ways users interact with a system and for defining the boundaries of system functionality. emphasizes that use case diagrams must clearly distinguish between different user types and their respective privileges to ensure that the system design reflects actual organizational roles. support this by explaining that understanding actor roles is fundamental to designing information systems that align with organizational structures and workflows. In the context of café and food service operations, identified administrators and cashiers as primary actors in their POS system, assigning distinct functions that matched their operational responsibilities. [12] defined admin and staff roles in their web-based inventory management system, ensuring that each actor had access only to functions relevant to their duties. [13] analyzed actor roles in his POS usability assessment, noting that clear definition of admin and staff functions was essential for both security and operational efficiency. The actor definitions in Operlytics thus align with established practices in systems analysis and design.

Actors:

Admin — The Admin represented the café manager or authorized administrator who held full access to the system's operational and managerial functions. The Admin was responsible for overseeing the overall use of the system and had the highest level of access privileges among all system users.

Review of Related Literature. The administrator role in information systems is consistently defined across the literature as the user with comprehensive system access and managerial oversight. describe system administrators as users responsible for configuring, managing, and overseeing system operations, including user account management, security settings, and system maintenance. identify the administrator as a critical actor who ensures system integrity, manages data access policies, and generates reports for strategic decision-making. In small business POS systems, defined the admin role as having full access to sales data, inventory records, user management, and report generation functions. similarly granted administrators comprehensive control over their food service POS system, including menu management, inventory oversight, and staff account administration. [2] structured their POS system so that administrators managed system configuration, user accounts,

and transaction monitoring. The Admin role in Operlytics thus reflected standard practice, ensuring that managerial oversight and system governance were centralized under authorized personnel.

Staff — The Staff represented the café employees who were granted limited but essential access to the system for the purpose of carrying out their daily operational tasks, particularly those related to sales transactions and inventory movement.

Review of Related Literature. The staff or operational user role is defined in the literature as having restricted access focused on daily transactional activities. explains that limiting operational user access to essential functions reduces the risk of accidental data corruption and simplifies the user interface for non-technical personnel. support this by noting that operational staff require access only to functions directly related to their job responsibilities, which both enhances security and improves usability. emphasize that restricting staff access to core operational tasks—such as transaction processing and basic data entry—is a standard security practice in business information systems. In POS system implementations, [13] found that limiting staff to sales processing and basic inventory functions improved system usability and reduced training requirements. restricted their café staff to POS order processing and inventory viewing, while administrators retained control over system configuration and reporting. [2] similarly limited HTM Department staff to transaction processing and card reader operations. The Staff role in Operlytics thus followed established conventions, ensuring that operational employees could perform their duties efficiently without exposure to sensitive administrative functions.

Use Cases:

Admin:

Admin can Log in

The login function provided secure authentication for the Admin, ensuring that only authorized personnel could access the system. This use case established the identity of the user and determined the appropriate access privileges based on their role.

Review of Related Literature. Secure user authentication is a foundational requirement for information systems. describe login mechanisms as the first line of defense in system security, verifying user identity before granting access to system functions. emphasize that authentication procedures must be robust yet user-friendly to balance security with usability. In web-based business systems, implemented secure login functionality for both admin and staff users, ensuring that each session was properly authenticated before granting role-appropriate access. required all users to authenticate through a login screen that validated credentials against stored records before permitting system access. [2] incorporated login authentication as a prerequisite for all system functions in their POS system. The login use case in Operlytics thus aligns with standard security practices, ensuring that system access is properly controlled from the point of entry.

Admin can Manage menu items

This use case allowed the Admin to add, edit, delete, and categorize menu items in the system. The Admin could set prices, descriptions, and ingredient associations for each menu item, ensuring that the POS system reflected current café offerings.

Review of Related Literature. Menu management is a critical administrative function in food service POS systems. included menu item management as a core admin function in their food service POS system, enabling administrators to update prices, descriptions, and availability status. explain that product or service catalog management is essential for maintaining accurate transaction records and ensuring that sales data reflects current business offerings. support this by noting that administrative control over product databases ensures data consistency and prevents unauthorized modifications that could affect pricing or inventory calculations. In café and restaurant contexts, [13] found that centralized menu management by administrators was essential for maintaining consistency across multiple service points. similarly granted administrators exclusive control over product catalog management in their retail system. The menu management use case in Operlytics thus addresses a well-established requirement for food service business systems.

Admin can Process POS order

This use case enabled the Admin to process customer orders through the Point of Sale interface, including item selection, quantity specification, payment handling, and receipt generation.

Review of Related Literature. POS order processing is the core transactional function of any retail or food service system. identified POS order processing as the central use case in their food service system, through which all sales transactions were recorded and inventory was automatically updated. describe transaction processing as a fundamental information system activity that must be accurate, efficient, and reliable to support business operations. explain that POS systems must handle item selection, pricing calculation, payment processing, and receipt generation as integrated workflow steps. [2] structured their POS system around order processing as the primary staff and admin function, ensuring that all sales were properly recorded. [13] evaluated POS order processing usability as a critical factor in system adoption, finding that streamlined transaction workflows significantly improved staff efficiency. The POS order processing use case in Operlytics thus reflects the central operational requirement identified across food service system literature.

Admin can Auto-deduct ingredients

This use case triggered automatic deduction of ingredient quantities from inventory whenever a menu item was sold through the POS system. The Admin could configure ingredient associations and deduction rules for each menu item.

Review of Related Literature. Automated ingredient deduction is a key feature that distinguishes advanced POS systems from basic transaction recorders. implemented automatic inventory deduction in their food service POS system, linking menu

items to their component ingredients so that each sale automatically reduced inventory quantities proportionally. explain that automated data updates across related records reduce manual entry errors and ensure real-time data consistency. support this by noting that automated workflow triggers—such as inventory updates following sales transactions—improve operational efficiency and data accuracy. [12] incorporated automatic stock adjustment features in their inventory management system, ensuring that inventory records remained current without manual intervention. [8] emphasizes that automated data flows are essential for real-time dashboard accuracy, as manual updates introduce delays and inconsistencies. The auto-deduct ingredients use case in Operlytics thus addresses a recognized need for automation in food service inventory management.

Admin can Manage inventory

This use case provided the Admin with comprehensive inventory management capabilities, including viewing current stock levels, adding new inventory items, updating quantities, setting reorder points, and reviewing inventory movement history.

Review of Related Literature. Inventory management is a core function of business analytics and monitoring systems. identify inventory control as a critical business process that information systems must support through real-time tracking, automated alerts, and historical reporting. explain that inventory management systems must maintain accurate records of stock levels, track item movements, and support reorder decisions to prevent stockouts and overstock situations. developed comprehensive inventory management functionality for their web-based system, enabling administrators to monitor stock levels, update quantities, and generate inventory reports. [12] centered their web-based system on inventory management, providing administrators with tools for stock monitoring, quantity updates, and movement tracking. integrated inventory management with their POS system, ensuring that sales and inventory data remained synchronized. [13] found that centralized inventory management was essential for maintaining accurate stock records in multi-point food service operations. The inventory management use case in Operlytics thus reflects established best practices for business system design.

Admin can View sales report

This use case allowed the Admin to access and review sales reports generated by the system, including daily summaries, period comparisons, and trend analyses presented through the dashboard interface.

Review of Related Literature. Sales reporting is a fundamental analytics function that supports managerial decision-making. [5], in their seminal work on business intelligence and analytics, emphasize that reporting capabilities transform transactional data into actionable insights for management. [8] explains that dashboard-based sales reports must present data at a glance, enabling managers to quickly assess performance without extensive analysis. [15] highlight that real-time sales reporting is essential for business intelligence dashboards, providing managers with current performance indicators. identify reporting as a critical output of management information systems, enabling data-driven decision-making at the operational and strategic levels. included sales reporting as a core admin function in their POS system, generating

daily and periodic summaries for managerial review. developed sales reporting functionality that enabled administrators to view transaction summaries and identify trends. The view sales report use case in Operlytics thus addresses a well-established requirement for managerial oversight in business systems.

Admin can Generate / export report

This use case enabled the Admin to generate detailed reports in various formats and export them for external use, such as sharing with accountants, investors, or regulatory authorities.

Review of Related Literature. Report generation and export functionality extends the utility of business analytics systems beyond internal use. explain that information systems must support data export in standard formats to facilitate information sharing with external stakeholders and compliance with reporting requirements. note that export capabilities enable integration with other business tools, such as accounting software and spreadsheet applications. [13] emphasizes that business analytics systems must support flexible reporting to meet diverse stakeholder information needs. [7] identify report generation as a key capability that enables organizations to compete on analytics by distributing insights across the enterprise. In small business contexts, included report export functionality in their web-based system, allowing administrators to generate and share sales and inventory reports. [12] developed export features for their inventory management system, enabling data sharing in standard formats. The generate/export report use case in Operlytics thus supports both internal analysis and external information sharing requirements.

Admin can Manage users

This use case allowed the Admin to create, modify, and deactivate user accounts, assign roles, and configure access permissions for all system users.

Review of Related Literature. User management is a critical administrative function that maintains system security and operational control. describe user account management as essential for maintaining system security, enabling administrators to control who can access the system and what functions they can perform. emphasize that user management is a core responsibility of system administrators, ensuring that access privileges align with organizational roles and security policies. notes that user management functions must be restricted to authorized administrators to prevent unauthorized privilege escalation. In POS system implementations, granted administrators exclusive control over user account creation, modification, and deactivation. [2] included user management as an admin-only function in their POS system, ensuring that staff could not create or modify accounts. [13] found that centralized user management improved system security perceptions and simplified administration in multi-staff environments. The managed users use case in Operlytics thus follows established security practices for business information systems.

Staff:

Staff can Log in

This use case provided secure authentication for Staff members, verifying their identity and granting access to their limited set of authorized functions.

Review of Related Literature. Staff login functionality, while similar in mechanism to admin login, operates within a restricted privilege framework. explain that authentication systems must verify user identity and then apply role-based access controls to determine available functions. emphasize that all users, regardless of role, must authenticate before accessing system functions to maintain security audit trails. implemented separate login credentials for staff members, ensuring that their system access was properly authenticated and logged. required all users to authenticate through a common login interface that then directed them to role-appropriate functions. [2] incorporated staff login as the entry point to their restricted POS functionality. The staff login use case in Operlytics thus maintains consistent security standards while enabling appropriate function access.

Staff can Process POS order

This use case enabled Staff members to process customer orders through the Point of Sale interface, performing the core transactional functions required for daily café operations.

Review of Related Literature. Staff access to POS order processing reflects the operational reality that frontline employees require efficient transaction tools. [13] found that restricting staff to core transactional functions improved system usability and reduced training time, as employees were not overwhelmed with unnecessary features. granted their café staff access to POS order processing while restricting administrative functions, enabling efficient service delivery while maintaining security. explain that operational staff require streamlined interfaces focused on their primary tasks to maximize efficiency and minimize errors. support this by noting that function restriction based on role improves both security and usability by presenting users only with relevant options. limited their retail staff to sales processing functions, finding that this focused access improved transaction speed and accuracy. The staff POS order processing use case in Operlytics thus aligns with established principles of role-appropriate system design.

Staff can Auto-deduct ingredients

This use case enabled the automatic deduction of ingredient quantities from inventory when Staff processed sales through the POS system, ensuring that inventory records remained current without requiring manual updates.

Review of Related Literature. Extending automated ingredient deduction to staff-level transactions ensures consistent inventory accuracy regardless of which user processes the sale. implemented automatic inventory updates triggered by all POS transactions, whether processed by admin or staff, ensuring that inventory data remained synchronized across all sales channels. explain that automated workflows should operate consistently across user roles to maintain data integrity. support

this by noting that business rules—such as inventory deduction triggers—should be enforced uniformly regardless of which user initiates the transaction. [12] designed their inventory system so that stock adjustments occurred automatically upon transaction completion, without requiring manual intervention from operational staff. [8] emphasizes that consistent automation across all users is essential for real-time dashboard accuracy. The staff auto-deduct ingredients use case in Operlytics thus ensures that inventory automation operates comprehensively across all transactional users.

System Design Rationale

The defined use cases of the Operlytics system reflected a deliberate and structured approach to role-based access control, ensuring that each actor within the system was granted only the functions necessary to fulfill their specific responsibilities. By assigning comprehensive managerial functions to the Admin and limiting Staff access to core transactional tasks, the system maintained operational security while minimizing the risk of unauthorized data modification or accidental system misuse.

Review of Related Literature. The principle of least privilege—granting users only the minimum access necessary to perform their duties—is a cornerstone of information security. describe this principle as essential for protecting sensitive data and preventing accidental or malicious system misuse. support this by explaining that role-based access control implements the principle of least privilege by restricting users to functions relevant to their organizational roles. notes that this approach simplifies user interfaces, reduces training requirements, and improves system security. In POS and inventory systems, applied the principle of least privilege by restricting staff to transactional functions while reserving administrative controls for managers. similarly limited operational staff access to essential sales and inventory functions. found that adherence to the principle of least privilege significantly improved system security perceptions and reduced operational errors in his POS usability assessment. structured their system access based on role-specific minimum requirements. The use case structure in Operlytics thus reflects established security principles and industry best practices.

The inclusion of key functionalities such as automated ingredient deduction, real-time inventory management, and report generation directly addressed the operational deficiencies previously identified at Empress Café Boracay. Together, these use cases formed the functional backbone of Operlytics, establishing a clear and practical framework through which the system supported efficient daily operations, accurate record-keeping, and informed managerial decision-making across all levels of the café's organizational structure.

Review of Related Literature. The integration of automated processes, real-time inventory tracking, and reporting capabilities addresses well-documented operational challenges in small food service businesses. identify automation of routine tasks, real-time data availability, and managerial reporting as essential capabilities for management information systems in small businesses. explain that these features reduce manual workload, minimize errors, and provide managers with timely information for decision-making. found that automated ingredient deduction and real-time inventory tracking eliminated manual reconciliation tasks and reduced stock discrepancies in their café implementation. demonstrated that web-based inventory management with automated updates significantly improved stock control accuracy. emphasizes that dashboard-based reporting transforms raw data into actionable insights that support at-a-glance decision-making. highlight that real-time dashboard functionality is essential for effective business intelligence in operational environments. showed

that integrated sales and inventory systems with reporting capabilities improved managerial oversight in small retail businesses. The combined functionality of Operlytics thus addresses a comprehensive set of operational requirements established in the literature for small business information systems.

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CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

Operlytics is a comprehensive web-based business analytics and monitoring system meticulously developed to address the multifaceted operational challenges confronting Empress Café Boracay. Situated within the highly competitive and tourism-driven market of Boracay Island, the café historically relied upon fragmented, manual, and paper-based methods for recording sales transactions, monitoring inventory levels, and compiling operational reports. These antiquated practices resulted in persistent data inconsistencies, delayed managerial oversight, frequent stock discrepancies, and an overall inability to leverage operational data for strategic decision-making. Operlytics was conceived as a targeted, integrated digital solution to systematically eliminate these deficiencies by providing a centralized platform that unifies Point-of-Sale (POS) transaction processing, real-time inventory tracking, automated reporting, and interactive data visualization within a single, cohesive environment.

The system architecture of Operlytics is deliberately designed to accommodate the unique operational rhythms of a small food service establishment serving a high-volume, seasonal tourist clientele. Unlike generic POS systems that merely record transactions without regard for downstream operational consequences, Operlytics embeds intelligent business logic that automatically correlates sales activities with inventory consumption, financial aggregation, and analytical reporting. When a staff member processes a customer order through the POS interface, the system does not simply store a transactional record; it simultaneously deducts ingredient quantities from inventory based on predefined recipe mappings, updates real-time dashboard metrics, recalculates sales summaries, and evaluates whether any stock items have fallen below critical reorder thresholds. This deep integration ensures that every transactional event propagates accurately and instantaneously across all functional domains, eliminating the information silos and reconciliation delays that characterize manual operational management.

Operlytics comprises five interconnected core modules, each engineered to address specific operational requirements identified during the needs analysis phase:

Point of Sale (POS) Module. This module serves as the primary operational interface for frontline staff, designed with an emphasis on speed, accuracy, and intuitive interaction during high-pressure service periods. The POS interface presents menu items through categorized grids with visual product identifiers, enabling rapid item selection via mouse click or touch interaction. Staff members can modify orders by adjusting quantities, applying percentage-based or fixed-amount discounts, accommodating special instructions or ingredient removals, and processing multiple payment methods including cash and digital wallets prevalent among Boracay tourists. Upon transaction confirmation, the system generates a digital receipt, logs the complete transaction audit trail including timestamp, attending staff member, table number, and payment type and triggers the automatic inventory deduction workflow. The POS module is optimized for responsiveness, ensuring that even during peak breakfast and dinner rushes, transaction processing does not introduce perceptible delays into customer service workflows.

Inventory Management Module. This module constitutes the operational backbone of the system's stock control capabilities. It maintains a comprehensive digital ledger of all ingredients and supplies utilized in café operations, ranging from perishable items such as fresh produce, dairy, and meats to non-perishable supplies including packaging materials, cleaning agents, and disposable utensils. Each inventory item is characterized by attributes including current stock quantity, unit of measurement, low-stock alert threshold, expiration date, supplier information, and last-updated timestamp. The module supports three categories of stock movements: automatic deductions triggered by confirmed POS transactions according to recipe specifications; manual adjustments for supplier deliveries, waste recording, spoilage documentation, and physical count reconciliations; and system-generated alerts when quantities breach defined minimum thresholds. Every adjustment is immutably logged with user identification, timestamp, and reason codes, creating a complete audit trail that enables management to trace inventory discrepancies to their source.

Dashboard and Real-Time Analytics Module. This module transforms raw operational data into immediately comprehensible visual intelligence for café administrators. The administrative dashboard serves as a centralized command center presenting key performance indicators through color-coded visual indicators, animated counters, trend charts, and exception alerts. Primary dashboard components include a real-time revenue monitor comparing current-day sales against historical baselines; an inventory status overview highlighting items below reorder thresholds in red, items approaching thresholds in yellow, and adequately stocked items in green; a top-performing products display ranking menu items by sales volume and revenue contribution; a staff activity summary showing transaction counts and average order values by employee; and an alerts panel surfacing operational exceptions such as unusual void patterns, low-stock conditions, or pending refund requests. The dashboard refreshes dynamically through asynchronous server communication, ensuring that management views reflect current operational reality without requiring manual page reloads.

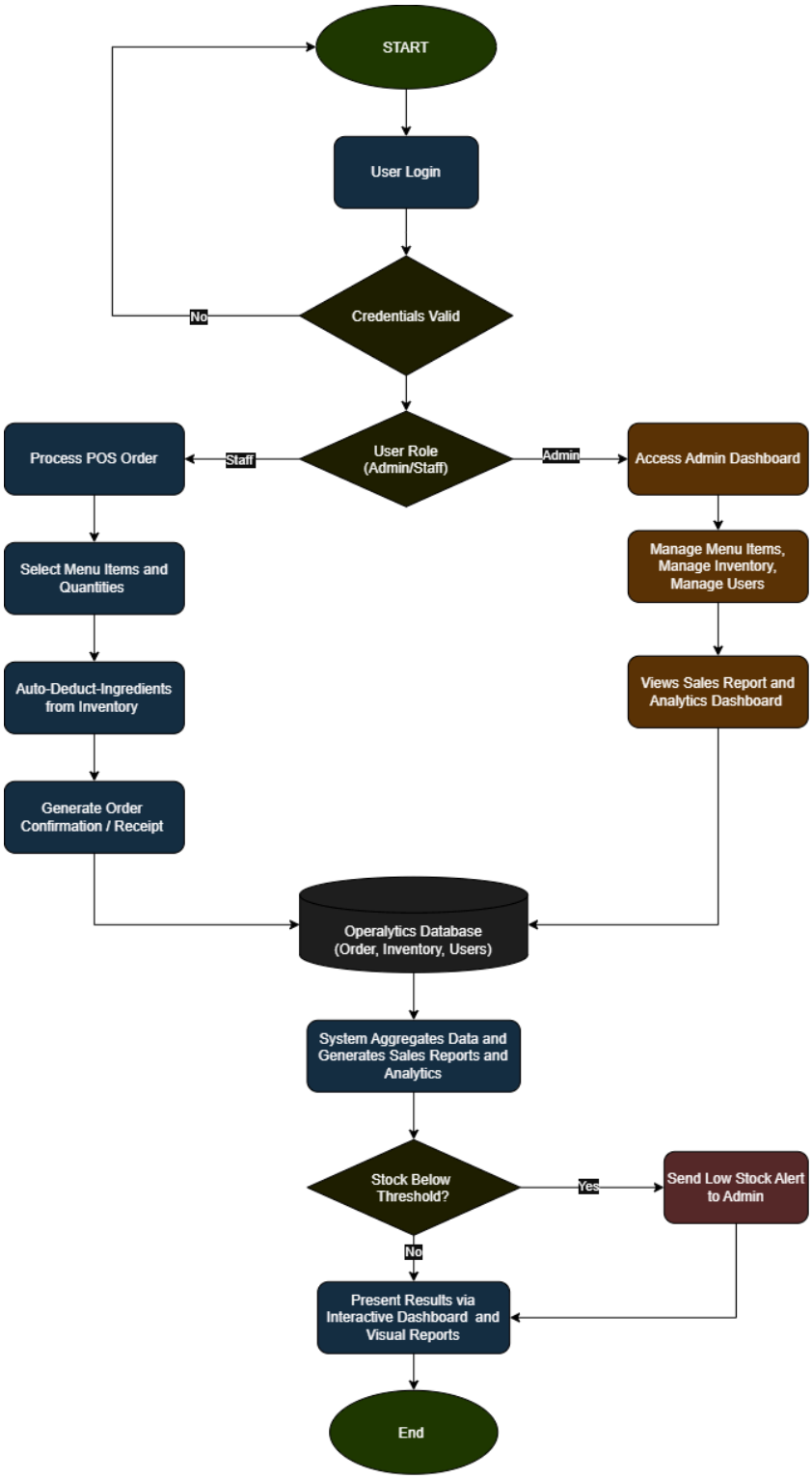
Automated Reporting Module. This module eliminates the laborious manual compilation processes previously required to produce operational summaries. It generates standardized reports across multiple temporal granularities, daily shift

summaries, weekly performance reviews, monthly financial overviews, and custom date-range analyses. Report categories include sales reports itemizing revenue by product, category, payment method, and time period; inventory reports detailing current stock levels, consumption rates, waste percentages, and reorder recommendations; financial reports calculating gross revenue, cost of goods sold, profit margins, and discount impacts; and operational reports analyzing transaction volumes, peak hour patterns, average order values, and customer throughput metrics. All reports are computed directly from live database records through optimized aggregation queries, ensuring mathematical accuracy and temporal relevance. Reports may be viewed within the browser interface, printed for physical distribution, or exported in standard formats for external sharing with accountants, suppliers, or regulatory authorities.

User Management and Security Module. This module implements a robust role-based access control framework distinguishing between Administrator and Staff user roles. Administrators typically the café owner or senior manager possess unrestricted system access including menu configuration, inventory oversight, comprehensive report generation, user account creation and modification, system parameter adjustment, and audit trail review. Staff members are granted restricted access limited to POS transaction processing, basic inventory quantity viewing, and personal profile management. The module enforces password-based authentication with encrypted credential storage, server-side session management with configurable timeout controls, and granular permission verification at every functional endpoint. All user activities including logins, transactions, inventory adjustments, and report accesses are logged to an immutable audit trail that supports security monitoring and operational accountability.

The development of Operlytics adhered to modern software engineering principles emphasizing modularity, scalability, and maintainability. The system is constructed upon a LAMP-stack foundation (Apache, MySQL, PHP) augmented with contemporary frontend technologies, ensuring broad compatibility with affordable hosting infrastructure while providing the performance characteristics necessary for real-time operational environments. The user interface employs responsive design methodologies that adapt seamlessly across desktop monitors, laptops, and tablet devices, acknowledging the diverse hardware contexts in which café staff and management interact with the system. Security considerations are embedded throughout the architecture, from input validation and SQL injection prevention at the application layer to password hashing and secure session management at the authentication layer.

Figure 4.1 illustrates the overall operational workflow of the developed system.



The workflow commences with user authentication, where the system verifies credentials and establishes a role-appropriate session. Upon successful login, administrators are directed to the main dashboard presenting high-level operational metrics, while staff members are presented with the streamlined POS interface. When staff process customer

orders, transactional data flows through the application server to the database, triggering cascading updates across inventory records, sales aggregations, and dashboard displays. Administrators may concurrently access inventory management functions to review stock levels, adjust quantities, or configure alert thresholds. At periodic intervals or on demand, the reporting engine aggregates accumulated transactional data into structured summaries accessible through the reports interface. Throughout all operations, the user management module enforces access controls and logs activities to maintain system security and operational transparency.

4.3 System Development Methodology

The development of Operlytics was structured around the Agile Software Development Methodology, an iterative and incremental approach that emphasizes adaptive planning, evolutionary development, early delivery, and continuous improvement through rapid and flexible response to change. Agile was deliberately selected over traditional linear models such as Waterfall because the operational environment of Empress Café Boracay presented requirements that were initially fluid and subject to refinement as stakeholders gained greater familiarity with digital operational possibilities. The methodology facilitated close collaboration between the development team and café personnel, ensuring that emergent operational insights such as the need for split-payment handling during peak tourist seasons or the importance of offline resilience during intermittent connectivity could be incorporated into the system without disrupting the overall development trajectory.

The Agile framework was operationalized through a series of time-boxed development sprints, each spanning two weeks, during which specific functional increments were designed, coded, tested, and demonstrated. This sprint-based structure provided regular milestones for stakeholder feedback while maintaining momentum toward the comprehensive system vision. The development process was organized into seven distinct phases, each contributing essential activities to the maturation of the Operlytics platform.

Phase 1: Requirements Gathering and Analysis.

This foundational phase involved intensive engagement with Empress Café Boracay stakeholders to establish a thorough empirical understanding of current operational practices, pain points, and functional aspirations. The research team conducted semi-structured interviews with the café owner and floor staff, performed direct observational sessions across multiple operational periods including weekday mornings, lunch rushes, and weekend peak hours, and analyzed existing operational documents including handwritten sales logbooks, paper inventory count sheets, supplier delivery receipts, and informal financial notes.

These multi-source data gathering techniques revealed critical insights: sales recording was decentralized across multiple staff-maintained notebooks with inconsistent formatting; inventory counts occurred sporadically with no systematic linkage to sales activity; report generation required several hours of manual arithmetic and transcription; and management lacked visibility into real-time stock conditions. These findings were synthesized into a prioritized product backlog

comprising user stories, functional requirements, and non-functional constraints that guided all subsequent development activities.

Phase 2: System Planning and Feasibility Analysis.

Following requirements consolidation, the development team established the project scope, defined measurable objectives, specified technical architectures, and constructed a detailed development schedule with resource allocation. Feasibility analysis encompassed technical feasibility confirming that the selected PHP-MySQL technology stack could support real-time operational demands within affordable hosting parameters; operational feasibility assessing whether café staff could realistically transition from paper-based to digital workflows given their current technical exposure; and economic feasibility demonstrating that development and deployment costs remained within constraints appropriate for a small café enterprise. The planning phase also identified risk factors including staff resistance to technological change, intermittent internet connectivity on Boracay Island, and hardware limitations, for which mitigation strategies were developed including comprehensive user training, offline contingency protocols, and responsive design accommodating older devices.

Phase 3: System Design and Modeling.

During this phase, the abstract requirements were translated into concrete technical specifications and architectural blueprints. Database design proceeded through normalization to Third Normal Form (3NF), resulting in the entity-relationship structure comprising eight core tables with defined primary keys, foreign key constraints, and indexing strategies. Process modeling produced Data Flow Diagrams (DFDs) at Level 0 and Level 1, illustrating the movement of transactional data from user inputs through processing engines to database storage and report generation. Unified Modeling Language (UML) diagrams including use case diagrams, activity diagrams, and sequence diagrams formalized user interactions and system behaviors. Wireframes and mockups for all major interface screens were developed and reviewed with café administrators to validate layout intuitiveness before coding commenced. The design phase explicitly addressed usability considerations for non-technical users, specifying large touch-targets for POS buttons, high-contrast color schemes for visibility in brightly lit café environments, and simplified navigation minimizing the number of clicks required for common tasks.

Phase 4: Iterative Software Development.

The coding phase was executed across five development sprints, each producing a demonstrable functional increment:

Sprint 1: Core Infrastructure and Authentication. This sprint established the foundational system architecture including database schema implementation, user account creation and login functionality, password encryption, session management, and role-based access control differentiation between Administrator and Staff interfaces.

Sprint 2: Menu Management and POS Foundation. This sprint developed the menu configuration interface enabling administrators to define menu items, categories, prices, descriptions, and ingredient associations. The basic POS interface was constructed allowing item selection, quantity adjustment, and order total calculation.

Sprint 3: Transaction Processing and Inventory Integration. This sprint implemented complete POS transaction workflows including payment handling, receipt generation, and the critical auto-deduct logic that reduces ingredient quantities upon transaction confirmation. The inventory management interface was developed supporting stock viewing, manual adjustments, and threshold configuration.

Sprint 4: Dashboard Visualization and Alerting. This sprint created the administrative dashboard with real-time metric displays, color-coded status indicators, and automated alert generation for low-stock conditions. Charting libraries were integrated to render sales trends and product performance visualizations.

Sprint 5: Reporting Engine and System Refinement. This sprint implemented the automated reporting module with multiple report types, date-range filtering, and export functionality. Interface refinements based on preliminary user feedback were incorporated, and comprehensive input validation, error handling, and security hardening were applied across all modules.

Each sprint followed a micro-cycle of task assignment, daily progress coordination, component development, unit testing, and sprint review, ensuring that defects were identified and resolved within the iteration in which they were introduced.

Phase 5: Comprehensive Testing.

Quality assurance was embedded throughout the development lifecycle rather than deferred to a terminal testing stage. Three progressive testing layers were executed:

Unit Testing isolated individual software components including database query functions, calculation algorithms, authentication routines, and inventory deduction logic to verify correct behavior under expected inputs and edge cases such as zero-quantity orders, expired sessions, or simultaneous database access requests.

Integration Testing evaluated the interactions between modules, confirming that POS transactions correctly triggered inventory updates, that dashboard displays accurately reflected database states, and that reporting engines properly aggregated transactional records without omission or duplication.

System Testing assessed end-to-end workflows under simulated operational conditions, including concurrent multi-user access emulating peak service periods, sustained operation over extended durations to detect memory leaks or performance degradation, and verification of responsive behavior across diverse device form factors and browser types.

User Acceptance Testing (UAT) constituted the final validation layer, wherein café administrators and staff performed realistic operational tasks including processing mock customer orders, checking inventory levels, generating daily reports, and interpreting dashboard metrics—under researcher observation. Structured feedback instruments captured quantitative usability ratings and qualitative experiential accounts.

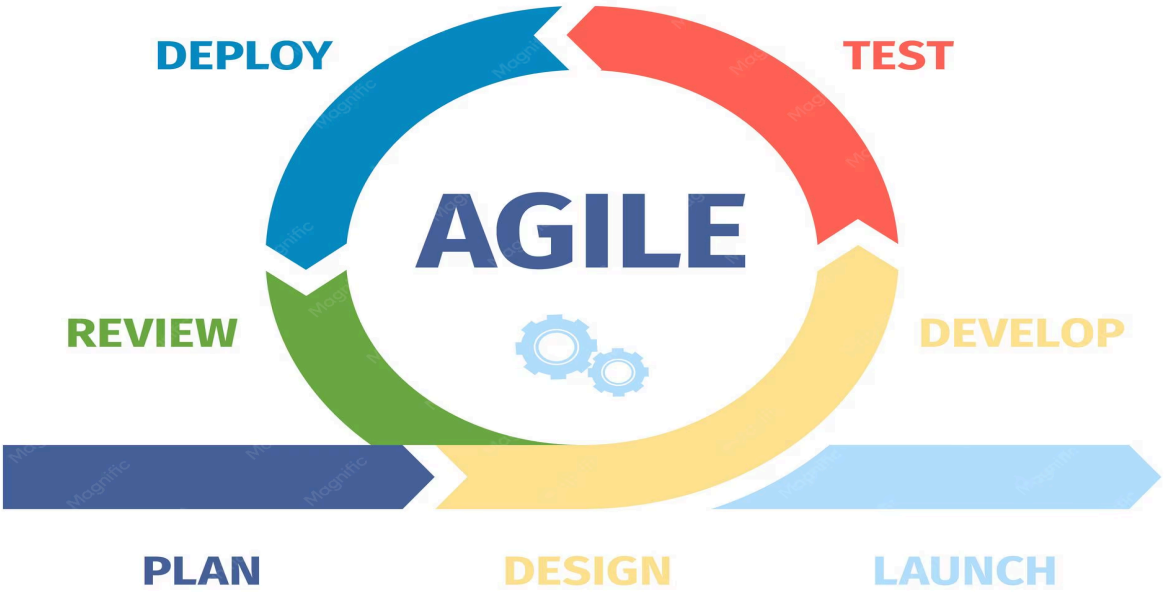
Phase 6: Deployment and Configuration.

Following successful testing, Operlytics was deployed to a production environment accessible to Empress Café Boracay. Deployment activities included web server configuration, database initialization with operational seed data (menu items, ingredient catalog, user accounts), SSL certificate installation for encrypted communication, and DNS configuration for domain accessibility. A parallel deployment strategy was initially employed, wherein Operlytics operated alongside existing manual methods during a brief transition period, allowing staff to verify system outputs against familiar paper records before committing fully to digital workflows. Data migration procedures transferred existing menu and inventory information into the new database schema, ensuring continuity of operational reference data.

Phase 7: Training, Handover, and Maintenance Planning.

The final phase encompassed comprehensive user training sessions conducted at the café premises. Administrators received detailed instruction on dashboard interpretation, report generation, inventory configuration, and user account management. Staff members participated in hands-on POS training emphasizing rapid transaction processing, discount application, and payment handling. Printed quick-reference guides were provided for common tasks. A maintenance plan was established specifying regular database backup schedules, periodic security updates, and a feedback channel for reporting operational issues or enhancement requests. The development team committed to a post-deployment support period to address emergent concerns and ensure smooth operational transition.

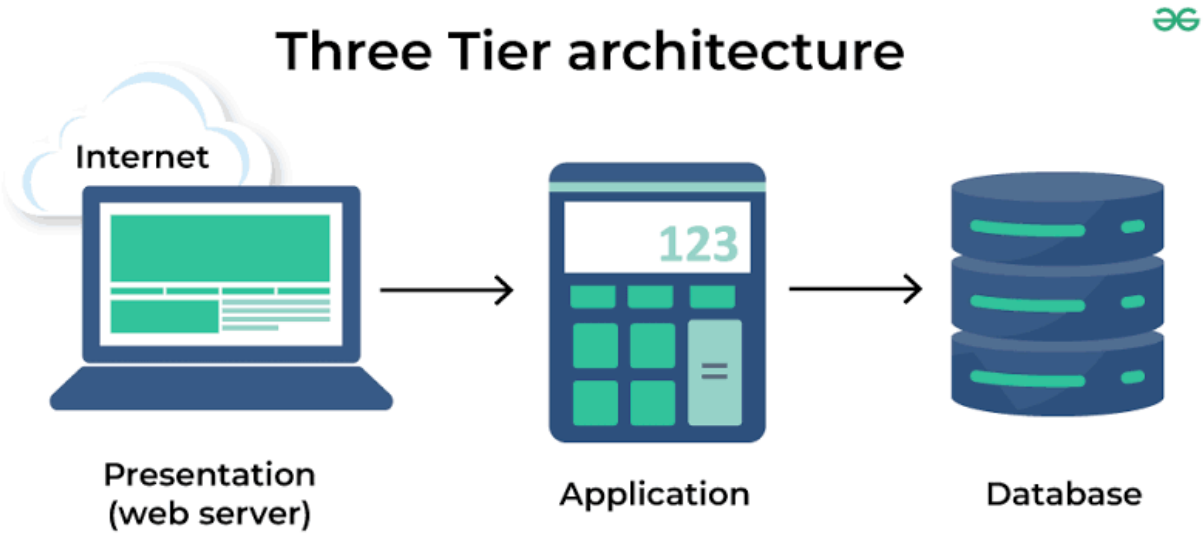
Figure 4.2 presents the Agile Software Development Life Cycle adopted in the study.



4.4 System Architecture

Operlytics is architected upon a Three-Tier Client-Server model that rigorously separates presentation concerns, business logic, and data persistence. This architectural pattern was selected for its proven reliability in web application contexts, its facilitation of independent component maintenance and scaling, and its alignment with the resource constraints of small business deployments. The tiered separation ensures that modifications to the user interface do not necessitate changes to database structures, and that business rule enhancements can be deployed without disrupting client-side operations.

Figure 4.3 illustrates the architecture of the developed system.



Presentation Tier (Client Layer).

The presentation tier constitutes the user-facing component of the system, responsible for rendering visual interfaces, capturing user inputs, and presenting analytical outputs. It is implemented entirely through standard, open web technologies requiring no proprietary plugins or local software installations beyond a modern web browser. HyperText Markup Language (HTML5) provides semantic structural markup for all interface screens, ensuring accessibility compliance and search engine interpretability. Cascading Style Sheets (CSS) govern visual presentation, implementing responsive grid layouts that adapt fluidly across viewport dimensions from large desktop monitors to compact tablet displays. CSS custom properties maintain consistent color schemes, typography, and spacing aligned with the café's branding while ensuring high legibility under varying ambient lighting conditions.

JavaScript enhances client-side interactivity through dynamic form validation, asynchronous data fetching, and real-time interface updates. The POS interface utilizes JavaScript event listeners to enable rapid item selection, automatic total recalculation upon quantity or discount changes, and immediate visual feedback confirming user actions. The

administrative dashboard employs asynchronous JavaScript techniques to poll the application server for updated metrics at regular intervals, enabling dynamic refresh of sales counters, inventory gauges, and alert banners without requiring disruptive full-page reloads. This asynchronous communication model is essential for maintaining the real-time monitoring capabilities central to the system's value proposition.

Application Tier (Business Logic Layer).

The application tier hosts the core processing engine of Operlytics, implemented in PHP (Hypertext Preprocessor), a mature server-side scripting language selected for its extensive web hosting support, robust MySQL integration, and broad developer community. This tier resides on the web server and serves as the exclusive intermediary between client requests and database operations, enforcing business rules, performing calculations, and coordinating data flows.

The application tier is organized into modular controllers corresponding to operational domains:

Authentication Controller manages user login verification through bcrypt password hashing comparison, session initialization with configurable timeout parameters, role identification, and secure logout procedures. It enforces access control gates that prevent unauthorized requests to administrative endpoints.

POS Controller processes order submissions from the client interface, validating item selections against current menu availability, calculating line-item subtotals and order totals, applying configured tax rates and discount rules, recording complete transaction records across the Orders and Order_Items tables, and invoking the inventory deduction service.

Inventory Controller handles stock quantity queries, executes automatic deduction procedures triggered by POS confirmations, processes manual adjustment requests with validation and logging, evaluates stock levels against configured thresholds, and generates alert notifications when replenishment is required.

Reporting Controller aggregates transactional data across specified date ranges, computes summary statistics including total revenue, average order value, product-level sales volumes, and inventory turnover metrics, and formats results for display within report templates or export to external file formats.

Dashboard Controller compiles real-time operational snapshots by querying current-day sales totals, active inventory statuses, recent transaction logs, and exception conditions, then structures this data into JSON responses consumed by the client-side visualization components.

The application tier implements comprehensive server-side validation for all incoming data, rejecting malformed requests before database interaction occurs. It also enforces the business logic governing recipe-based ingredient deduction, ensuring that each menu item sale triggers precise, proportionate stock reductions across all constituent ingredients.

Data Tier (Database Layer).

The data tier provides persistent storage and retrieval services through MySQL, an open-source relational database management system selected for its performance, reliability, and seamless PHP integration. The database schema implements the entity-relationship model described in Section 4.6, with tables structured according to normalization principles that minimize data redundancy while preserving referential integrity.

The data tier is responsible for maintaining ACID properties (Atomicity, Consistency, Isolation, Durability) across all transactions. When a POS order is processed, the database executes the sales recording and inventory deduction operations within a single transaction boundary, guaranteeing that either both operations complete successfully or both are rolled back, thereby preventing the data inconsistencies that would arise from partial updates. Foreign key constraints enforce relational integrity for instance, preventing the deletion of menu items referenced by existing order records, or ensuring that every order item corresponds to a valid menu entry.

Database indexes are strategically defined on frequently queried columns including order timestamps, user identifiers, inventory item names, and report date ranges to optimize query execution times and support the sub-second response requirements of real-time dashboard updates. Automated backup procedures protect operational data against loss, with scheduled exports stored securely to facilitate disaster recovery.

Communication and Security Protocols.

Communication between the presentation and application tiers occurs exclusively over HTTP/HTTPS protocols. For production deployment, Transport Layer Security (TLS) encryption is enforced to protect data in transit between client browsers and the server, preventing interception of sensitive operational or authentication information. The application tier never exposes database credentials or direct connection strings to the client; all database interactions are proxied through PHP controllers, ensuring that the data tier remains insulated from direct external access. Session identifiers are transmitted via secure, http-only cookies to mitigate cross-site scripting risks, and all user inputs undergo sanitization and parameterized query binding to prevent SQL injection attacks.

4.5 Hardware and Software Requirements

Operlytics is designed to run on modest hardware consistent with the resource constraints of a small café establishment, discussed in Chapter 1 as a defining characteristic of Empress Café Boracay's operating environment. The hardware requirements are organized into two categories corresponding to the client-server architecture described in Section 4.4: the server-side equipment that hosts the PHP application and the MySQL database, and the client-side devices used by Admin and Staff to access the system through a web browser.

On the server side, the system requires a computer or dedicated server capable of continuously running the PHP application server and the MySQL database service during the café's hours of operation. Given the transaction volume of a single café establishment rather than a multi-branch enterprise, this requirement can be met by a standard desktop-class machine with a multi-core processor, a minimum of 4 gigabytes of random-access memory, and at least 20 gigabytes of available storage to accommodate the operational database as it grows over time, together with a stable internet or local network connection through which client devices communicate with the server.

On the client side, Admin and Staff require only a device capable of running a modern web browser and maintaining a connection to the server, a requirement satisfied by the desktop computers, laptops, or tablets already available in most small café environments, without any need for specialized point-of-sale terminals or proprietary hardware. This design choice directly supports the accessibility goals discussed in Chapter 1, allowing Empress Café Boracay to adopt Operlytics without incurring the capital expense of dedicated hardware beyond what is already typically present in a small business setting.

These requirements are deliberately kept modest because Operlytics is intended to support a single café location with a level of transaction volume consistent with Empress Café Boracay's current scale of operations, rather than a multi-branch or enterprise-level deployment. The processing and memory requirements specified below are sufficient to run the Apache web server, the PHP interpreter, and the MySQL database service simultaneously while responding to requests from several client devices at once during peak service hours, such as the lunch and dinner periods when tourist traffic on Boracay Island is at its highest. Storage requirements are similarly modest at the outset, since transactional records, while accumulating continuously, consist primarily of structured text and numeric data rather than large binary files; the café's management can expect the storage footprint of the operational database to grow gradually over months and years of continued use rather than expanding rapidly enough to require immediate hardware upgrades.

It is also worth noting that the hardware supporting Operlytics does not need to be exclusively dedicated to the system. Because PHP, MySQL, and a standard operating system can coexist with other ordinary business software on the same machine, Empress Café Boracay is able to host the server-side components of Operlytics on a computer that may also be used for other administrative purposes, further reducing the capital investment required to adopt the system. Should transaction volume grow substantially in the future for instance, if the café expands its seating capacity or adds a second service counter the same architecture accommodates a straightforward hardware upgrade, such as increased memory or a faster processor, without requiring any change to the PHP application code or the MySQL database schema described in Section 4.4.

4.5.1 Hardware Requirements

Table 4.1 presents the recommended hardware specifications.

COMPONENT	MINIMUM SPECIFICATION	RECOMMENDED SPECIFICATION
Processor	Intel Core i3	Intel Core i5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	50 GB SSD	50 GB SSD or higher SSD
Monitor	1366 × 768 Resolution	Full HD Display
Internet Connection	Stable Internet or local area network	Stable Internet or Fast Internet
Client Device	Desktop, laptop, or tablet	Desktop, laptop, or tablet

The recommended hardware configuration provides sufficient processing capability for executing analytical operations and rendering graphical dashboards without significant delays.

4.5.2 Software Requirements

The software environment of Operlytics mirrors the tools and technologies identified in Chapter 3 and reflects the actual stack running in the deployed system. On the server side, the operating system hosting the application may be any platform capable of running PHP and MySQL, including Windows, provided that the required services can be installed and maintained on it. During development, the PHP and MySQL services are run locally through XAMPP, which bundles the Apache web server, PHP interpreter, and MySQL database engine into a single, consistently configured environment; this same combination of services continues to underlie the system once deployed for regular use at Empress Café Boracay.

The core software components required to run Operlytics are PHP, which executes the server-side business logic described in Section 4.4, and MySQL, which stores and manages the system's operational data across the USERS, ORDERS, ORDER_ITEMS, ORDER_REFUNDS, MENU, MENU_INGREDIENTS, INGREDIENTS, and SALE_REPORT tables. On the client side, Admin and Staff require only a modern, up-to-date web browser to access the system; Google Chrome is used as the primary browser throughout development and testing, and the interface built using HTML, CSS, and JavaScript is designed for compatibility with other widely used browsers as well. No additional client-side software installation is required beyond the browser itself, consistent with the accessibility goals discussed in Section 4.2.

For ongoing maintenance and future development of the system, Visual Studio Code continues to serve as the code editor through which the PHP, HTML, CSS, and JavaScript source files that make up Operlytics are edited and maintained, supporting any future enhancements identified through the feedback and retrospective process described in Section 4.3.

Maintaining currency across this software environment is a straightforward undertaking given the maturity and continued active support of each component. PHP and MySQL are both maintained through regular security and stability

updates, and the XAMPP distribution used during development packages compatible versions of Apache, PHP, and MySQL together, reducing the likelihood of version-mismatch issues when the environment is set up on a new machine or reinstalled following a hardware change. Because the client-side interface of Operlytics is built using standard HTML, CSS, and JavaScript rather than a proprietary or browser-specific technology, it remains accessible not only through Google Chrome, the browser used throughout development and testing, but also through other widely used modern browsers without requiring separate versions of the interface to be maintained for each one. This combination of a stable, well-supported server-side stack and a browser-based client-side interface is what allows Operlytics to be adopted by Empress Café Boracay without introducing software licensing costs or specialized technical maintenance requirements beyond what a small café establishment can reasonably sustain.

Table 4.2 presents the software requirements.

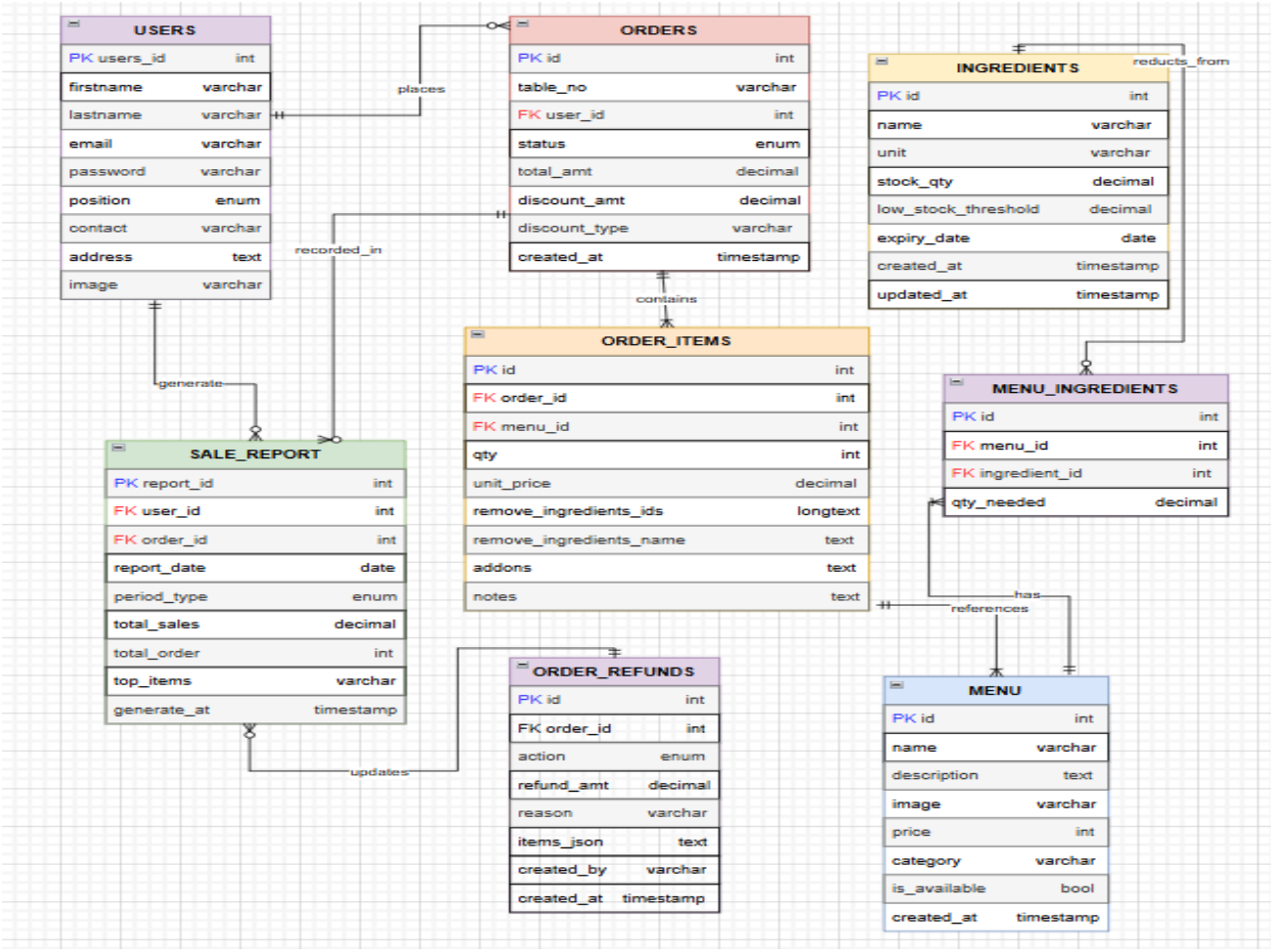
COMPONENT	REQUIREMENT	FUNCTION
Server Operating System	Windows	Hosts the application and database services
Local Development	XAMPP	Bundles Apache, PHP and MySQL for development and testing
Backend Language	PHP	Executes server-side business logic
Data Management System	MySQL	Store and manages operational data
Client Browser	Google Chrome or other modern browser	Accesses the Operlytics Web interface
Fronted Technologies	HTML, CSS, JavaScript	Renders and drives the client-side interface
Code Editor	Visual Studio Code	Supports ongoing development and maintenance

The selection of these software tools was based on their stability, community support, compatibility, and widespread adoption in modern web application development.

4.6 Database Design

The database serves as the structural foundation of Operlytics, organizing and storing all information related to user accounts, customer orders, menu offerings, ingredient inventory, refund transactions, and generated sales reports. A well-structured database design ensures efficient data retrieval, minimizes redundancy, and maintains the integrity of records throughout the system's daily operation.

Operlytics employs a relational database management system (MySQL) to structure and manage its data. [Note: confirm this matches the DBMS stated in your Chapter 3 Tools and Technologies section replace if different.] The database design follows normalization principles to reduce data duplication and preserve consistency across related tables, allowing the system to accurately track orders, deduct ingredient stock in real time, and generate reliable sales reports.



The primary entities of the database include:

- Users
- Orders
- Order Items
- Order Refunds
- Menu

- Menu Ingredients
- Ingredients
- Sale Report

Users Table

This table stores the account information of all system users, including administrators and staff members.

Field	Description
users_id	Primary Key
firstname	User's First Name
lastname	User's Last Name
email	User's Email Address
password	Encrypted Password
position	User Role (Admin / Staff)
contact	Contact Number
address	User Address
image	Profile Image

Orders Table

This table records all customer orders placed within the café.

Field	Description
id	Primary Key
table_no	Table Number
user_id	Foreign Key — References Users
status	Order Status
total_amt	Total Order Amount
discount_amt	Discount Amount Applied
discount_type	Type of Discount
created_at	Date and Time the Order Was Created

Order Items Table

This table stores the individual items included in each order.

Field	Description
id	Primary Key
order_id	Foreign Key — References Orders
menu_id	Foreign Key — References Menu
qty	Quantity Ordered
unit_price	Price per Unit
remove_ingredients_ids	IDs of Excluded Ingredients
remove_ingredients_name	Names of Excluded Ingredients

addons	Additional Items / Add-ons
notes	Special Order Notes

Order Refunds Table

This table tracks refund transactions associated with specific orders.

Field	Description
id	Primary Key
order_id	Foreign Key — References Orders
action	Refund Action Taken
refund_amt	Refund Amount
reason	Reason for Refund
items_json	Refunded Items (JSON Format)
created_by	User Who Processed the Refund
created_at	Date and Time of Refund

Menu Table

This table contains the list of available café products and their details.

Field	Description
id	Primary Key
name	Menu Item Name
description	Item Description
image	Item Image
price	Selling Price
category	Menu Category
is_available	Availability Status
created_at	Date the Item Was Added

Menu Ingredients Table

This table serves as a junction table linking menu items to their required ingredients.

Field	Description
id	Primary Key
menu_id	Foreign Key — References Menu
ingredient_id	Foreign Key — References Ingredients
qty_needed	Quantity of Ingredient Required

Ingredients Table

This table manages the café's inventory of raw ingredients and supplies.

Field	Description
id	Primary Key
name	Ingredient Name
unit	Unit of Measurement
stock_qty	Current Stock Quantity
low_stock_threshold	Minimum Stock Alert Level
expiry_date	Expiration Date
created_at	Date the Ingredient Was Added
updated_at	Date Last Updated

Sale Report Table

This table stores generated sales reports summarizing operational performance.

Field	Description
report_id	Primary Key
user_id	Foreign Key — References Users
order_id	Foreign Key — References Orders
report_date	Date of Report
period_type	Reporting Period Type
total_sales	Total Sales Amount
total_order	Total Number of Orders
top_items	Best-Selling Items
generate_at	Date and Time the Report Was Generated

The relationships among these tables allow Operlytics to trace each order back to the user who processed it, associate order items with their corresponding menu entries, and automatically deduct ingredient stock whenever a menu item is sold. A user account is linked to one or many orders and one or many generated sale reports; each order in turn contains one or many order items and may be linked to one or more refund records, while also feeding into the sales reports that summarize business performance. Each order item references a specific menu item, and each menu item is associated with one or many ingredients through the Menu Ingredients table, which in turn references specific entries in the Ingredients table. This chain of relationships is what enables the system's automatic ingredient deduction: when an order item is recorded, the ingredients tied to its corresponding menu item are reduced from stock accordingly.

Foreign key constraints and referential integrity rules ensure that sales data, inventory levels, and generated reports remain accurate and consistent, supporting reliable business analytics and informed managerial decision-making at Empress Café Boracay.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the summary, conclusion, and recommendations that emerge from the development and evaluation of Operlytics, the web-based business analytics and monitoring system built for Empress Café Boracay. It draws together the operational problems identified in Chapter 1, the theoretical and empirical grounding established in Chapter 2, the methodology and technical design described in Chapter 3, and the completed system presented in Chapter 4, in order to assess the extent to which the study achieves its general and specific objectives. The chapter is organized into six sections: the introduction presented here; a summary of findings that reviews the study in relation to its stated objectives and the five evaluation criteria of functionality, usability, accuracy, security, and efficiency; the conclusions drawn from these findings; recommendations directed toward Empress Café Boracay, future researchers, and future developers of the system; a discussion of the study's impact on its identified beneficiaries; and a final summary that closes the chapter and the study as a whole.

5.2 Summary of Findings

The findings of this study are organized around the five specific objectives established in Chapter 1, each of which corresponds to a concrete component of the system described in Chapter 4.

With respect to the first specific objective, the design and development of a web-based business analytics system, this study succeeds in producing Operlytics as a fully functional, integrated platform that unifies point-of-sale transaction recording, inventory management, and reporting for Empress Café Boracay within a single system. The Agile development methodology described in Chapter 3 and applied throughout Chapter 4 allows the system to progress through successive, testable increments, culminating in the four interdependent modules the POS module, the inventory management module, the reporting and dashboard module, and the user and access management module presented in Chapter 4.

With respect to the second specific objective, real-time monitoring of sales and inventory, the completed system demonstrates that a confirmed transaction at the point of sale is immediately reflected in inventory levels through the automatic ingredient deduction mechanism, without requiring any separate manual stock adjustment. This finding directly addresses the temporal gap between actual consumption and recorded inventory that Chapter 1 identifies as one of Empress Café Boracay's most persistent operational weaknesses, and it confirms that real-time monitoring, as designed, functions as intended within the deployed system.

With respect to the third specific objective, automated reports and dashboards, the study finds that the reporting and dashboard module is capable of compiling total sales, order counts, and top-selling items directly from stored transaction

records, without manual transcription or calculation on the part of café staff or management. This finding indicates that the system meets the objective of eliminating the labor-intensive, error-prone manual reporting process that previously consumed management's time at Empress Café Boracay.

With respect to the fourth specific objective, improvement of operational efficiency, the integration of the system's modules described in Chapter 4 indicates that a single sales transaction now triggers inventory deduction, report aggregation, and dashboard updates in one continuous process, removing the duplicate data entry and manual reconciliation that the café's previous manual practices required. This finding supports the objective of reducing recording errors and reporting delays, since standardized digital entry through the POS interface replaces the handwritten, inconsistently formatted records described in Chapter 1.

With respect to the fifth specific objective, evaluation of the system based on user feedback, this study relies on the five evaluation criteria of functionality, usability, accuracy, security, and efficiency established in Chapter 3 to assess the completed system. On functionality, the system demonstrates that each of its core features order processing, automatic inventory deduction, report generation, and dashboard visualization operates as specified and produces the outputs described in Chapter 4. On usability, the role-based separation between the Admin and Staff interfaces keeps the daily experience of Staff members focused on order entry, while giving the Admin complete access to configuration, inventory, and reporting functions, a design intended to minimize the training burden and technical difficulty that café personnel encounter when adopting the system. On accuracy, the study finds that because reports and dashboard figures are generated directly from the same transaction records used to process orders, rather than from a separately maintained summary, the outputs of the reporting module correspond exactly to recorded sales activity. On security, the system enforces authentication and role verification on the server side for every request, ensuring that Staff accounts cannot reach Admin-only functions such as report export or user management regardless of what is requested from the client interface. On efficiency, the centralization of business logic on the server, together with the client-server architecture described in Chapter 4, allows the system to process orders and refresh dashboard figures without requiring the client device to perform any business logic of its own, keeping the interface responsive during regular café operations.

It is important to note that the specific, quantitative results ordinarily reported in this section such as user satisfaction ratings, task completion times, response time measurements, and the number of respondents involved in usability and acceptance testing depend on the actual evaluation activities conducted with Empress Café Boracay's administrators and staff. As this document was prepared from the completed system description in Chapter 4 rather than from a separate results dataset, these figures are not fabricated here; the specific scores, response rates, and testing outcomes obtained during actual user evaluation should be inserted into this section once that evaluation has been conducted and documented.

5.3 Conclusion

Based on the findings summarized above, this study concludes that Operlytics successfully addresses the operational problems identified in Chapter 1 and achieves the general objective of developing and evaluating a comprehensive business analytics and monitoring system for Empress Café Boracay. The system replaces the manual, paper-based recording practices that previously exposed the café to inconsistent sales records, delayed and inaccurate inventory tracking, and labor-intensive manual reporting, with a single, integrated, web-based platform that performs these functions automatically and continuously.

The client-server architecture and the relational database structure implemented in Chapter 4 provide a reliable technical foundation for the system's core functions, ensuring that transaction recording, inventory deduction, and report generation remain consistent with one another as transaction volume accumulates over time. The Agile methodology adopted throughout development allows the system to be refined progressively in response to feedback from café stakeholders, resulting in features such as the refund and adjustment workflow that emerged directly from the practical experience of café personnel using the system rather than from the initial requirements analysis alone.

Taken as a whole, the study concludes that a purpose-built, web-based system of this kind is both a technically feasible and an operationally appropriate solution for a small food-service establishment such as Empress Café Boracay, and that the specific design decisions made throughout this study the modular separation of POS, inventory, and reporting functions; the role-based distinction between Admin and Staff access; and the automatic linkage between sales and inventory data collectively fulfill the objectives that motivated the study from its outset.

5.4 Recommendations

Based on the conclusions drawn from this study, the following recommendations are offered to Empress Café Boracay, to future researchers, and to future developers who may extend the Operlytics system.

For Empress Café Boracay, it is recommended that management formally schedule the user evaluation activities referenced in Section 5.2 involving both Admin and Staff personnel, so that the specific usability, accuracy, security, and efficiency results needed to complete this chapter's findings are documented under actual operating conditions rather than assumed from the system's design alone. It is further recommended that the café establish a routine schedule for reviewing the low-stock and expiry alerts generated by the inventory management module, so that the real-time monitoring capability described in Chapter 4 translates into consistent reordering practice rather than remaining an underused feature of the system. Management should establish and document standard operating procedures (SOPs) that define precise protocols for POS transaction processing, inventory adjustment authorization, report review schedules, and user account management. These SOPs will ensure consistency across staff transitions and operational periods. Regular database backups should be automated and verified, with backup archives stored in a geographically separate location or cloud storage service to protect against localized disasters. A contingency protocol should be developed addressing potential connectivity interruptions, including provisions for offline transaction queuing or paper-based emergency procedures that can be synchronized once connectivity resumes.

For future researchers, it is recommended that subsequent studies building on this work incorporate a structured, quantitative evaluation phase gathering task completion times, error rates, and satisfaction ratings from Admin and Staff users to empirically validate the usability and efficiency claims associated with the completed system. It is further recommended that future work examine the system's performance under higher transaction volumes than those currently observed at Empress Café Boracay, in order to confirm that the server-side architecture described in Chapter 4 continues to perform reliably as the business scales or as the system is adapted for other food-service establishments.

For future developers of Operlytics, it is recommended that subsequent versions of the system extend the reporting module to support additional analytical views, such as ingredient-level cost tracking and supplier performance comparisons, building on the MENU_INGREDIENTS and INGREDIENTS structures already established in the current database design. For automated financial reconciliation, supplier portals for electronic purchase ordering, and digital payment gateways (e.g., GCash, Maya, credit card processors) for streamlined transaction settlement. Advanced analytical visualizations, such as predictive inventory depletion curves, profit-margin sensitivity analyses, and comparative performance benchmarking against historical baselines or industry standards, would elevate the system's decision-support capabilities from operational monitoring to strategic intelligence. Multi-location architecture enabling centralized oversight of multiple café branches or franchise locations would support business growth and scalability without requiring fundamental system restructuring. It is further recommended that authentication mechanisms be periodically reviewed and updated in line with current security practices, ensuring that the access controls described in Section 4.4 continue to protect the café's operational data as the system remains in active use.

5.5 Impact of the study

The impact of this study is most immediately evident at Empress Café Boracay itself, where the developed system replaces manual sales recording, inventory tracking, and report compilation with a unified digital platform. Administrators are able to review sales and inventory information through the reporting and dashboard module rather than relying on fragmented paper records, and Staff members process customer orders through a standardized POS interface rather than through the manual methods described in Chapter 1. This shift is intended to reduce the recording errors, reporting delays, and inventory discrepancies that motivated the study, giving café management a more accurate and current view of daily operations than was previously available.

Beyond its immediate application at Empress Café Boracay, this study contributes to the broader body of work on information systems adoption among small, tourism-dependent food-service enterprises. The system design documented in Chapters 3 and 4 the modular separation of POS, inventory, and reporting functions, the automatic linkage between sales and ingredient consumption, and the role-based access structure separating Admin and Staff functions offers a practical reference point for other small businesses considering similar digital transformation efforts, particularly those operating under comparable resource constraints.

The study also holds academic relevance for the field of information technology and information systems research, illustrating how an Agile-based development process, informed by direct engagement with a real business stakeholder, can produce a system that is grounded in the specific operational realities of a small enterprise rather than in generic, one-size-fits-all assumptions about business software. In this respect, the study extends the literature reviewed in Chapter 2 by providing a documented, complete case of business analytics system development and evaluation within a Philippine small business and tourism context.

5.6 Summary

This study sets out to design, develop, and evaluate Operlytics, a web-based business analytics and monitoring system intended to resolve the operational challenges of manual sales recording, inventory tracking, and reporting at Empress Café Boracay. Guided by the general and specific objectives established in Chapter 1 and supported by the literature reviewed in Chapter 2, the study applies an Agile development methodology, described in Chapter 3, to design and build a client-server system organized around four interdependent modules: the POS module, the inventory management module, the reporting and dashboard module, and the user and access management module, as presented in Chapter 4.

The findings summarized in this chapter indicate that the completed system fulfills each of the five specific objectives set out at the beginning of the study, and that it is evaluated against the five criteria of functionality, usability, accuracy, security, and efficiency established in Chapter 3. The study concludes that Operlytics offers a feasible and appropriate solution to the operational problems documented at Empress Café Boracay, and it recommends that the café proceed to formal user evaluation, that future researchers pursue further empirical validation of the system's usability and performance, and that future developers continue to extend the system's reporting and security capabilities. Taken together, the chapters of this study document the complete progression of Operlytics from an identified operational problem, through a grounded review of related literature and methodology, to a working system whose design and evaluation are intended to benefit Empress Café Boracay directly and to inform similar efforts elsewhere.

End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical considerations, user safety, and data privacy.